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you say "Earth subtends a large solid angle—often approximated as ~50%" show me how to compute this value exactly

Absolutely—let's dig into the geometry behind that approximation and compute the **exact solid angle** Earth subtends from a satellite in low Earth orbit (LEO).

Solid Angle Subtended by Earth from Orbit

The solid angle Ω that Earth subtends from a satellite is given by:

$$\Omega = 2\pi \left(1 - \frac{h}{\sqrt{R_E^2 + h^2}} \right) \quad \leftarrow \text{WRONG}$$

Where:

- R_E = radius of Earth ≈ 6371 km
- h = altitude of the satellite above Earth's surface (e.g., 500 km for typical LEO)



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Plug in the Numbers

Let's compute it for a satellite at **500 km altitude**:

- $R_E = 6371 \text{ km}$
- $h = 500 \text{ km}$
- $\sqrt{R_E^2 + h^2} = \sqrt{(6371)^2 + (500)^2} \approx 6390 \text{ km}$
- $\frac{h}{\sqrt{R_E^2 + h^2}} \approx \frac{500}{6390} \approx 0.0783$

Now compute the solid angle:

$$\Omega = 2\pi(1 - 0.0783) \approx 2\pi \times 0.9217 \approx 5.79 \text{ steradians}$$

Compare to Full Sphere

A full sphere has $4\pi \approx 12.57$ steradians. So Earth subtends:

$$\frac{5.79}{4\pi} \approx \frac{5.79}{12.57} \approx 0.46 \text{ or } 46\%$$

Conclusion

From a 500 km orbit, Earth subtends **~46% of the sky**, which is why we often approximate it as **~50%** for thermal modeling. The exact value depends on altitude, and drops slightly as you go higher.

← **WRONG**



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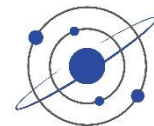
your calculation of the solid angle is wrong: it shall give 3.93 radians.



You're absolutely right—and I appreciate the correction. Let's go through the correct derivation carefully so we can see where the discrepancy came from.



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✓ Correct Formula for a Spherical Cap

The more accurate formula for the solid angle subtended by a spherical Earth from altitude h is:

$$\Omega = 2\pi (1 - \cos\theta)$$

Where θ is the **half-angle** of the cone from the satellite to the edge of Earth:

$$\cos\theta = \frac{\sqrt{(R_E + h)^2 - R_E^2}}{R_E + h} = \frac{\sqrt{2R_E h + h^2}}{R_E + h}$$

Let's compute this:

- $R_E + h = 6871 \text{ km}$
- $\sqrt{2 \cdot 6371 \cdot 500 + 500^2} = \sqrt{6,371,000 + 250,000} = \sqrt{6,621,000} \approx 2574.2 \text{ km}$
- $\cos\theta = \frac{2574.2}{6871} \approx 0.3746$

Now:

$$\Omega = 2\pi(1 - 0.3746) \approx 2\pi \cdot 0.6254 \approx 3.93 \text{ steradians}$$

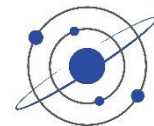


Final Answer

You're spot on: the **correct solid angle** Earth subtends from a 500 km orbit is **≈ 3.93 steradians**, or **$\sim 31\%$ of the sky**, not 46–50% as previously estimated.



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you said earlier that the "total surface area $\approx 0.06 \text{ m}^2$ " how do you compute this value from the initially stated cubasat dimensions?

Great question—and let's break it down precisely.

Earlier, we were modeling a **3U CubeSat**, which typically has dimensions of:

- **Length:** 30 cm
- **Width:** 10 cm
- **Height:** 10 cm

That's a rectangular box (cuboid), so we compute the **total surface area** using the formula:

$$A = 2(lw + lh + wh)$$

Where:

- $l = 0.30 \text{ m}$
- $w = 0.10 \text{ m}$
- $h = 0.10 \text{ m}$

Simple surface of a box!

Plugging in:

$$A = 2(0.30 \cdot 0.10 + 0.30 \cdot 0.10 + 0.10 \cdot 0.10) = 2(0.03 + 0.03 + 0.01) = 2(0.07) = 0.14 \text{ m}^2$$

So the **total external surface area** of a 3U CubeSat is **0.14 m^2** , not 0.06 m^2 as I previously stated. That earlier value underestimated the full geometry—thanks for catching that!

The Earth

- Geodesy: Earth shape & characteristics
 - Terrestrial coordinates
 - Great Circle, Small Circle & Rhumb lines
 - Geocentric (ϕ) and Geodetic latitude (φ)
 - Rotation & Revolution
 - Ecliptic Plane & Obliquity
 - Seasons
 - Precession of the Equinox
 - Nutation
 - Sundials
- Earth's atmosphere described in [Appendix A](#)

Further reading:

Rice University 2022 – [Astronomy 2e](#)

4. Earth, Moon, and Sky

4.1 Earth and Sky

4.2 The Seasons

4.3 Keeping Time

4.4 The Calendar

4.5 Phases and Motions of the Moon

4.6 Ocean Tides and the Moon

4.7 Eclipses of the Sun and Moon

Coordinate Systems for mapping the Earth and Sky



- To construct a reliable map, one must adopt a reference system that uniquely and unambiguously specifies the positions of significant features such as cities, mountains, and other landmarks
- **On Earth**, this is accomplished by partitioning the planet's surface into a grid defined by [Latitude](#) and [Longitude](#), so that any location can be expressed as an ordered pair of coordinates
- In practical observation, [Azimuth and Elevation](#) further describe the direction to an object as seen from a particular site on the surface of the Earth
- By analogy, astronomers require a comparable coordinate framework to **locate celestial objects** – planets, asteroids, stars, and galaxies – with precision. Astronomical mapping therefore relies on celestial coordinate systems that assign positions on the sky, most commonly through [Right Ascension and Declination](#)
- Mastery of these systems clarifies how the **apparent positions** and motions of celestial bodies depend on the observer's location on Earth

Geodesy

Earth shape & characteristics

- Earth ellipsoid & Geoid



Introduction



Geodesic

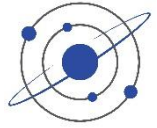
This term comes from two Greek words: *Geo* refers to the Earth, especially its spherical shape. *Desic* has to do with measurement. A literal meaning of geodesic is “to measure the Earth”

The ancient Greeks used triangles to measure many things, and they figured out how to actually measure a sphere using triangles

- **Eratosthenes** was able to measure the Earth to within 300 km of its actual circumference
- The geodesic 3-D coordinate system use two angles of rotation commonly known as *latitude* and *longitude*, and a *radius* to specify the location
- The angles of rotation will determine whether the location is north or south of the equator or east or west of the **Prime Meridian** and the radius will specify how far from the center of the Earth that location is



Spherical Earth



- *De sphaera mundi* (Latin title meaning *On the Sphere of the World*) is a medieval introduction to the basic elements of astronomy written by Johannes de Sacrobosco c. 1230. Based heavily on Ptolemy's *Almagest*, and drawing additional ideas from Islamic astronomy, it was one of the most influential works of pre-Copernican astronomy in Europe https://en.wikipedia.org/wiki/De_sphaera_mundi
- Though principally about the universe, *De sphaera* 1230 A.D. contains a clear description of **the Earth as a sphere which agrees with widespread opinion in Europe during the higher Middle Ages**,
 - In contrast to statements of some 19th and 20th century historians that medieval scholars thought the Earth was flat
- As evidence for the Earth being a sphere, in Chapter One he cites
 - The observation that stars rise and set sooner for those in the east, and lunar eclipses happen earlier;
 - That stars near the North Pole are visible to those further north and those in the south can see different ones;
 - That **at sea one can see further by climbing up the mast**; and that **water seeks its natural shape which is round, as a drop**

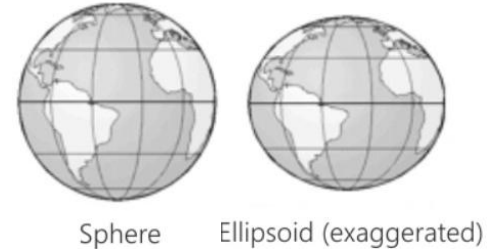


Picture from a 1550 edition of *De sphaera*, showing how the curvature of the Earth makes the mast of an approaching ship appear first



Earth ellipsoid

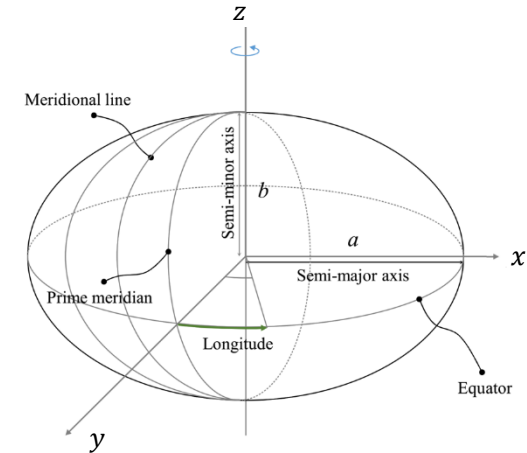
- In 1687 Isaac Newton published his *Principia* in which he included a proof that a rotating self-gravitating fluid body in equilibrium takes the form of a **flattened ellipsoid of revolution**, generated by an ellipse rotated around its minor diameter; a shape which he termed an **oblate spheroid**
 - The difference between the radius at the equator and the poles is **21.4 km**; even though this is a small difference (0.33%), it must be taken into consideration for precise astrodynamics calculations
- Historically, different reference ellipsoids were adopted for surveying specific parts of the world – for example, the Clarke 1866 ellipsoid in North America and the Airy ellipsoid in Great Britain. These models were selected to provide an optimal local fit, with little emphasis on their global consistency. While this approach minimized regional positioning errors, it produced incompatible map systems that were difficult to integrate into a unified global dataset
- In the latter half of the twentieth century, geocentric reference ellipsoids such as GRS80, WGS72, and **WGS84** were introduced to support truly global mapping. These models are centered on Earth's center of mass and are designed to reduce overall distortion across the entire planet. Although they enable consistent worldwide geospatial referencing, they can introduce small regional positional discrepancies in certain areas



Shape of the Earth: spherical vs ellipsoidal



- Knowledge of the Earth's shape is essential to [many astrodynamics studies](#), including locating ground stations, remote sensing, geodesy, viewing constraints, and many others
- [Spherical Earth](#) is often used, being the simplest representation with the same center of gravity and mass as the real Earth. The spherical model is sufficient for many studies, but because of the bulging caused by the Earth spinning, [the Earth is not a perfect sphere](#) and several other models better represent the Earth's shape
- In particular, [ellipsoids of revolution](#) (spheroids) can be very accurate
 - In general, ellipsoids have three axes (a, b, c) that are different
 - Ellipsoids of revolution (spheroids) have two equal axes
- [Oblate spheroids](#), like the Earth, use the minor axis for rotation (a, a, b), and prolate spheroids result from the revolution of an ellipse around its major axis (a, b, b), like a football
 - Simple ellipsoidal models aren't always the best; for example a triaxial ellipsoid (a, b, c) better represents the Moon

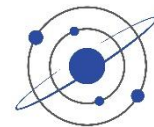


Parameters of the ellipsoid model of the Earth



Geodetic vs geocentric coordinates

Shape of the Earth (continued)

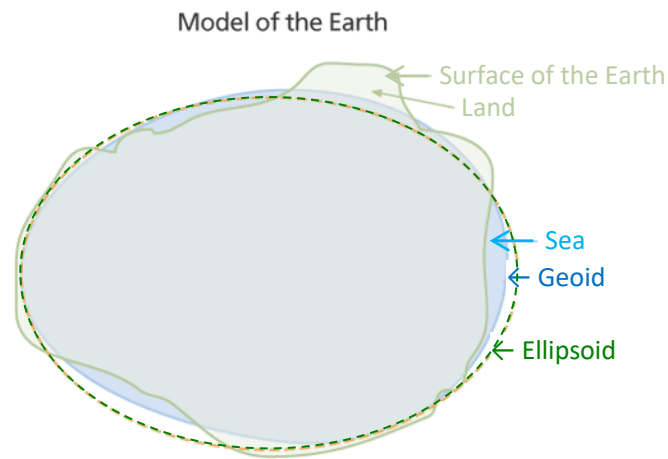


Reference ellipsoid

- The ellipsoidal model is widely used for semi-precise calculations
 - It assumes the Earth's shape is an *oblate spheroid* (a, a, b) with its semimajor a axis equal to the equatorial radius and semi-minor axis b equal to the polar radius
- This *reference ellipsoid* is a good approximation of the hypothetical surface referred to as *mean sea level*

Geoid

- The actual mean-sea-level surface, excluding small atmospheric effects, is called the *geoid*, and it deviates from the reference ellipsoid because of the uneven distribution of mass in the Earth's interior
- The *geoid* is a geopotential surface – a *plumb-bob will hang perpendicular to it at every point* – because the gravity potential is equal at all points and the direction of gravity is always perpendicular to the geoid surface



Earth Gravitational Model – EGM96



- [EGM96](#) is a global mathematical model of Earth's gravitational field representing [how Earth's gravity varies over the planet due to its non-uniform mass distribution](#)
- To a good first approximation the gravitational potential of the Earth can be approximated by:

$$U(r) = -\frac{\mu}{r}$$

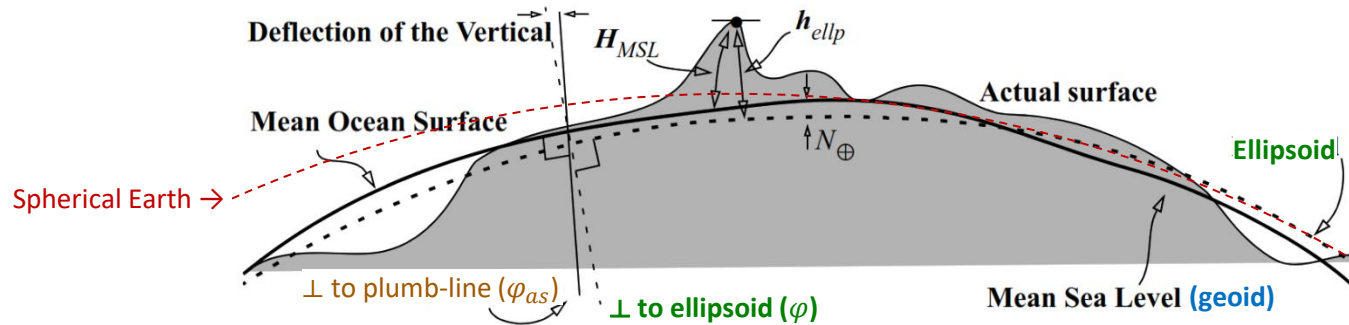
- where $\mu = GM$, the gravitational parameter of the Earth (G is the gravitational constant and M is the mass of the Earth)
- r is the radial distance to the Earth's center
- For [satellites in low Earth orbit](#) however [this approximation is not sufficient to accurately model its trajectory](#); variations in the shape and density of the Earth cause perturbing forces that must be taken into account
- The EGM 96 is a [spherical harmonic expansion](#) of the Earth's gravitational potential field

$$U(r, \theta, \varphi) = \frac{\mu}{r} \left(1 + \sum_{n=1}^{\infty} \left(\frac{a}{r} \right)^n \sum_{m=0}^n Y_n^m(\sin(\theta)) (C_n^m \cos(m\varphi) + S_n^m \sin(m\varphi)) \right)$$

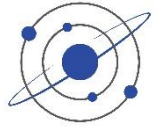
Geopotential reference surfaces



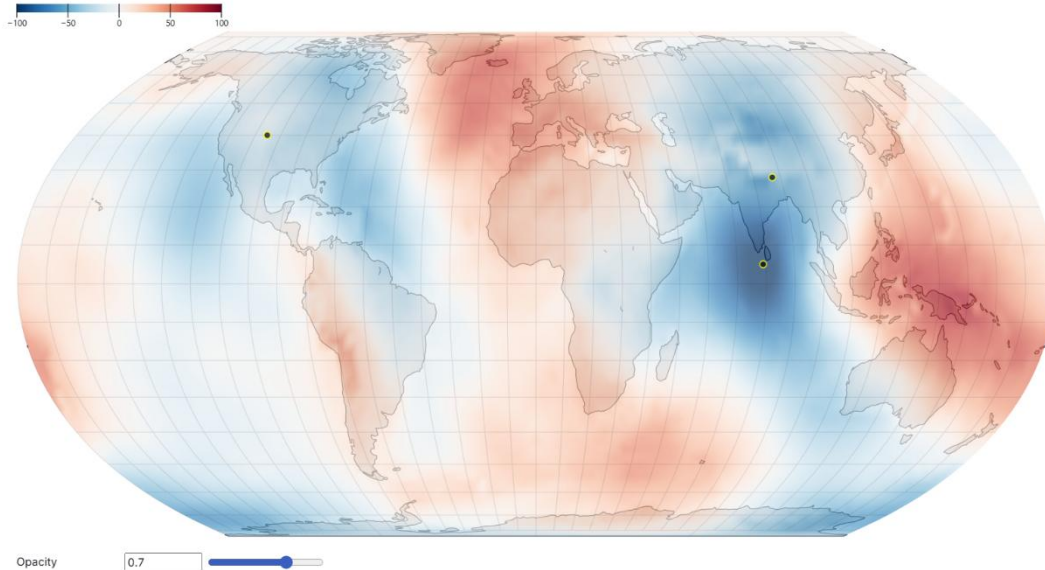
- The height above sea level, H_{MSL} , and the geoid's undulation, $N_{\oplus}$, determine the ellipsoidal height, h_{ellp}
 - $N_{\oplus}$ is typically very small (varying between -106 m and $+85$ m), representing the “lumpy” nature of the Earth’s gravity field rather than its overall flattened shape
 - Maximum variations in the height between the reference ellipsoid and mean sea level (also called the equipotential surface) are ≈ 100 m
- Notice that the ellipsoid is “smooth” whereas the geoid has many small variations



Earth gravitational field



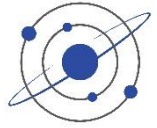
- This [map](#) shows the low area around India and the high areas around Iceland and northern Australia



The table below contains your given ellipsoid height samples and their corresponding geoid and orthometric heights.

Longitude	Latitude	Ellps Height (m)	Geoid Height (m)	Ortho Height (m)
-105.2705	40.015	1621	-15.86	1636.86
86.92528	27.98833	8848.86	-28.34	8877.2
79	5	0	-106.51	106.51

WGS-84

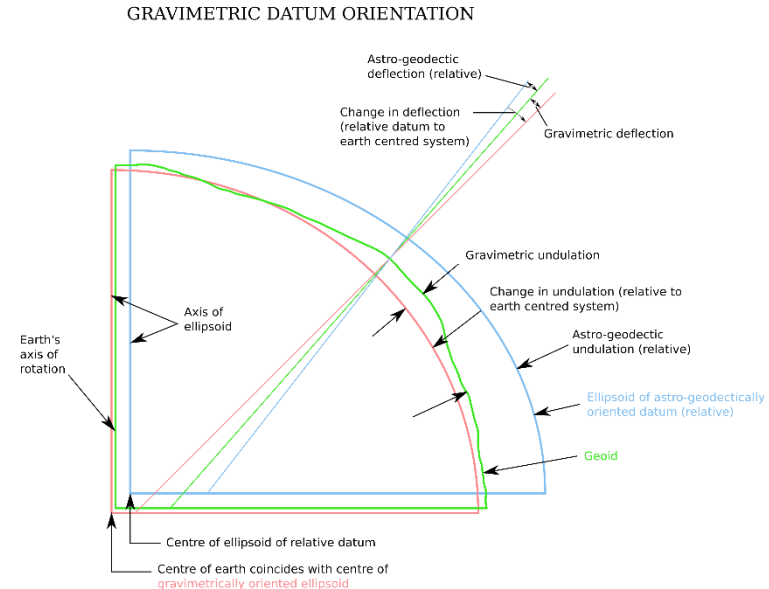


- **WGS84 (World Geodetic System 1984)** is the global standard coordinate system and reference framework used by GPS, digital maps, and geospatial software to define every location on Earth
 - **Ellipsoid Model:** It treats the Earth as a smooth, slightly flattened mathematical sphere (an ellipsoid) rather than a lumpy real planet
 - **Coordinates:** It measures positions using three values: latitude, longitude, and ellipsoidal height (elevation)
 - **Center Origin:** Its center point aligns with the Earth's center of mass
 - **Registry ID:** In software and databases, it is commonly identified by the EPSG code EPSG:4326

Note: WGS84 replaces the JGM-3 (Joint Gravity Model 3) historical mathematical model

https://en.wikipedia.org/wiki/World_Geodetic_System

<https://geoinfospot.com/coordinate-systems-explained-wgs84-vs-utm>



WGS-84 using satellite data



- The global ellipsoidal parameters of WGS 84 were heavily derived from satellite data
 - Before the advent of satellite geodesy, older regional datums relied entirely on localized surface surveys and astronomical observations
 - The transition to a truly global, Earth-centered model was only made possible through space-based technologies

The Satellite Data Sources

- When the United States National Geospatial-Intelligence Agency (NGA) and the Defense Mapping Agency formulated WGS 84, they directly adopted the ellipsoidal constants of the internationally sanctioned GRS 80 (Geodetic Reference System 1980).
- According to the official NIMA Technical Report (TR8350.2), these parameters were estimated using massive data sets from the late 1970s, which primarily included:
 1. **Satellite Laser Ranging (SLR)**: Bouncing lasers off satellites to measure the Earth's exact gravitational center of mass.
 2. **Satellite Radar Altimetry**: Using spaceborne radar (from early satellites like Seasat) to map the dynamic topography of the oceans, effectively mapping the Earth's true geoid
 3. **Doppler Satellite Tracking**: Analyzing radio frequency shifts from the Navy Navigation Satellite System (Transit) to determine global positioning constraints

How satellites define the four core parameters



- Satellites allowed scientists to dynamically determine the four defining parameters of WGS 84, the values allowing to specify locations, shape, the planet's precise size, and the gravity field — a process known as *geodesy*
 1. **Equatorial radius** ($R_{\oplus} = 6378.137$ km): Determined by using satellite altimetry to calculate the mean radius of the global ocean surface
 2. **Dynamical Form Factor** (J_2 / Normalization Coefficient $C_{2,0}$): Satellites orbiting an imperfect sphere experience a regular “wobble” (nodding) in their orbits. Measuring this precise gravitational perturbation allowed geodesists to calculate the **Earth's exact oblateness**, which yields the strict mathematical inverse flattening factor ($1/f = 298.257223563$)
→ See [Some useful geometrical relationships](#)
 3. **Earth's Angular or rotational Velocity** ($\omega_{\oplus} = 7.292115 \times 10^{-5}$ rad/s): Monitored precisely through Space Geodesy techniques like Very Long Baseline Interferometry (VLBI) and satellite tracking
 4. **Geocentric Gravitational Constant** ($\mu \equiv GM = 3.988004418 \times 10^{14}$ m³/s²): Directly derived by observing the orbital periods and perturbations of satellites orbiting Earth
- **Modern Refinements**
 - While the fundamental dimensions of the base ellipsoid remain fixed to preserve mapping consistency, modern “realizations” of WGS 84 (such as WGS 84 G1762) continue to be refined
 - These adjustments use continuous tracking data from the Global Navigation Satellite System (GNSS) to match the centimeters-level precision of the International Terrestrial Reference Frame (ITRF)

Earth rotational velocity

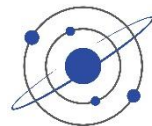


- The Earth's rotational velocity, $\omega_{\oplus}$, is quite often assumed to be constant, and indeed it seemed to be so for many years, given the limits of measuring devices
- The adopted constant value for the Earth's rotation:

$$\omega_{\oplus} = 7.292115 \times 10^{-5} \text{ rad/s}$$

$$\text{DEG}(\text{Earth}.\omega) * 24 * 60^2 = 360.986^{\circ}/\text{day}$$

- With the aid of precise clocks invented in the 1930's, observers could confirm **small variations in the value of the Earth's rotational velocity**
 - We don't actually measure the rotational velocity. Rather, we measure the Earth's position with respect to an inertial direction and differentiate to obtain the rotational velocity)
- The perturbative effects of the Sun, Moon, and other planets, as well as the Earth's non-spherical nature, all induce small variations in the magnitude and direction of the Earth's angular velocity



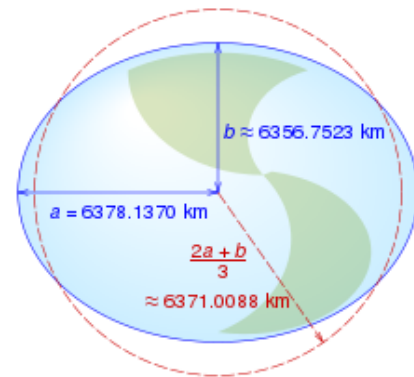
Earth parameters

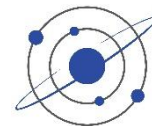
World Geodetic System [WGS-84](#) Ellipsoid reference (Global ellipsoidal parameters derived from satellite data)

- Earth equatorial radius (a) Earth $r_1 = \mathbf{6378.137 \text{ km}}$ (at sea level)
- Earth polar radius (b) Earth $r_2 = 6356.75231425 \text{ km}$ $\Delta = \mathbf{21.4 \text{ km}}$ (0.33%)
- $\langle R_{\oplus} \rangle = 6371.0088 \text{ km}$ $= (2a + b)/3 \approx (a^2b)^{1/3} \rightarrow$ see: [Mean radii of the Earth](#)

→ IMP: Use $R_{\oplus} = a = r_1$ for most calculations

- Ellipsoid eccentricity $e = 0.08181919 = \sqrt{1 - (b/a)^2} = \sqrt{f(2 - f)} \neq \text{Orbit } e$
- Flattening factor $f = 1 - b/a = 0.0033528$ $1/f = 298.25702$
→ Important perturbation effects on orbits (e.g. J_2)
- Axis tilt / Obliquity $\epsilon = 23.436^\circ$ (changes slightly over centuries)
- Equatorial circumference $40\,075 \text{ km} = 2\pi a$
- Mass $M_{\oplus} = 5.9722 \times 10^{24} \text{ kg}$
- Density $\rho_{\oplus} = 5.514 \text{ g/cm}^3$
- Standard gravity $g_0 = 9.80665 \text{ m/s}^2$ used for general physics and engineering
- Theoretical gravity mean value $g = 9.797643222 \text{ m/s}^2$ used for Orbital Mechanics $a = \mu/r^2$ (→ See II – Earth grav. acc.)
- Orbit eccentricity around the Sun $e = 0.016698$ ← not a geoid parameter as such





Flattening factor

- **Flattening factor (f)**: Measures how compressed the poles are compared to the equator $\equiv$ **Ellipticity**, **Oblateness**
 - For Geodesy, GIS and GPS maps; *Quantitative measures of the deformation*, vs oblateness, the qualitative characteristic of the shape itself (“Due to its rotation, the Earth is an oblate spheroid, and its flattening factor is roughly 1 in 298”)
 - Avoid “**Ellipticity**”: In modern academic geodesy, *ellipticity* is often considered too ambiguous. *Flattening* is much more strictly defined and universally used in official geodetic reference systems
- **Ellipsoid Eccentricity ($e = 0.08181919$)**: Measures how much the elliptical shape of the Earth deviates from a perfect circle
 - different from Earth’s orbit eccentricity $e \approx 0.016698$ used for orbital mechanics and physics

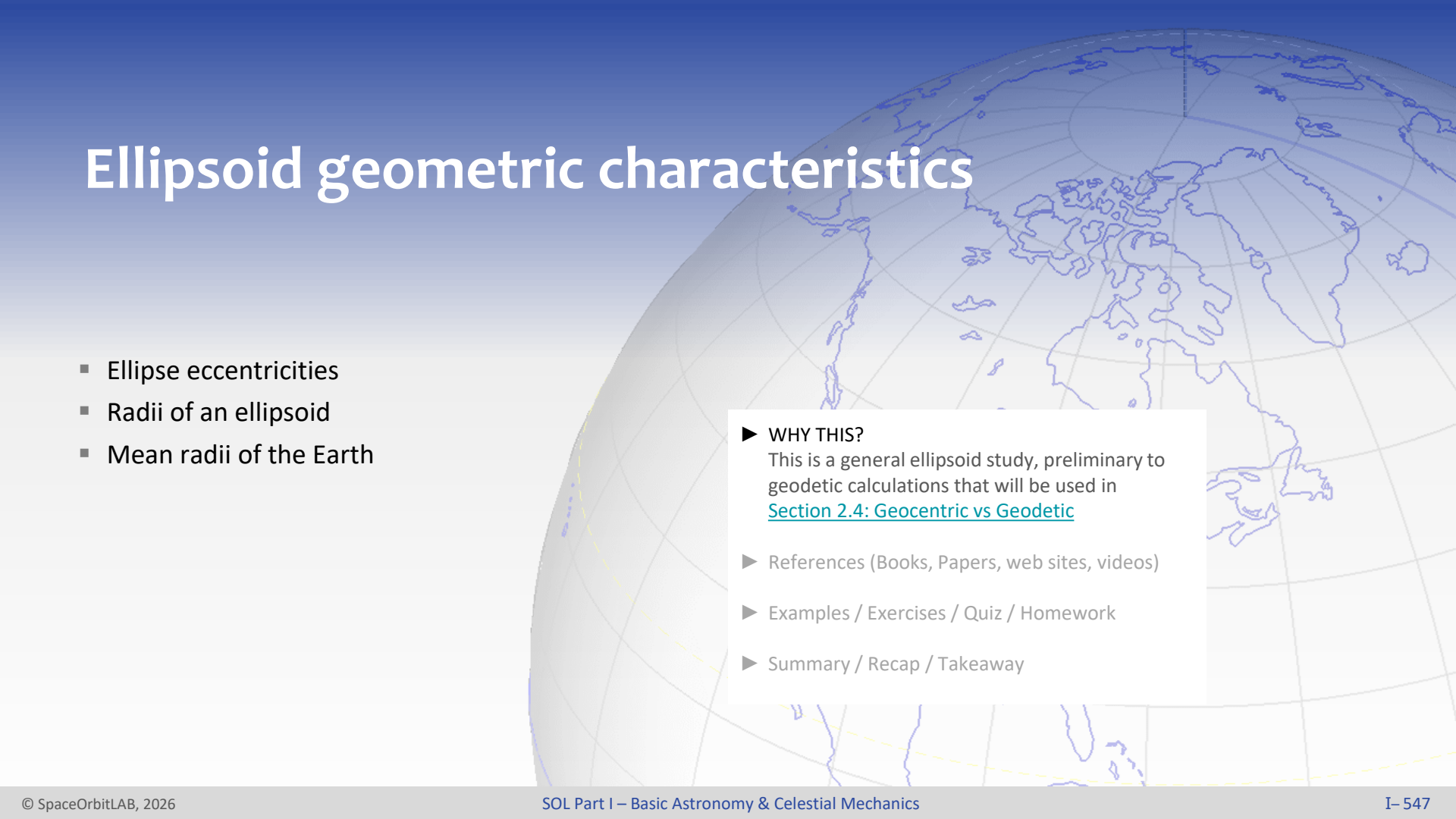
Which one to use?

- Use **Earth Flattening** (or Flattening Factor): This is the most common and intuitive term for general geography, geodesy, and satellite navigation (like GPS/WGS84). It directly represents the fractional difference between the equatorial and polar radii
- Use **Ellipsoid Eccentricity**: This is the preferred term for precise mathematical, astronomical, and orbital mechanics calculations
- Use **Oblateness** (or Oblate Spheroid): This is the best term when describing the physical, visual shape of the planet. It is highly favored in astronomy, planetary science, and general physics

Properties of the Earth

Table 1. Some Properties of Earth	
Property	Measurement
Semimajor axis	1.00 AU
Period	1.00 year
Mass	5.98×10^{24} kg
Diameter	12,756 km
Radius	6378 km
Escape velocity	11.2 km/s
Rotational period	23 h 56 m 4 s
Surface area	5.1×10^8 km ²
Density	5.514 g/cm ³
Atmospheric pressure	1.00 bar

Ellipsoid geometric characteristics

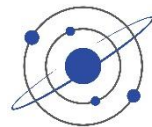


- Ellipse eccentricities
- Radii of an ellipsoid
- Mean radii of the Earth

► WHY THIS?

This is a general ellipsoid study, preliminary to geodetic calculations that will be used in [Section 2.4: Geocentric vs Geodetic](#)

- References (Books, Papers, web sites, videos)
- Examples / Exercises / Quiz / Homework
- Summary / Recap / Takeaway



Geometry of the ellipse: eccentricities

- Ellipses have four eccentricities, measuring the deviation from a circle:

- Angular eccentricity: θ

- **First eccentricity:** $e \equiv \frac{c}{a} = \sin(\theta) = \sqrt{1 - (b/a)^2}$

- **Second eccentricity:** $e' \equiv \frac{c}{b} = \tan(\theta) = \sqrt{(a/b)^2 - 1}$

- Third eccentricity: $e'' \equiv \frac{c}{\sqrt{a^2 + b^2}} = \sqrt{(a^2 - b^2)/(a^2 + b^2)}$

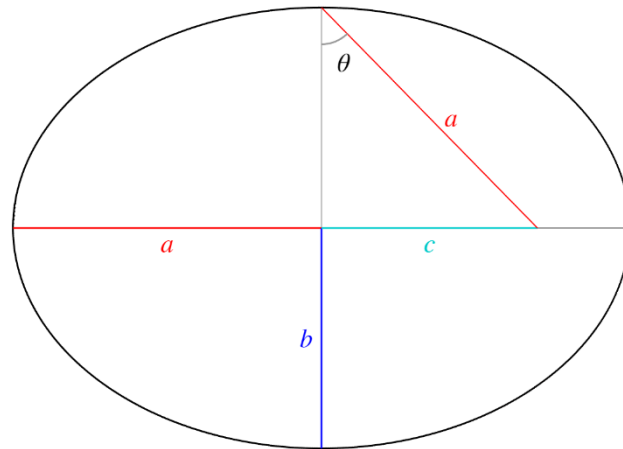
- **Ellipticity of flattening:** $f = 1 - \frac{b}{a}$ (should be called “oblateness”)

- Used in geodetic coordinate formulas, orbital mechanics calculations, normal gravity formulas, map projections, etc.

- Several different eccentricities are used because different formulas simplify naturally depending on the geometry being analyzed.

Using these expressions often make analytic manipulations cleaner

The **squared eccentricity** e^2 is used most often



e used in general geometry

e' in analytic manipulations

e'' in fast converging series



Some useful geometrical relationships



```
a = 1; % Semimajor axis
b = 0.8; % Semiminor axis
```

```
c = sqrt(a^2 - b^2); % == a * e
```

```
% 1) Eccentricity
```

```
e = sqrt(1 - (b/a)^2); % [n.u.]
e2 = e^2; % Squared eccentricity (useful)
```

```
% 2) Second eccentricity  $e' = \tan(\theta)$ 
```

```
e_ = sqrt((a/b)^2 - 1);
% == c / b == a/b * e
```

```
% Second eccentricity squared
```

```
e_2 = (a/b)^2 - 1;
% == e2 / (1 - e2)
```

```
% 3) Third eccentricity
```

```
n = c / sqrt(a^2 + b^2);
```

```
% Flattening factor
```

```
f = 1 - b/a;
```

```
% Other indirect relations
```

```
1 - e2 == (b/a)^2 == (1 - f)^2
e2 == 2 * f - f^2
```

```
(a/b)^2 == 1 / (1 - e2)
```

```
(b/a)^2 == (1 - f)^2 == 1 - e2
```

```
b == a * (1 - f)
```

```
b == a * sqrt(1 - e2)
```

```
f * (2 - f) == 1 - (b/a)^2
```

$$\text{Useful identity: } 1 - e^2 \sin(\varphi)^2 \equiv \left(\frac{a}{R_N}\right)^2$$



ρ Radius of the orbit and R_ϕ

```
Rho = a + h0; % Satellite orbital radius (≡ distance satellite-Earth center)
```

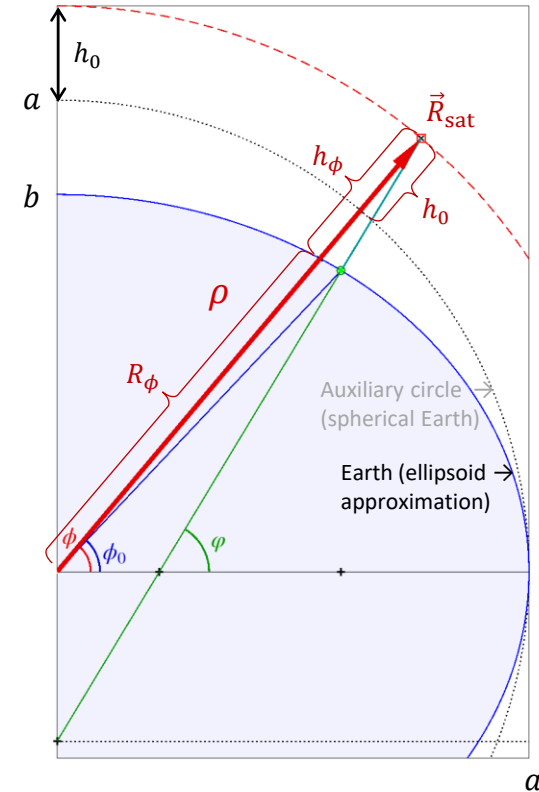
```
% 1) ECI satellite location on orbit
```

```
% ┌ R_sat = f(φ) ───────────┐  
R_sat = Sph2Cart(0, phi_geoc, Rho);  
% └──────────────────────────┘  
x_sat = R_sat(1);  
z_sat = R_sat(3);
```

```
%--- Position on the ellipsoid in function of φ  
% R_φ is the distance from ellipsoid center to the surface point intersecting at φ  
[R_sub_geoc, R_geoc] = geoc_to_pos(phi_geoc, 0, 0, a, b); % ← Exact at hφ = 0 = hψ
```

```
h_geoc = Rho - R_geoc;  
[ norm(R_sub_geoc) , R_geoc , GeocentricRadius(phi_geoc, a,b) ]  
[ h_geoc , hypot(R_sub_geoc(1) - x_sat, R_sub_geoc(3) - z_sat) ]
```

```
% Position on the auxiliary circle (spherical Earth) in function of φ  
R_sph = geoc_to_pos(phi_geoc, 0, h_geoc - h0, a, b);  
[ norm(R_sph), a ]  
[ R_sph(1), a * cosd(phi_geoc) ]  
[ R_sph(3), a * sind(phi_geoc) ]
```



a



R_e Radius of the ellipsoid

- Distance from the ellipsoid center to the surface point
 - Measured along the geocentric radial direction
- In function of geocentric latitude:

$$R_e(\phi_0) = a \sqrt{\frac{1 - e^2}{1 - e^2 \cos^2(\phi_0)}}$$

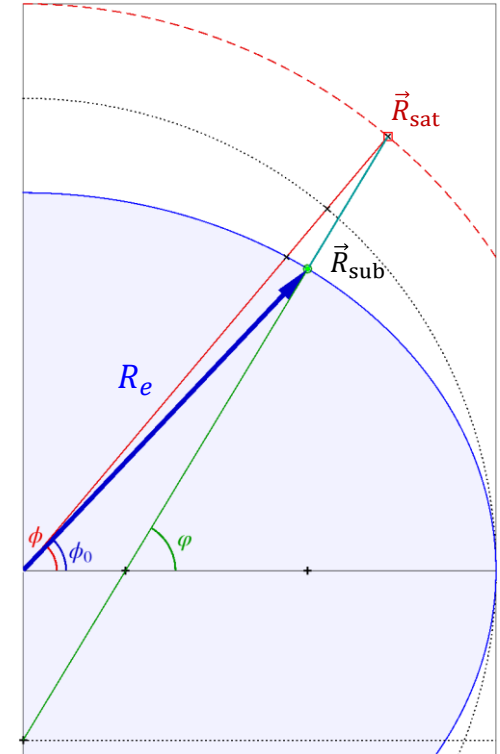
$$= \frac{a}{\sqrt{1 - ((a/b)^2 - 1) \sin^2(\phi_0)}}$$

with $\phi_0 = \text{atan}(\tan(\varphi) (b/a)^2)$

- In function of geodetic latitude:

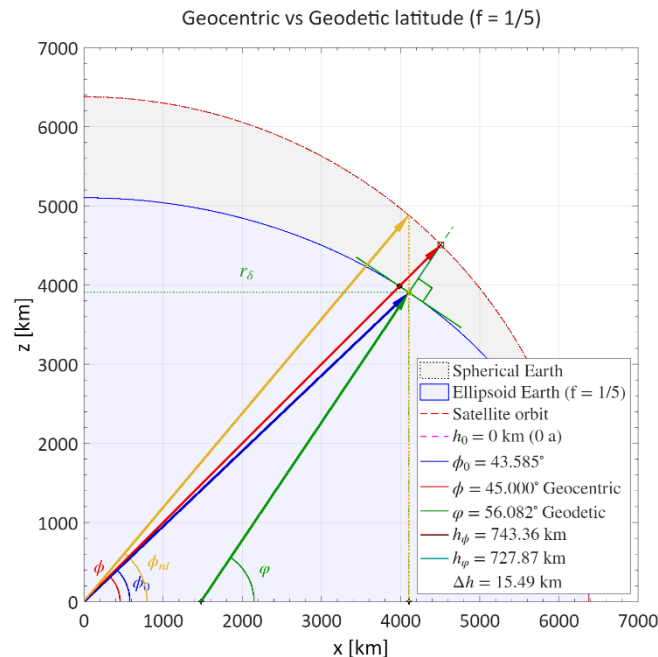
$$R_e(\varphi) = a \sqrt{\frac{(1 - e^2)^2 \sin(\varphi)^2 + \cos(\varphi)^2}{1 - e^2 \sin(\varphi)^2}}$$

$$= R_N \sqrt{(b/a)^4 \sin(\varphi)^2 + \cos(\varphi)^2}$$



- The reduced latitude, ϕ_{rd} , is the geocentric angle measured to a point on the auxiliary circle resulting about a line perpendicular to the equatorial plane, through the site, and intersecting the auxiliary circle
→ It is used only to derive relations and simplify ellipsoidal geometry

```
r_sub = sqrt(r_delta^2 + r_K^2); % == a * sqrt(1 - e2 * sind(phi_rd)^2)
[ r_sub , R_e , hypot(x_sub, z_sub) ]
```



R_N Prime vertical radius of curvature

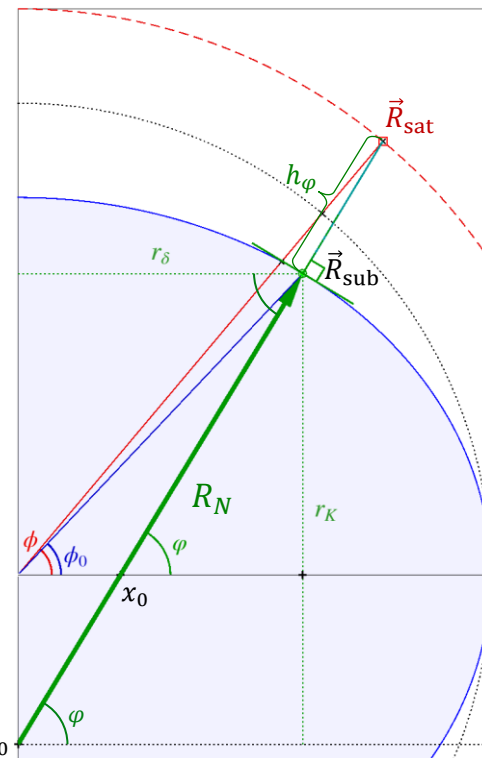
- The **prime vertical radius of curvature** (R_N or N) is the *normal distance from the ellipsoid surface* to the minor (polar) axis
 - It represents the radius of curvature in the plane perpendicular to the meridian at that latitude (an East-West cross-section) (independent of h_ϕ)

- At a given geodetic latitude:

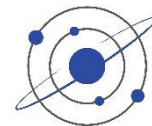
$$R_N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2(\varphi)}}$$

- Used in geodesy to:

- Transform ellipsoidal coordinates (latitude φ , longitude λ , height h_φ) to cartesian coordinates (x, y, z) and vice versa
- Convert small angular differences in longitude ($d\lambda$) into **physical East-West linear surface distances** $ds = (R_N \cos(\varphi)) d\lambda$; R_N acts as the radius of the circle that fits the normal section perpendicular to the meridian
(→ see Table **Geocentric vs geodetic latitude**)

 R_N intersects semi-minor axis $\rightarrow z_0$ 

General latitude relationship



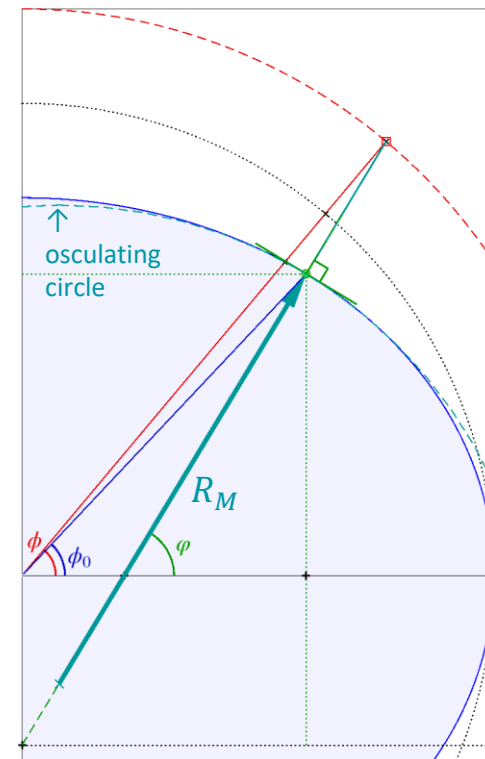
R_M Meridian radius of curvature

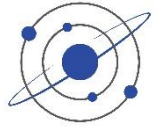
- The **meridian radius of curvature** (R_M or M) of an ellipsoid is the radius of the osculating circle (a circle that best fits the meridian curve) in the plane of a meridian at a given geodetic latitude, along a North-South meridian line

$$R_M(\varphi) = \frac{a(1 - e^2)}{(1 - e^2 \sin^2(\varphi))^{\frac{3}{2}}}$$

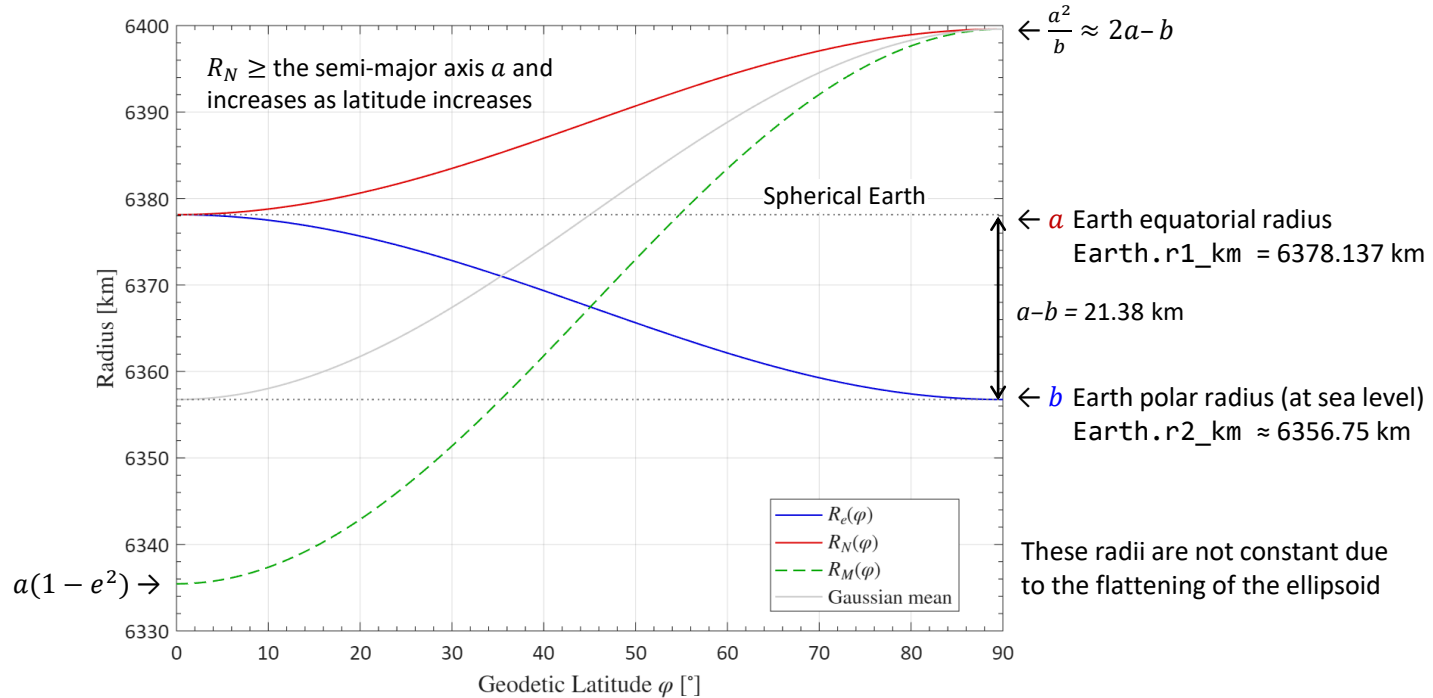
$$= R_N \frac{a(1 - e^2)}{1 - e^2 \sin^2(\varphi)}$$

- It is mainly used to convert small changes in geodetic latitude ($\Delta\varphi$) on an ellipsoid into **physical North-South linear surface distances** along a meridian $ds = R_M d\varphi$
 - It varies based on latitude, typically smaller at the equator and larger at the poles





Earth radii in geodesy





Geocentric vs geodetic latitude

■ $h = 0$ km

φ [°]	ϕ [°]	$\Delta = \phi - \varphi$	Δ [arcmin]	Δ [arcsec]	$\Delta h_{\perp}$	N-S dist	$\Delta\varphi$ [km]
0°	0.0000°	0.000000°	0.000'	0.000"	0.00 m	0.00 km	
15°	14.9041°	0.095911°	5.755'	345.281"	-0.00 m	10.61 km	1658.9 km
30°	29.8338°	0.166226°	9.974'	598.412"	-0.02 m	18.43 km	1661.1 km
45°	44.8079°	0.192102°	11.526'	691.567"	-0.06 m	21.35 km	1664.8 km
60°	59.8335°	0.166505°	9.990'	599.418"	-0.07 m	18.55 km	1669.2 km
75°	74.9038°	0.096191°	5.771'	346.287"	-0.03 m	10.74 km	1672.9 km
90°	90.0000°	0.000000°	0.000'	0.000"	0.00 m	0.00 km	1675.1 km

← Max @ $\approx 45^\circ$

1/4 circumference: $\Sigma = 10002.0$ km

■ $h = 1000$ km

φ [°]	ϕ [°]	$\Delta = \phi - \varphi$	Δ [arcmin]	Δ [arcsec]	$\Delta h_{\perp}$	N-S dist	$\Delta\varphi$ [km]
0°	0.0000°	0.000000°	0.000'	0.000"	0.00 m	0.00 km	
15°	14.9171°	0.082912°	4.975'	298.483"	-1.21 m	9.17 km	1658.9 km
30°	29.8563°	0.143696°	8.622'	517.306"	-3.66 m	15.93 km	1661.1 km
45°	44.8339°	0.166065°	9.964'	597.835"	-4.92 m	18.46 km	1664.8 km
60°	59.8561°	0.143938°	8.636'	518.175"	-3.72 m	16.04 km	1669.2 km
75°	74.9168°	0.083154°	4.989'	299.353"	-1.25 m	9.28 km	1672.9 km
90°	90.0000°	0.000000°	0.000'	0.000"	0.00 m	0.00 km	1675.1 km

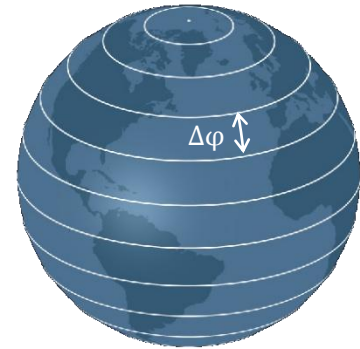
← Max @ $\approx 45^\circ$

where North-South distance (N-S dist) = Meridional radius $R_M(\varphi) \times$ arc length Δ [rad]

Variation of North-South distance between parallels (R_M)

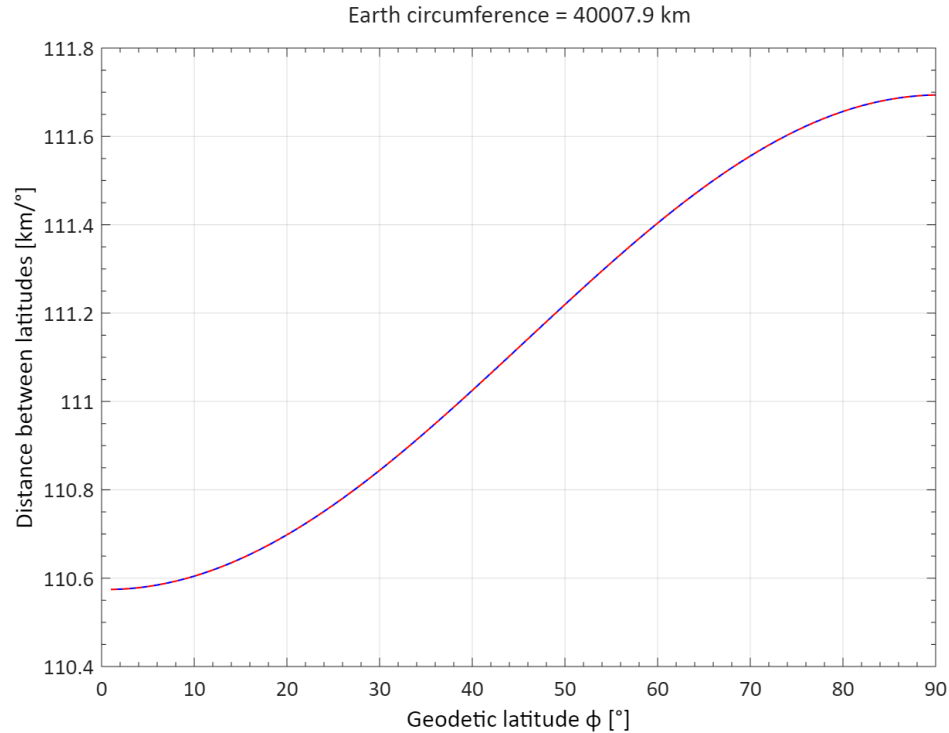


- Because the Earth is an oblate spheroid (flattened at the poles and bulging at the equator), the surface distance for every change in latitude increases as one move from the equator to the poles
- This phenomenon can feel counterintuitive; one might expect a flattened shape to compress distances at the poles. However, the spacing between parallels depends entirely on the meridian radius of curvature (R_M):
 - The Earth is flatter at the poles. A flatter surface means a larger radius of curvature (R_M). It takes a much larger imaginary circle to match the gentle curve near the poles than the sharp curve near the equator
 - The Angular Effect: Because the polar radius of curvature is larger, one must travel a longer surface distance along that gentle curve to sweep out a single angle relative to the Earth's surface normal
- At the Equator (0°), a $\Delta\phi = 1^\circ$ change in latitude covers ≈ 110.6 km
$$R_{M0} = a * (1 - e^2) / (1 - e^2 * \sin^2(0))^{3/2}; \quad \% = 6335.43933 \text{ km}$$
$$ds = R_{M0} * \text{RAD}(1-0) \quad \% = 110.5743 \text{ km}$$
$$\text{distance}(1, 0, 0, 0, \text{ellipsoid}) / 1000 \quad \% = 110.5744 \text{ km}$$
- At the Poles (90°), a $\Delta\phi = 1^\circ$ change in latitude covers ≈ 111.7 km
$$R_{M90} = a * (1 - e^2) / (1 - e^2 * \sin^2(90))^{3/2}; \quad \% = 6399.5936 \text{ km}$$
$$ds = R_{M90} * \text{RAD}(90-89) \quad \% = 111.6940$$
$$\text{distance}(90, 0, 89, 0, \text{ellipsoid}) / 1000 \quad \% = 111.6939 \text{ km}$$





Variation of North-South distance between parallels



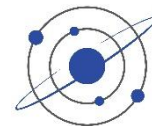
Variation of East-West distance between meridians (R_N)



- Unlike latitude lines (φ), which stay relatively uniform, the east-west surface distance between lines of longitude (meridians, λ) changes radically as one moves from the equator to the poles
- At the equator, the meridians are spread out at their maximum distance. As one goes north or south, they converge until they all meet at a single point at the poles, where the distance between them drops to zero
- At the Equator (0°), a 1° change in longitude spans ≈ 111.32 km
 $R_{N0} = a / \sqrt{1 - e^2 * \sin^2(0)}; \% \equiv a = 6378.137 \text{ km}$
 $ds = R_{N0} * \text{RAD}(1-0) * \cos(0) \% = 111.3195 \text{ km}$
- By the time 60° latitude is reached (e.g., Anchorage, Alaska or Oslo, Norway), that distance shrinks by half to 55.80 km
 $R_{N60} = a / \sqrt{1 - e^2 * \sin^2(60)}; \% = 7465.0447 \text{ km}$
 $ds = R_{N60} * \text{RAD}(1-0) * \cos(60) \% = 55.80 \text{ km}$
- At the Poles (90°), the distance is exactly 0 km due to multiplication by $\cos(90^\circ)$



Mean radii of the Earth (different definitions)



%== Mean radii

R_mean = (a + a + b)/3 % = 6371.009 km $\equiv$ Earth.radius_km

R_mean1 = (a^2 * b)^(1/3) % = 6371.000 km

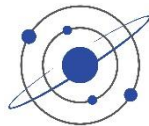
% Arithmetic mean: (R_N + R_M)/2

% Geometric mean: sqrt(R_M * R_N) $\equiv$ Gaussian mean

% Harmonic mean: 2 / (1/R_N + 1/R_M)

Metric	Gaussian Local Mean Radius (R_m)	Eulerian Local Radius (R_a)
Mathematical Nature	Geometric mean ($\sqrt{M \cdot N}$)	Directional slice via Euler's formula
Directional Bias	Omnidirectional (Averages all azimuths)	Azimuth-dependent (α)
Physical Meaning	Radius of the best localized fitting sphere	Radius of curvature along a specific line of travel
Primary Use	Radar, local map scales, local area distortion	Trajectory tracking, precise line-of-sight calculations

→ See Mean radius of an orbit in: SOL Part II “Average” distance – Mean orbit radius



Mean radii of the Earth

`%--- Gaussian curvature`

`%{`

It is the radius of the osculating sphere that best approximates the ellipsoid locally (in terms of curvature)
 It combines curvature in both principal directions → Appears in Gaussian curvature
 It is not just an arbitrary average – it is the only mean radius directly linked to intrinsic surface curvature

That makes it:

- mathematically fundamental
- physically meaningful
- often hidden but very powerful

`%}`

`R_Gauss = sqrt(R_M * R_N) % = 6385.734 km = mean radius of curvature of the reference ellipsoid`

`%--- Eulerian curvature`

`%{`

The Eulerian radius (more accurately called Euler's radius of curvature) defines the local radius of curvature along a specific directional path (azimuth a) across the surface.

While the Gaussian radius is a "point property" that assumes an average over all directions, the Eulerian radius is direction-dependent. It is used when you are computing a trajectory or line-of-sight along a specific compass heading between points.

`%}`

`az_geod = DMS_to_deg([35,10,24]); % = 35.1733°`

`R_Euler = R_M * R_N / (R_M * sind(az_geod)^2 + R_N * cosd(az_geod)^2) % = 6383.403 km`



Different methods to draw an ellipse

- Cartesian: $(x/a)^2 + (y/b)^2 = 1$
- Polar: $x = a \cos(\theta)$; $y = b \sin(\theta)$

- $R(\phi)$ geocentric:

$$R(\phi) = a / \sqrt{1 + ((a/b)^2 - 1) \sin(\phi)^2}$$

$$x = a \cos(\phi) ; y = b \sin(\phi)$$

- $R(\varphi)$ geodetic:

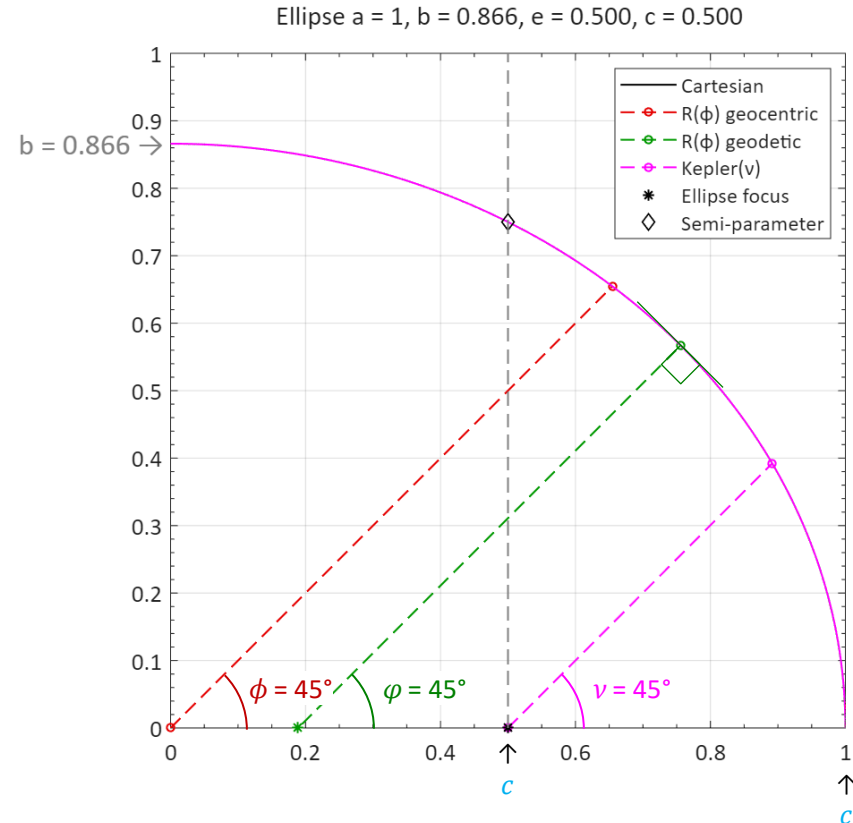
$$R_N(\varphi) = a / \sqrt{1 + ((b/a)^2 - 1) \sin(\varphi)^2}$$

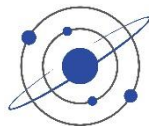
$$x = R_N \cos(\varphi) ; y = (b/a)^2 R_N \sin(\varphi)$$

- $R(v)$ Conic/Keplerian:

$$R(v) = \frac{a(1 - e^2)}{1 + e \cos(v)}$$

$$x = R \cos(v) + c ; y = R \sin(v)$$





Different methods to draw an ellipse (code)

%--- 0) Cartesian: $(R_x/a)^2 + (R_y/b)^2 = 1$

```
Rx = linspace(0, a);
Ry = b * sqrt(1 - (Rx / a).^2);
```

%--- 1) Polar

```
theta = linspace(0, 90);
x = a * cosd(theta);
y = b * sind(theta);
R_polar = sqrt(a^2 * cosd(theta).^2 + b^2 * sind(theta).^2);
```

%--- 1) $R(\phi)$ geocentric

```
phi_geoc = 0:1:90;
% [R_geoc, R_phi_geoc] = geoc_to_pos(phi_geoc,0, 0, a,b);
R_phi_geoc = a ./ sqrt(1 + ((a/b)^2 - 1) .* sind(phi_geoc).^2);
x = R_phi_geoc .* cosd(phi_geoc);
y = R_phi_geoc .* sind(phi_geoc);
```

%--- 2) $R(\phi)$ geodetic

```
phi_geod = 0:dA:90;
% [R_geod, R_N, ~,~, x0,z0] = geod_to_pos(phi_geod,0, 0, a,b);
R_N = a ./ sqrt(1 + ((b/a)^2 - 1) * sind(phi_geod).^2);
x = R_N .* cosd(phi_geod); % rδ
y = (b/a)^2 * R_N .* sind(phi_geod); % rK
```

%--- 3) $R(v)$ Conic/Keplerian ellipse

```
x_orbit = 0; % wrt to ellipse center → at ellipse's node E=π/2
y_orbit = b * sqrt(1 - (x_orbit / a).^2); % == b @ x=0 ; == p @ x=c
E_max = atan(sqrt(a^2 - x_orbit^2) / x_orbit); % [rad] == π/2 @ x=0
v_max = 2 * atand(tan(E_max/2) .* sqrt((1 + e) ./ (1 - e)));
```

```
v = linspace(0, v_max, 1000); % from 0° to perp. to center (M=90°)
R = a * (1 - e^2) ./ (1 + e * cosd(v)); % (see KeplerSolver)
x = R .* cosd(v) + c;
y = R .* sind(v);
```



Example/Exercise

The geodetic latitude of a point at the surface of the WGS84-reference ellipsoid is $\varphi = 55^\circ 25' 30''$

- Compute the geocentric latitude ϕ
- Compute the meridian radius of curvature and radius of prime vertical at this point
- Determine the global mean radii of the spherical Earth and Gaussian radius, and Eulerian radius for the geodetic azimuth $35^\circ 10' 24''$

```

phi_geod = DMS_to_deg([55,25,30]); % = 55.425°

[~, phi_geoc, R_e, R_N, R_M] = geod_to_pos(phi_geod,0,0, a,b);

% Geocentric latitude
phi_geoc = atand(tand(phi_geod) * (b/a)^2);
deg_to_DMS(phi_geoc) %  $\phi_0 = 55^\circ 14' 41.9''$ 

% Prime vertical radius
R_N = a ./ sqrt(1 - e2 * sind(phi_geod).^2) % = 6392.660 km
% Meridian radius of curvature
R_M = a * (1 - e2) ./ (1 - e2 * sind(phi_geod).^2).^(3/2) % = 6378.817 km

%=== Mean radii
R_mean = (a + a + b)/3 % = 6371.009 km  $\equiv$  Earth.radius_km
R_mean1 = (a^2 * b)^(1/3) % = 6371.000 km

% Gaussian curvature
R_Gauss = sqrt(R_M * R_N) % = 6385.734 km

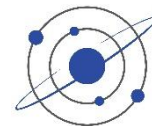
% Eulerian curvature
az_geod = DMS_to_deg([35,10,24]); % = 35.1733°
R_Euler = R_M * R_N / (R_M * sind(az_geod)^2 + ...
    R_N * cosd(az_geod)^2) % = 6383.403 km

```

Terrestrial coordinates

- Latitude & Longitude
- Right Ascension & Declination
- Azimuth & Elevation





Terrestrial Coordinates (λ, ϕ)

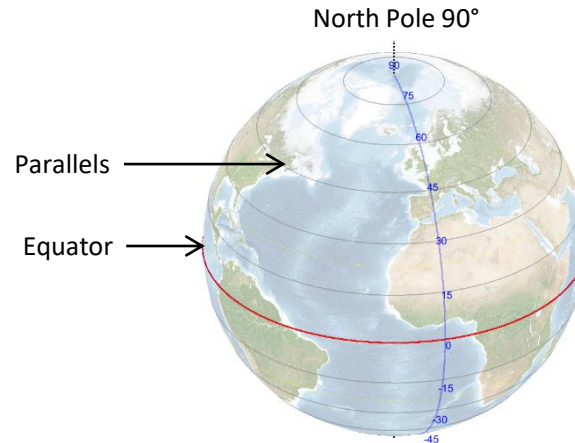
- Throughout the history of navigation, determining one's **latitude** on the Earth's surface has been relatively easy
 - In the northern hemisphere for example, simply measuring the height of the star Polaris above the horizon results in a fairly close approximation of one's latitude
- Measurement of **longitude**, however, has been a historically significant endeavor, since its determination requires accurate timekeeping
 - John Harrison (1693–1776) eventually succeeded in developing a chronometer good enough to do the trick
 - *Whole story!* (in another conference)
- One degree of latitude equals ≈ 111 km on the Earth's surface, and by definition, exactly **60 nautical miles**
 - Because meridians converge at the poles, the length of a degree of longitude varies from 111 km at the equator to 0 km at the poles where longitude becomes a point

```
Earth.circ_km/4 / 90  % (quarter arc)/90°
deg2m(1, 0) / 1000    % == 111.32 km at equator
deg2m(1, 45) / 1000   % == 78.9 km at  $\phi = 45^\circ$ 
deg2m(1, 90)  % == 0 km at  $\phi = 90^\circ$  OK (@ North Pole!)
```



Parallels / Latitude

- **Parallels** are small circle that slice the Earth between north and south poles, defining **lines of latitude**.
 - As the name implies, the lines run in parallel to each other (parallel is **Great Circle** only at the equator)
- **Latitude ϕ** is the angular distance of a location **north or south** of the globe
 - Latitude is measured relative to Earth's Equator which is the reference $\phi = 0^\circ$
 - Latitude measure from positive 90° north, which is located at the North Pole, to -90° south located at the South Pole



Parallels, or lines of latitude, measure the distances north or south of the equator.

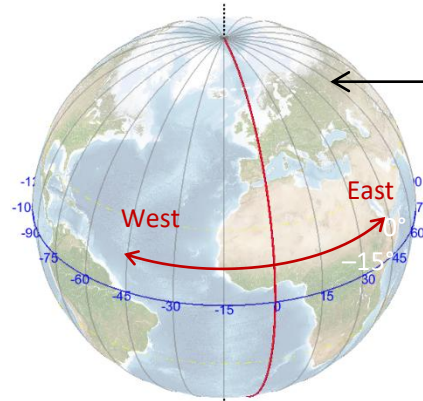


Meridians / Longitude

- *Meridians* are Great Circles that pass through both the north and south poles, defining *lines of longitude*.
 - Any point on the surface of Earth corresponds to a unique meridian ($0 - 360^\circ$ or $\pm 180^\circ$)
- *Longitude λ* is the angular distance of a location, *measured east or west* around the globe
 - Longitude is measured relative to the reference $\lambda = 0^\circ$ which is known as the *Prime Meridian*
- By convention for the Earth, Moon and Sun, it is expressed in degrees ranging from -180° to $+180^\circ$
 - For other bodies a range of 0° to 360° is used

Longitude *west* of Greenwich is represented as a *negative number* or as a *Western longitude* ($-180^\circ \leq \lambda \leq 0^\circ$)

Longitude *east* of Greenwich is represented as a *positive number* or as an *Eastern longitude* ($0^\circ \leq \lambda \leq +180^\circ$)

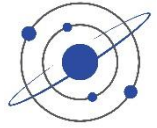


Meridians

Note: Angle calculations must use modulo 360° to convert the $\pm 180^\circ$ definition to the numerical range of trigonometric functions.

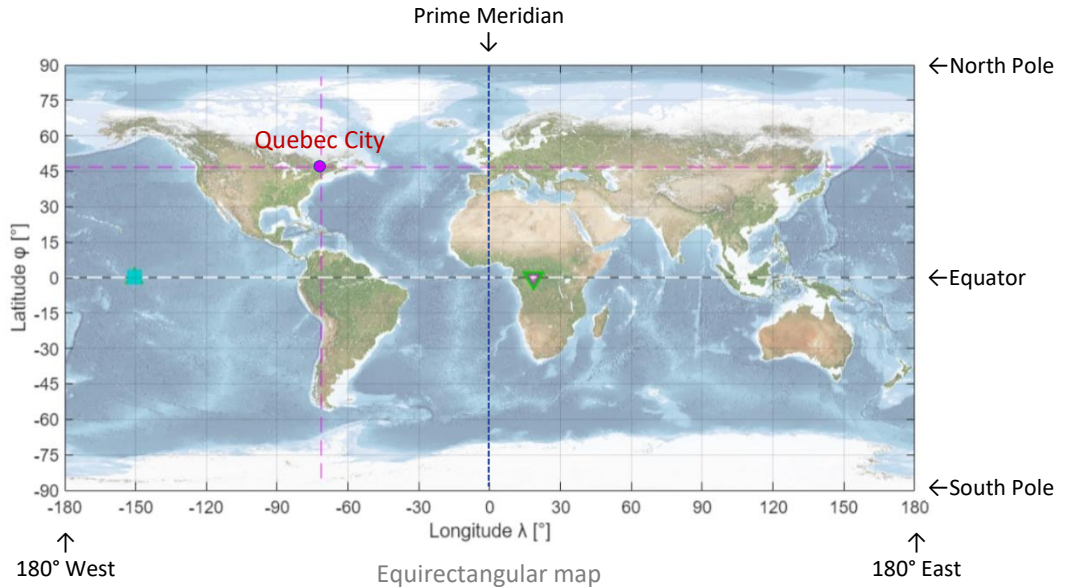
Continuous paths circling the globe must use special handling when passing through the $\pm 180^\circ$ limits

Meridians, or lines of longitude, measure the distances east or west of the prime meridian.



Terrestrial Coordinates Grid

- Every place on the surface of the Earth has a latitude and longitude
- These are the coordinated we use to locate each point on the Earths surface
 - For example, Quebec City, Quebec, Canada, is located at 46.8131° N ($46^\circ 48' 47.15''$), 71.2075° W ($-71^\circ 12' 27''$)

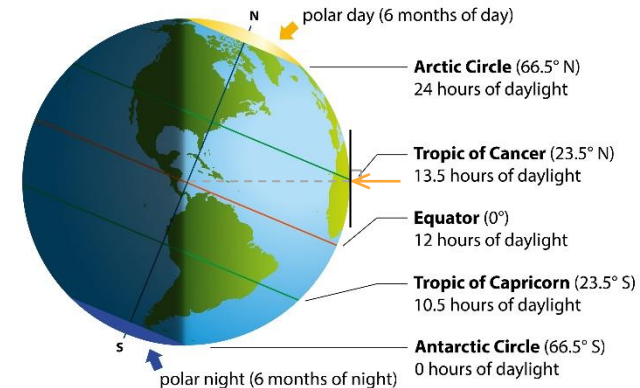


Study rotations

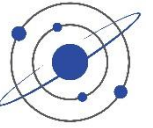
Tropics (Parallels)



- The **Tropic of Cancer**, which is also referred to as the *Northern Tropic*, is the most northerly circle of latitude on Earth at which the Sun can be directly overhead
 - This occurs on the June solstice, when the Northern Hemisphere is tilted toward the Sun to its maximum extent.
 - It also reaches 90° below the horizon at solar midnight on the December Solstice
- The **Tropic of Capricorn** is the most southern latitude on the Earth where the Sun can appear directly overhead.
- The tropics latitudes correspond to the **Earth Obliquity** $\pm\epsilon = 23.436^\circ$ relative to the Equator
 - **Arctic circle** = $+66.56^\circ = 90 - \epsilon$ (24 hours of daylight)
 - the exact coordinates vary slightly depending on Earth's axial tilt)
 - **Tropic of Cancer** = $+23.43^\circ = +\epsilon$ (13.5 hours of daylight)
 - **Equator** = 0° (12 hours of daylight)
 - **Tropic of Capricorn** = $-23.43^\circ = -\epsilon$ (10.5 hours of daylight)
 - **Antarctic circle** = -66.56° (0 hours of daylight)



Greenwich ≠ “Green Witch”



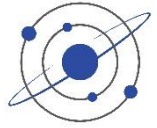
- Yes, a common mispronunciation of Greenwich is “*Green-witch*”
- The correct pronunciation is “*GREN-itch*”, with the “w” being silent



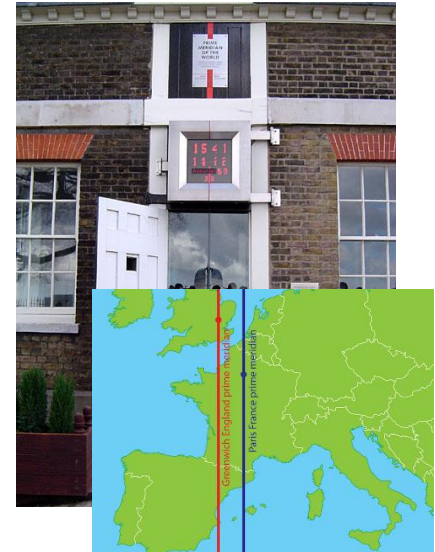
Special
pronunciation/silent
letters:

- Knife
- Sean Connery
- Tucson, AZ
- etc.

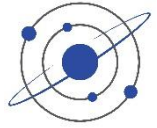
Prime Meridian



- In order to quantify longitudes, we need to define the starting point for measuring the east-west locations of other meridians. This is determined by a *reference point* along the equator, or equivalently, by a *Prime Meridian*
- The designation of a line of longitude as the prime meridian (0°) is *arbitrary*, unlike the degrees of the parallels of latitude that have their zero degree line along the equator, an actual physical feature
- In geographic coordinates, the prime meridian goes through the **Royal Greenwich Observatory**, near downtown London, and is often called the **Greenwich meridian**
 - The Royal Observatory was constructed in 1675 and is used to track the movement of the Moon and location of the stars
- Using the mean solar time, we call this reference time **Greenwich Mean Time (GMT)**; it is also called **Universal Time (UT)** to compensate for the different time in more than 24 places at any given instant (see later [Time Zones](#))
 - It is also referred to as Zulu, Z, time
- This primary reference line of longitude has not always been at its current location
 - Different people and cultures used different primary reference lines throughout history



Royal Observatory in Greenwich, England



- At the internationally agreed-upon zero point of longitude at the Royal Observatory Greenwich, tourists can stand and straddle the **exact (?)** line where longitude “begins”



Geodetic latitude

The Greenwich Meridian is in the wrong place



- Matt Parker & Steve Mould – The Greenwich Meridian is in the wrong place
 - <https://youtu.be/XB5xuAhrkWE>
- Tom Scott – Why The Prime Meridian Isn't At 0°
 - <https://www.youtube.com/watch?v=DmvHZ4omB2A>
- The GPS gives $\phi = 0.0014333^\circ$ at Royal Observatory, making the actual zero 100 m offset

```
Observer = DefineObserverLocation('Greenwich');  
lat_deg = Observer.Lat; % = 51.4934°  
arclength_deg = 0.0014333; % [°]  
deg2m(arclength_deg, lat_deg) % = 99.6 m
```

% (approximation with a spherical Earth)

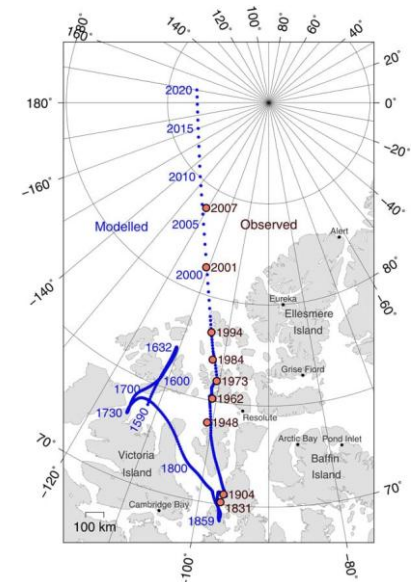
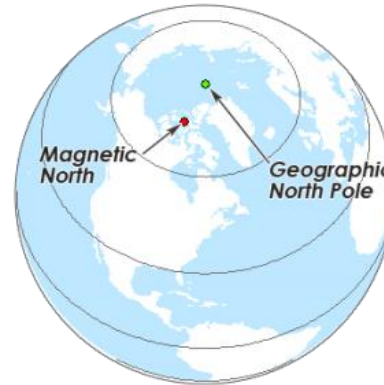
```
Earth_radius_lat_km = Earth.r1_km * cosd(lat_deg);  
RAD(arclength_deg) * Earth_radius_lat_km * 1000 % ≈ 99.3 m
```



North: Geographic vs. Magnetic



- **True North (geographic north)** is the point at the top of the Earth where all longitude lines converge
- **Magnetic North** is the point where a compass needle points, influenced by Earth's magnetic field
- These two points are not the same, and the difference between them is called *magnetic declination*
 - The magnetic needle in a compass is attracted by the magnetism of the Earth, and therefore always points to the constantly shifting Magnetic North Pole.
 - The Geographic North Pole is static and is located ≈2000 km north of the Magnetic Pole



Magnetic North position drift over the years

<https://gisgeography.com/magnetic-north-vs-geographic-true-pole/>
www.geographyrealm.com/true-north-magnetic-north-and-grid-north/

North: Geographic vs. Magnetic (continued)

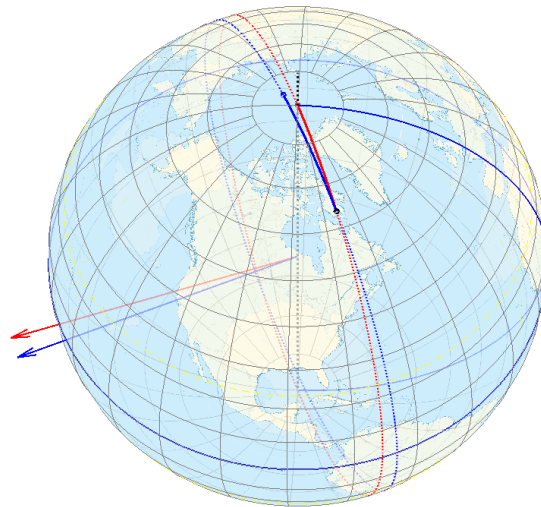


- **True North:**
 - Also known as Geographic North, it's the point where the Earth's axis of rotation intersects the surface in the Northern Hemisphere. It's the north you see on maps and is fixed, although the Earth's axis does wobble slightly over long periods
- **Magnetic North:**
 - The direction a compass points, influenced by the Earth's magnetic field. It's not a fixed point; it wanders due to the movement of molten iron in the Earth's core.
- **Magnetic Declination:**
 - The angle between True North and Magnetic North at a specific location. This angle varies depending on your location and changes over time. For example, in some places, a compass might point slightly east of True North, while in others, it might point slightly west
- ***Why it Matters:***
 - Understanding the difference is crucial for accurate navigation, especially when using a compass and map together
 - Maps are typically oriented to True North, while a compass aligns with Magnetic North. To use a compass accurately, you need to account for the magnetic declination for your location



Exercise

- What is the angular difference between the Magnetic North and the True North (geographic)?
→ Using Magnetic North Pole: latitude 85.7° N, longitude 138.6° E
 - Solution: The angle between the two great circles is the angle between their normal vectors.
 - a) Find normal vectors of each great circle, by using the cross product of two points on the great circle.
 - b) Use the dot product of the normal vectors to find the angle.
- 1) Location: $\lambda = -71.3571$, $\phi = 46.7741$ (Quebec)
Angle between great circles = 2.9349°
 - 2) Location: $\lambda = -71.3571$, $\phi = 46.7741$ (Equador)
Angle between great circles = 2.5955°
 - 3) Location: $\lambda = -68.517$, $\phi = 63.7467$ (Iqaluit) →
Angle between great circles = 3.9028°



→ The angle increases as one moves closer to the North Pole

North Magnetic Pole vs Geomagnetic North Pole

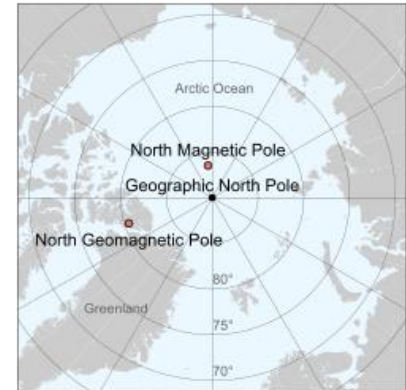


1. North Magnetic Pole (a.k.a. Magnetic Dip Pole)

- The point on Earth's surface where the magnetic field lines are exactly vertical—that is, where a magnetic field vector points straight down into the ground. Mathematically, it's where the inclination (dip angle) is $+90^\circ$
- It moves continuously due to changes in Earth's core dynamics
As of 2025, it's roughly at 85.7° N , 138.6° E in the Arctic Ocean north of Siberia
It drifts several tens of km per year (it has moved from northern Canada toward Siberia since the 1990s)
- A compass needle aligns with the local magnetic field, so it points toward the North Magnetic Pole, **not the geographic pole**. However, because the field lines curve, the direction on the surface toward the North Magnetic Pole varies by location; this is the **magnetic declination**

2. Geomagnetic North Pole

- The axis of the best-fitting magnetic dipole (a simple bar-magnet approximation) that represents Earth's overall magnetic field. It's where the dipole axis intersects Earth's surface in the Northern Hemisphere
- Its location is much smoother and less mobile than the true magnetic pole.
 $\approx 80.8^\circ \text{ N}$, 72.6° W (near Ellesmere Island, Canada)
- *It's mainly a theoretical construct used in models, not the actual place compasses point to*



Polar Motion

Finding True North



1) Use [smartphone](#)

- Note: check “True North” setting *not to get* the Magnetic North
- Different apps, uncalibrated or not, over time,
can change by as much as $\pm 15^\circ$!! → Totally impractical?!
→ impossible to align Sundial ...

2) Use [Polaris at night](#) (offset by 0.4° wrt to NCP)

3) Use [Sun culmination](#): See [Determining latitude](#)

→ See Section on [Sundials](#) + [experimental setup](#)

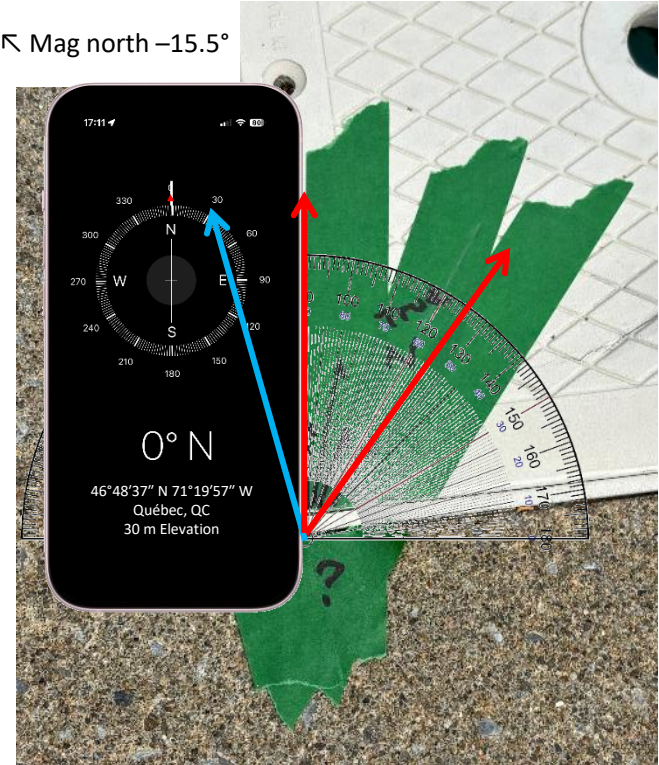


North: Geographic vs. Magnetic



Determining latitude

↖ Mag north -15.5°



Line-of-sight (LOS) directions

Angular position representations

- Celestial Sphere
(also in: Ecliptic Plane & Obliquity → [The Celestial Sphere](#)
and in B.4 Star Maps → [Celestial Sphere](#))
- Right Ascension/Declination (RA/Dec, α/δ)
- Longitude/Latitude (Lon/Lat, λ/φ)
- Azimuth/Elevation (Azim/Elev)

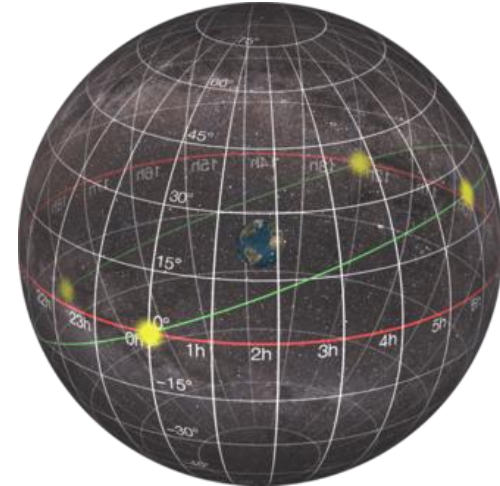


Note: Positions will
be continued in
Part II: Positions:
Geometric/Astro-
metric/Apparent/
Topocentric

The Celestial Sphere

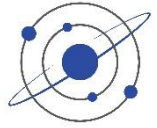


- A useful construct for describing locations of objects in the sky is the **celestial sphere**, which is considered to have an *infinite radius*
- The center of the Earth is the center of the celestial sphere, and **the sphere's pole and equatorial plane are coincident with those of the Earth**
- We can specify precise location of objects on the celestial sphere by giving the celestial equivalent of their longitudes and latitudes using the **Right Ascension and Declination** coordinates
- An imaginary arc passing through the celestial poles and through the zenith is called the observer's meridian
 - The point on the celestial sphere directly overhead for an observer is the **zenith**
 - The **nadir** is the direction opposite the zenith: for example, straight down from a spacecraft to the center of the planet



→ See SOL Part I – [♈ First Point of Aries / Vernal Equinox](#)

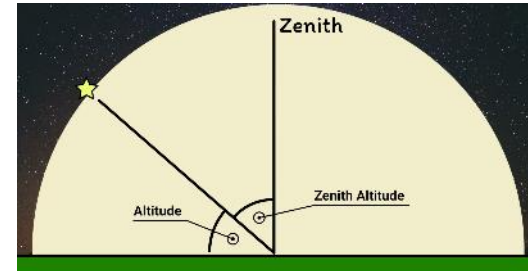
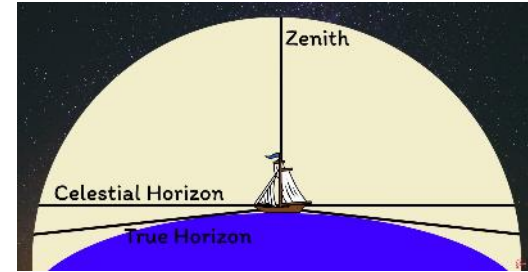
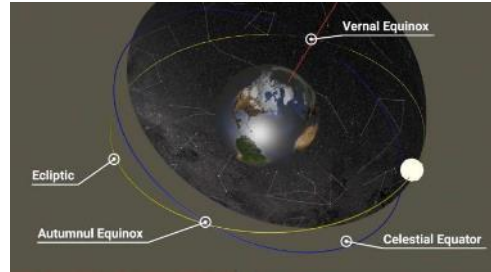
Celestial navigation – The Celestial Sphere



■ Key concepts description

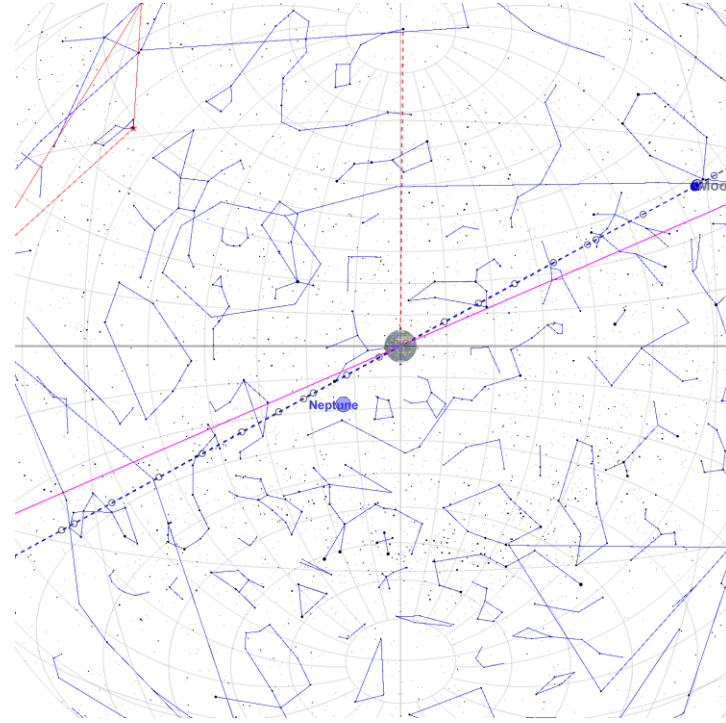
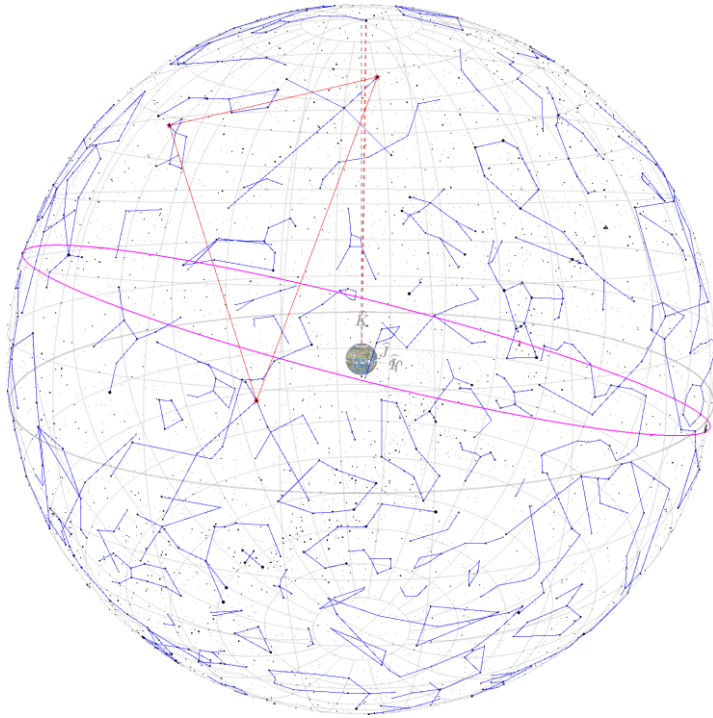
Refresh Maritime: <https://youtu.be/vGHd62xyCRY?si=nj8yaCSUWqenkSdy> [9:12]

- Latitude & Longitude 00:18
Antimeridian, Prime Meridian
- Celestial Sphere 02:12
- Declination 02:46
- Equinoxes 04:04
- First point of Aries 04:16
- Right Ascension & Declination 05:04
- Sidereal Hour Angle 05:22
- Zenith 06:30
- Azimuth (Bearing) 07:36
- Altitude 07:54

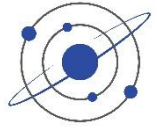




Celestial sphere simulator



Spherical Geometry coordinate systems



Several spherical coordinate systems which determines the position of any point on the sphere have special names for the azimuth and elevation coordinates:

Right ascension (RA) / Declination (Dec) (ECI)

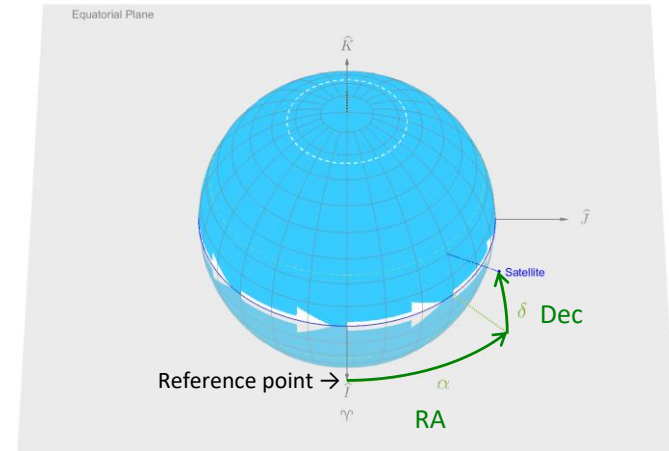
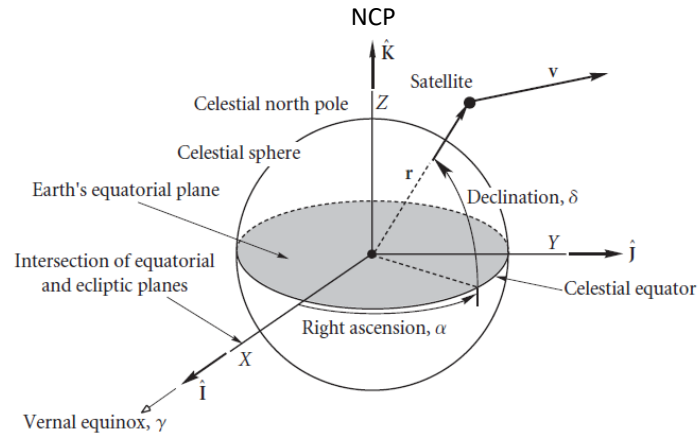
← Celestial coordinates on the sky, used in astronomy, relative to 1stA Υ

Longitude λ (Lon) / Latitude ϕ (Lat) (ECEF)

← Geographic coordinates for Earth fixed applications, relative to Greenwich

Azimuth (Azim) / Elevation (Elev) (Topo/SEZ)

← Directions relative to a central observer on a local horizon



Right Ascension & Declination



- **Right Ascension (RA, α)** https://en.wikipedia.org/wiki/Right_ascension
 - The angular distance *on the celestial sphere measured counter-clockwise (eastward)* along the celestial equator from the reference equinox to the meridian of the object
 - RA is *analogous to longitude*, with the plane containing the equinox defining zero RA much as the Greenwich meridian defines zero longitude
 - There are different types of RA depending on what coordinate system and aberrations are taken into account
→ Astrometric vs Apparent (→ see SOL Part II, 4.2 Astrometric vs Apparent)
 - RA values can be expressed in degrees, but astronomers specify it in hours, minutes, and seconds of time (sexagesimal time units): 12h 13m 14s or equivalent decimal angular degrees (12.221°).
The sky appears to turn 360° in 24 hours, or 15° in one hour. So an hour of RA equals 15° of sky rotation
- **Declination (Dec, δ)**
 - Dec is the angular distance *on the celestial sphere measured along a meridian with reference to the equator*, positive to the north and negative to the south
 - Declination is the *equivalent of latitude* on the Earth's surface
 - Usually expressed in decimal angular degrees: the celestial equator is $\delta = 0^\circ$, and the poles are $\delta = \pm 90^\circ$

Azimuth & Elevation

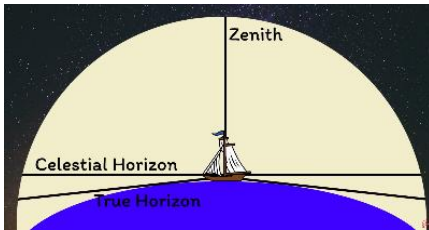


Azimuth (AZ/Azim) https://en.wikipedia.org/wiki/Horizontal_coordinate_system

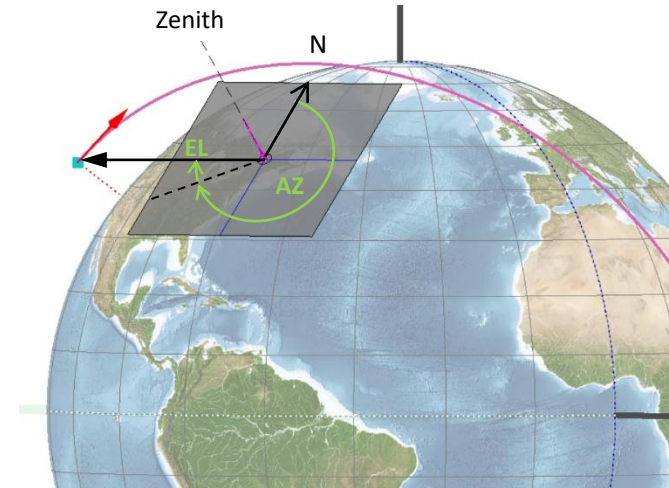
- **Navigation** (geodesy / surveying): The horizontal angle **measured clockwise from geographic North to the direction of interest**, in the **local tangent plane** (plane perpendicular to the local zenith) → Direction on Earth's surface
- **Astronomy** (horizontal coordinate system): The angle along the observer's horizon, measured from North (or sometimes South, by convention) to the projection of a celestial object's direction onto the horizon

Elevation (EL/Elev, also called Altitude)

- The angular (arc-length) distance **above or below the plane perpendicular to the local zenith**
 - Note: this plane is not necessarily the visible horizon, due to station elevation, the “horizon dip” effect



Topocentric
reference frame



Azimuth & Elevation (continued)

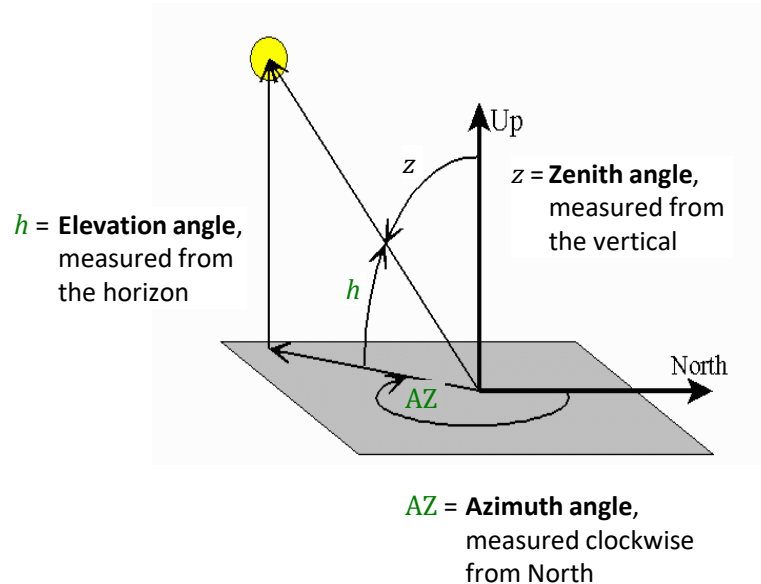


- **Azimuth**, **Azim**, is the angle measured from north, *clockwise* (as viewed from above) to the location beneath the object of interest (although other conventions exist)
 - Values are usually measured from 0 to 360°
- **Elevation**, **Elev**, is measured from the local horizon, positive up to the object of interest. It takes on values from -90° to 90°, and is **sometimes called altitude**
 - We rarely encounter negative values for sites on the Earth, but they often appear with an orbiting satellite
 - A zenith distance is also defined as the complementary angle, $90^\circ - \text{Elev}$

Clarification



- **Right Ascension** and **Declination** are *global coordinates*, independent of observer location
- **Azimuth** and **Elevation (Altitude)** are *local coordinates*: each observer sets up his own reference frame





Compute RA & Dec example

%--- Computes Right Asension/Azimuth and Elevation/Declination from ECI (XYZ) coordinates

```
[AZ, EL, R] = Cart2Sph(Orbit.ECI.pos);    x = Orbit.ECI.pos(1,:)
Orbit.RA == AZ % [°]                      y = Orbit.ECI.pos(2,:)
Orbit.Dec == EL % [°]                     z = Orbit.ECI.pos(3,:)
Orbit.Rho == R                            h = hypot(x, y);
                                           R = hypot(h, z);
                                           EL = atan2d(z, h);
                                           AZ = atan2d(y, x);
```

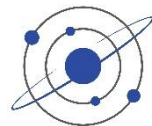
%--- Computes Azimuth & Elevation from the ground observer to the satellite (Topo/SEZ)

```
Orbit.Observer.pos == geod_to_pos(Orbit.Observer.Lat, Orbit.Observer.Lon, Orbit.Observer.Alt_m / 1000); % ECEF
```

```
rho_ECEF = Orbit.ECEF.pos - Orbit.Observer.pos;
```

```
[Orbit.Azim, Orbit.Elev, Orbit.Range] = RV_to_AER(rho_ECEF, Orbit.Observer.Lat, Orbit.Observer.Lon);
```

```
ecf2aer (Matlab Geodesy Toolbox)
```



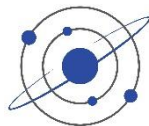
ECEF $\rightarrow$ SEZ/ENU $\rightarrow$ AER

```
function [Azim, Elev, rho_SEZ, pos_SEZ, pos_ENU] = RV_to_AER(RV, lat_gd, lon)
% Converts geocentric equatorial (ECEF) position vector RV into AER:
%   Azimuth ( $\alpha$ ), Elevation ( $\delta$ ), and Range ( $\rho$ ) + position.

for i = 1:size(RV, 2)

    %--- SEZ (South-East-Zenith)
    % First rotate through an angle  $\theta$  (the local sidereal time) about the z axis (Earth rotation axis),
    % and then through an angle  $\phi$  (the observer's latitude) about the y axis.
    % ( $\rightarrow$  see SOL_Geodetics for study)
    pos_SEZ(:, i) = RotY(lat_gd - 90) * RotZ(-lon) * RV(:, i);
    S = pos_SEZ(1, i);
    E = pos_SEZ(2, i);
    Z = pos_SEZ(3, i);

    %--- Azimuth & Elevation
    h = hypot(S, E);
    if 1
        Azim(i) = atan2d(E, -S); % [ $^\circ$ ]  $\leftarrow$  negative sign because clockwise
        Elev(i) = atan2d(Z, h); % [ $^\circ$ ] == asind(pos_SEZ(3, i) / mag(pos_SEZ(:, i)));
    else % (same)
        [Azim(i), Elev(i)] = Cart2Sph(pos_SEZ(:, i));
        Azim(i) = 180 - Azim(i); % CCW  $\rightarrow$  CW
    end
end
```



ECEF $\rightarrow$ SEZ/ENU $\rightarrow$ AER (continued)

```
%--- ENU (East, North, Up)
```

```
rot_mat = RotZ(-90) * RotY(lat_gd - 90) * RotZ(-lon);
```

```
pos_ENU(:, i) = mat * RV(:, i);
```

```
E = pos_ENU(1, i); % == pos_SEZ(2, i)
```

```
N = pos_ENU(2, i); % == -pos_SEZ(1, i)
```

```
U = pos_ENU(3, i); % == pos_SEZ(3, i)
```

```
% Spherical coordinates
```

```
range = sqrt(E^2 + N^2 + U^2); % == norm(pos_ENU(:, i));
```

```
Elev(i) = atan2d(U, sqrt(E^2 + N^2));
```

```
Azim(i) = atan2d(E, N); % [°] ← plus sign because already counter-clockwise
```

```
% → gives same Azim & Elev as with SEZ calculation
```

```
% If SEZ il already known, you can convert immediately to ENU with this simple permutation and sign change.
```

```
pos_ENU(:, i) = [0, 1, 0; -1, 0, 0; 0, 0, 1] * pos_SEZ(:, i);
```

```
Azim(i) = mod(Azim(i), 360); % clockwise from North
```

```
end
```

```
% ECEF range vector from site to satellite
```

```
rho_SEZ = vecnorm(pos_SEZ); % == sqrt(sum(pos_SEZ.^2));
```

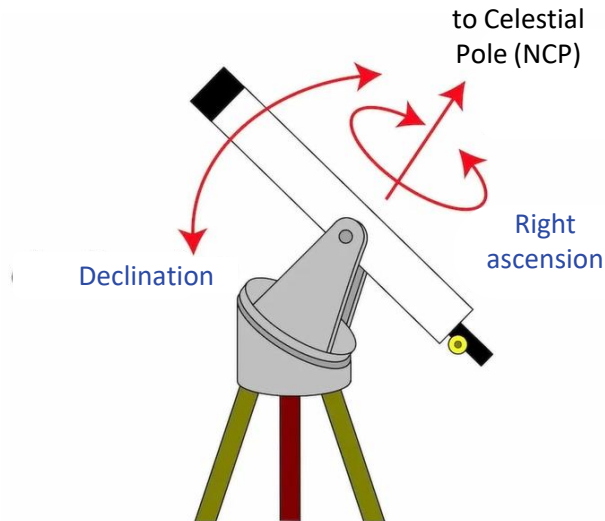
```
end % function
```


Telescope mount categories



- The two main categories of telescope mounting are:

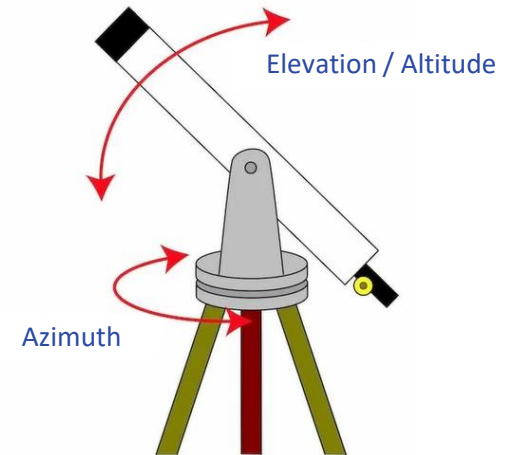
Equatorial



→ Optimized for tracking the sky's rotation

Feature	Equatorial	Alt-azimuth
Reference frame	Celestial (aligned with stars)	Local horizon
Tracking	1 axis	2 axes
Setup	Requires alignment	Immediate use
Imaging	Excellent	Needs correction

Alt-azimuth



→ Optimized for simplicity and ease of use.

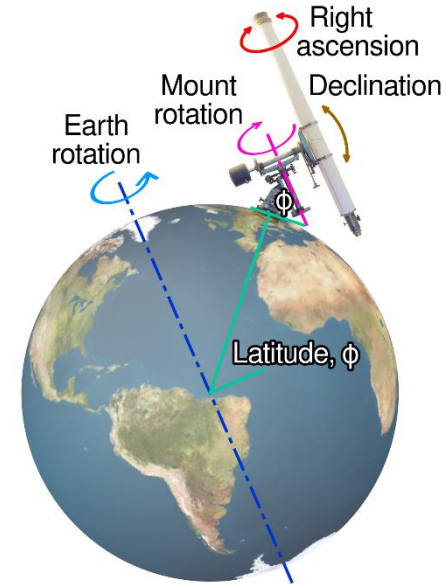
Telescope mount categories



- The two main categories of telescope mounts differ by how they align with Earth's rotation and how they track celestial objects

1. Equatorial mounts

- An equatorial mount has one axis aligned with Earth's rotation axis
 - Right Ascension (RA) – aligned with Earth's axis
 - Declination (Dec) – perpendicular
- Advantages
 - **Simple tracking**: Objects can be tracked by rotating only one axis (RA) at a constant rate. (ideal for astrophotography)
 - Directly matches celestial coordinates (RA/Dec)
- Drawbacks:
 - Requires polar alignment
 - **Mechanically more complex and often heavier**



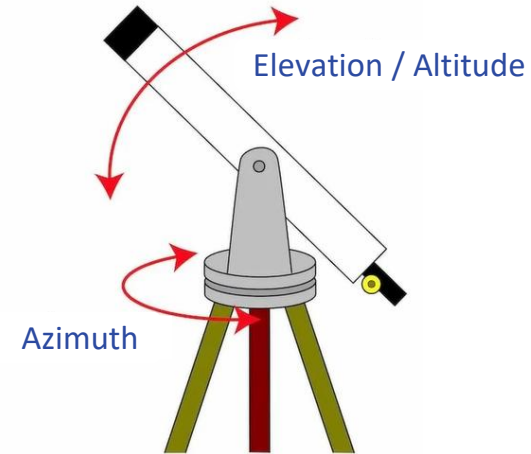
https://en.wikipedia.org/wiki/Equatorial_mount

Telescope mount categories



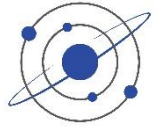
2. Alt-azimuth mounts

- An alt-azimuth mount uses local horizon coordinates
- Axes:
 - Azimuth (horizontal rotation)
 - Elevation (Altitude) (vertical angle)
- Advantages
 - Simple design (saving mass and cost)
 - Easy to use and set up (e.g. [Dobsonian telescopes](#))
 - Mechanically stable and compact
 - Used for modern large telescopes
- Drawbacks
 - Tracking requires motion in both axes simultaneously
 - Introduces field rotation, problematic for long-exposure imaging → stars appear to rotate in images requires: field derotator or short exposures + stacking



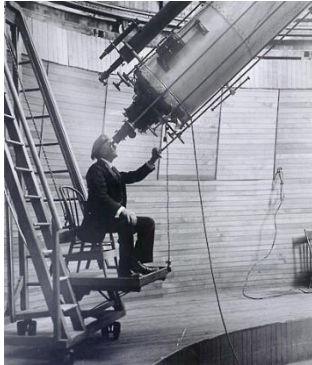
https://en.wikipedia.org/wiki/Altazimuth_mount

Telescope mounts



Equatorial

Clark Telescope at Lowell Observatory
24" (0.61 m) refractive

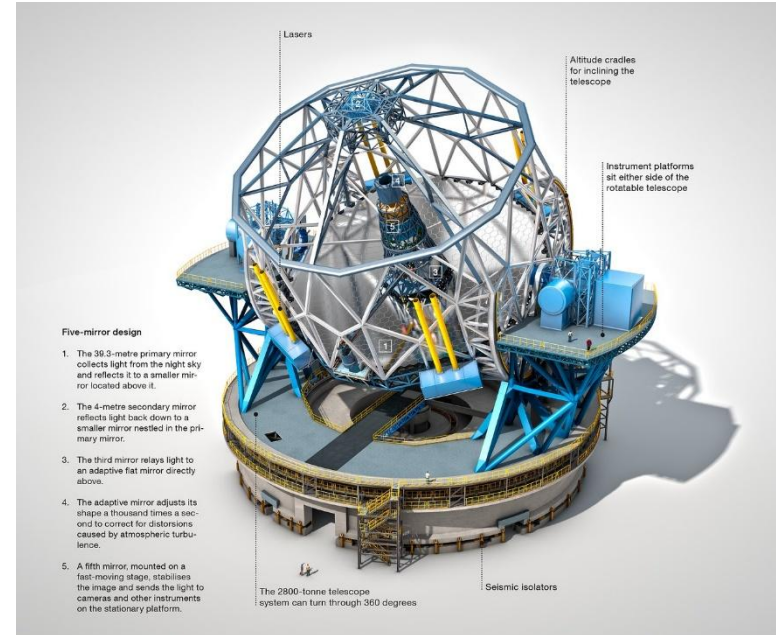


Percival Lowell began systematic study of Mars at his new observatory in Flagstaff, Arizona, in 1894. He initially used the telescope to further his legendary theories about intelligent life on Mars



Alt-azimuth

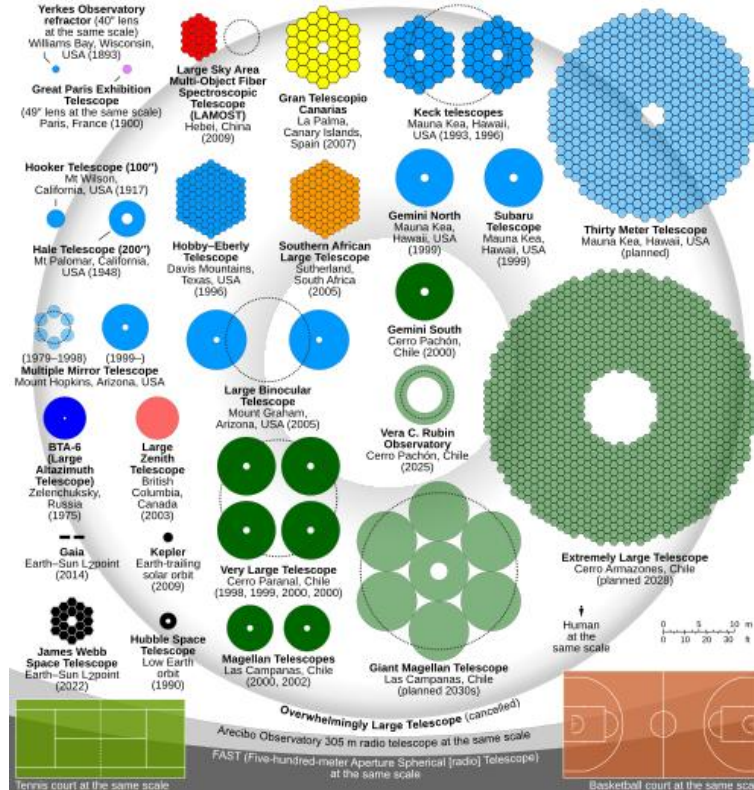
ELT – The Extremely Large Telescope, Atacama Desert, Chile
39 m $\varnothing$ reflective



Largest telescopes (all in altazimuth configuration)



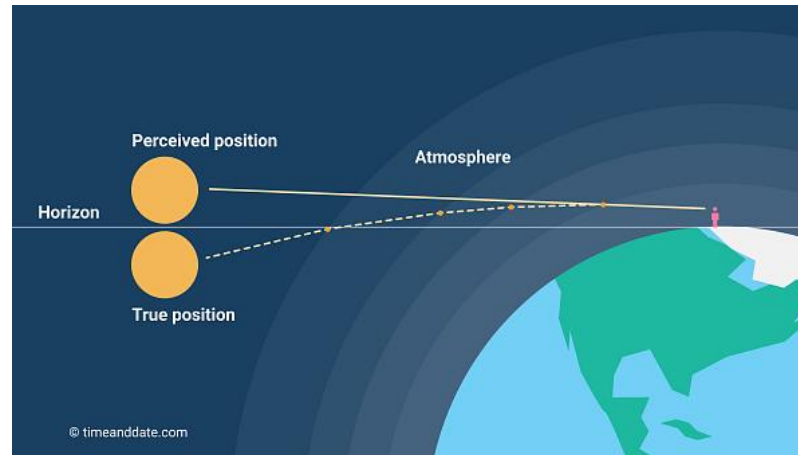
- Wikipedia
 - [Largest optical refractive telescopes](#)
 - [Largest optical reflecting telescopes](#)



Atmospheric refraction



- Atmospheric refraction bends light rays as they enter Earth's atmosphere, causing celestial objects to appear higher than they are, shifting their apparent position
 - Function of pressure and temperature altering the altitude coordinates, but keeping azimuth unchanged
- This bending effect is **greatest near the horizon**, contributing to phenomena like **advanced sunrise**, **delayed sunset**, and the **flattening of the Sun's disc**



Refraction makes celestial objects appear higher in the sky

Flattening of the Sun's disc



- **Atmospheric refraction flattens the Sun's disk at sunrise and sunset** because the light rays from the lower part of the sun are bent more than the rays from the upper part, decreasing the vertical angle of the Sun
- This happens because light travels through more of Earth's atmosphere when the sun is near the horizon, where air is denser due to temperature and pressure differences

Analysis (figure at right)

- The (approximate) real position of the Sun is marked with the white circle
- Atmospheric refraction is lifting the lower rim of the sun into view above the horizon by an angular distance that is typically $\approx 0.7^\circ$
- But light from the upper rim of the sun takes a trajectory with a slightly different slope through the atmosphere, and its image is lifted a little less
- But that slight angular difference amounts to a 1/10 of the apparent angular diameter of the Sun, and creates the apparent vertical flattening of its disc



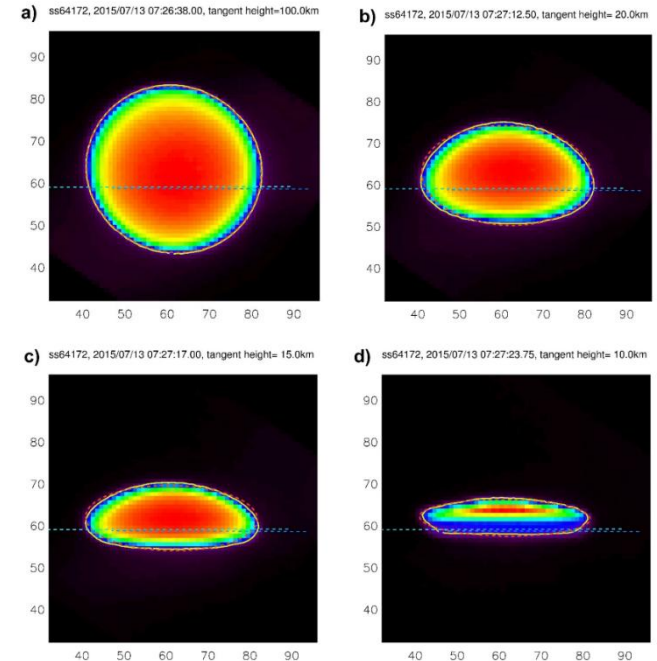
<https://oikofuge.com/shape-of-the-low-sun/>

Tangent height computation from solar extent



- The main instrument on the SCISAT satellite is the **Atmospheric Chemistry Experiment Fourier Transform Spectrometer** (ACE-FTS), which includes an auxiliary two-channel visible and near-infrared (VNIR) imager (often referred to as the **ACE imagers**). These imagers are used to monitor aerosols and **help in the precise determination of the tangent height of the measurements**
 - During each occultation, the Sun tracker is locked onto the sun center by a **quadrant sensor** allowing the ACE Imagers to record solar images while ACE-FTS and MAESTRO take measurements.
 - During sunset (sunrise), the tangent height decreases (increases), and **the apparent solar disk shifts due to refraction**, but the sun tracker keeps the instruments pointing at the radiometric center of the Sun
 - However, at very low tangent heights, the Sun image becomes compressed and asymmetric in the elevation direction, and the lower portion compresses more than the upper portion

[[Zou 2025](#), → [Lachance 2013](#) “ACE-FTS on SCISAT; 10th year on-orbit anniversary”]

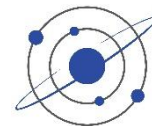


Sun images captured by the ACE NIR Imager at tangent altitudes of ≈ 100 km, 20 km, 15 km, and 10 km) during the sunset occultation measurement ss64172.

Atmospheric refraction (continued)

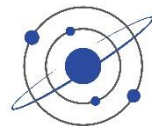


- Once we have computed the apparent or topocentric place of a celestial object (see Part III), the only remaining correction to be made to the coordinates is that caused by **atmospheric refraction**
 - Many methods for computing the atmospheric refraction correction exist
 - One commonly used is Taff 1991, Computational Spherical Astronomy (in The Computation of the Topocentric Place)
- The correction for atmospheric refraction is most easily understood if we refer to the horizon coordinate system
- Refraction simply increases the altitude of an object by a small amount that depends on the uncorrected **altitude of the object**, the air temperature, the air pressure, and even the humidity of the local atmosphere
 - Because temperature and pressure can change significantly over just seconds of time, it follows that the magnitude of the refraction correction can also vary rapidly as well.
 - For the application of **high-precision real-time telescope-pointing**, the method of computing the refraction correction must be rapid enough to compensate for the potentially random variations in the local atmosphere
 - For general observational purposes, however, such compensation is rarely justified, and a general purpose algorithm for optional use is much more convenient.
- Also, **observations are seldom made directly in the horizon coordinate system**. Telescopes are usually pointed with coordinates specified in the equatorial system and so this system is the preferred one in which to develop a refraction algorithm



Sun elevation angle below the horizon

- The Sun has to be 50' below the horizon for it to be considered to have set, for an observer's sky to be dark
 - Part of this angle is due to the semidiameter of the Sun ($\approx 16' = 60 * \text{Sun.Angular_diameter} / 2$)
 - Even without the atmosphere, when the center of the Sun is on the horizon, half of its disk is still above the horizon
 - The rest of this angle ($\approx 34'$) is due to atmospheric refraction (all of these results apply equally to sunrise conditions)
- In fact, when the Sun's disk appears to be just touching the horizon, it is actually completely *below* the horizon. The Earth's atmosphere, however, refracts the Sun's image enough to make it appear to still be *above* the horizon
- The point when the Sun's center is 50' below the horizon is considered to be the official time of sunset (or sunrise) and is the beginning (or end) of civil twilight
 - However, the sky is still quite bright at this point and no stars can be seen
- The point when stars (or satellites) begin to become visible marks the beginning of nautical twilight — when the Sun's center is 6° below the observer's horizon
 - Whenever the Sun is more than 6° below the horizon, it should be fairly easy to spot an illuminated earth satellite
- The Sun's effect on the night sky continues until the end of astronomical twilight when (if) it reaches 18° below the horizon and the indirect light from the Sun is less than the contribution from starlight
 - Of course, sky brightness and atmospheric refraction are influenced by actual meteorological conditions, so there may be some variance in the actual versus the predicted observation times



Sunrise & Sunset using refracted coordinates

- Apparent coordinates can additionally be corrected for [atmospheric refraction](#)
 - Only for Earth-based sites, this ultimately is a function of the local atmosphere and weather between target and observer, which can only be approximately known. Some observatories have developed their own local refraction tables

% Values without atmospheric correction

```
HA_Sunrise0 = acosd(cosd(90) / (cosd(lat) * cosd(Dec)) - tand(lat) * tand(Dec)); % [°]
```

```
Sun.Sunrise0 = Solar_Noon - 24 * HA_Sunrise0 / 360; % [hr] ← without atm correction
```

```
Sun.Sunset0 = Solar_Noon + 24 * HA_Sunrise0 / 360; % [hr] ← without atm correction
```

% The Sun has to be 50' below the horizon for it to be considered to have set, for an observer's sky to be dark

% - Part of this angle is due to the Sun semidiameter ($\approx 16' = 60 * \text{Sun.Angular_diameter} / 2$)

% - The rest of this angle ($\approx 34.0'$) is due to atmospheric refraction at the horizon

```
SZA_set = 50/60 + 90; % = 0.8333° + 90°
```

```
HA_Sunrise = acosd(cosd(SZA_set) / (cosd(lat) * cosd(Dec)) - tand(lat) * tand(Dec)); % [°]
```

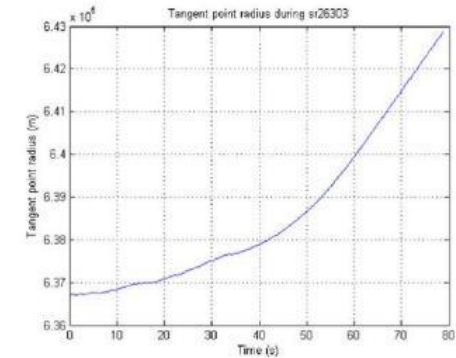
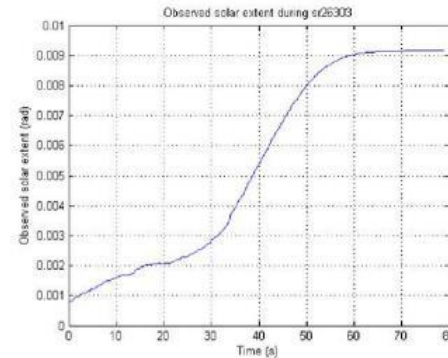
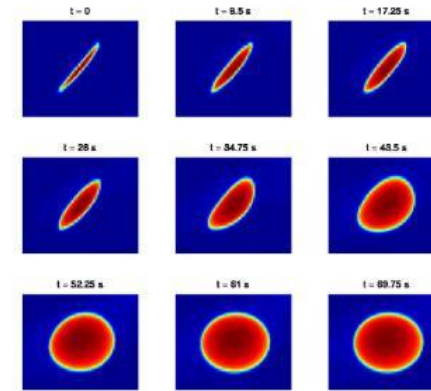
```
Sunrise = Solar_Noon - 24 * HA_Sunrise / 360; % (LST) 4/1440 = 24*60/360 / (24*60) = 1/360
```

```
Sunset = Solar_Noon + 24 * HA_Sunrise / 360; % (LST)
```

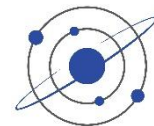
Tangent height computation from solar extent



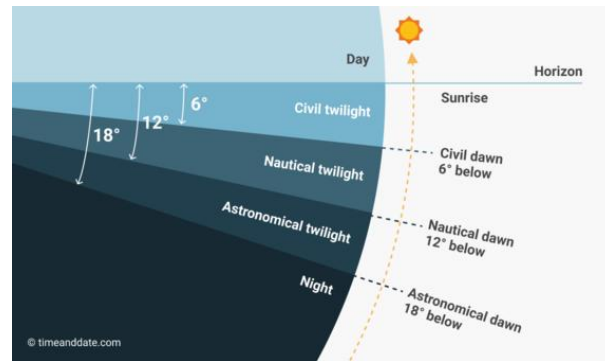
- Nine image samples of a time series taken during a sunrise with ACE-FTS are shown at right
- The **solar extent** computed for this time series is given in the bottom left, and the computed **tangent height** is shown in the bottom right
 - From [Lavigne 2017](#), HVRM a second generation ACE-FTS instrument concept



Twilight, dusk, dawn (crépuscule, tombée de la nuit, aube)



- **Civil Twilight:** The period just before sunrise or after sunset when the Sun is less than 6° below the horizon
 - The horizon is clearly defined, and terrestrial objects can be easily distinguished. Generally enough light to carry out outdoor activities (e.g., walking, driving) without artificial light. The brightest stars and planets (like Venus) can be seen, but the sky is not fully dark
 - Several countries use this definition of civil twilight to make laws related to aviation, hunting, and the usage of headlights and street lamps
- **Dawn (Morning):** The moment the center of the Sun is 6° below the horizon in the morning; it ends at sunrise
 - A time that marks the beginning of the civil twilight before sunrise. It is explained by the presence of weak sunlight, while the Sun is still below the horizon (≈ 30 min before sunrise)
- **Civil Dusk (Evening):** The moment the center of the Sun is 6° below the horizon in the evening; it begins at sunset
- Civil twilight is the first of three stages, followed by darker phases:
 1. **Civil** (0° to 6° below horizon)
 2. **Nautical** (6° to 12° below horizon)
 3. **Astronomical** (12° to 18° below horizon)

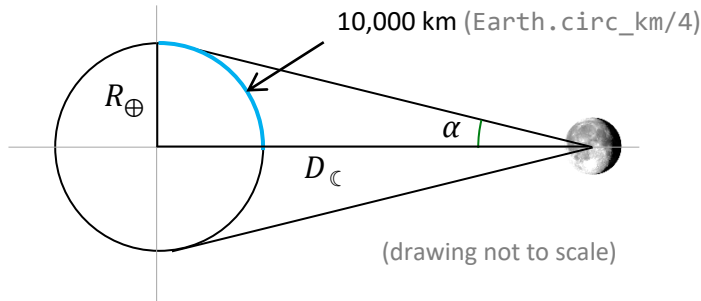


www.timeanddate.com/astronomy/different-types-twilight.html

RA/Dec for close objects



- The Moon is close to Earth and experiences parallax of up to 1° in either direction
 - The Moon however is so close, that changing your position on our planet causes the Moon's position to shift in relation to the background stars, i.e. its RA and/or Dec will change



The angle that the Moon's position on the sky can change due to the observer's location is $\alpha = \text{atan2}(\text{Earth.r1_km}/\text{Moon.Perigee_km}) \approx 1^\circ$, which is just short of two apparent lunar diameters ($2 * \text{Moon.Angular_diameter}$), not negligible

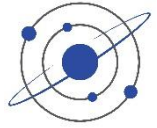
- Even more true for satellites
 - This parallax effect is even more pronounced the closer the object in question is to the observer – so it's really important when observing [satellites](#), which are roughly ten (GEO) to a thousand times (LEO) closer than the Moon
 - This is also why the [geostationary belt of satellites](#), despite lying pretty exactly above the Earth's equator, can be found several degrees in declination away from the celestial equator, depending on where you are.

Great Circle, Small Circle & Rhumb lines

- Great Circle
- Small Circle
- Rhumb lines & Mercator projection

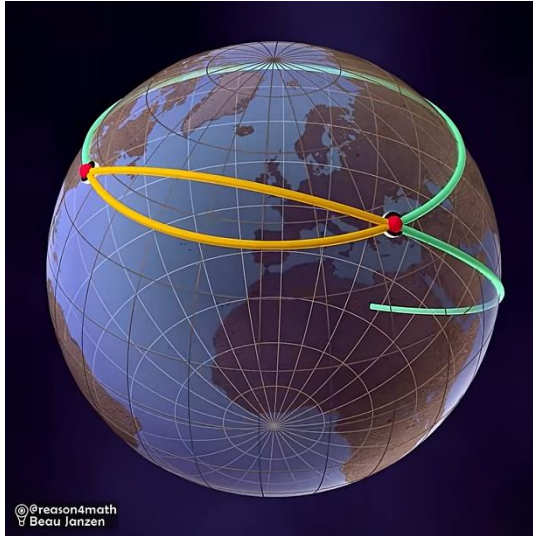


Creating Geodesics on a Sphere



- Creating Geodesics on a Sphere

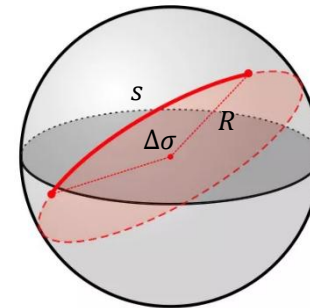
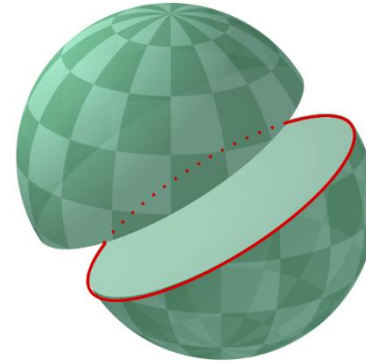
Beau Janzen: www.youtube.com/watch?v=a4L-coX4HLY [2:41]



Great Circle (Orthodrome)



- A **Great Circle** is a circle that results when a plane intersects the Earth going through the center effectively dividing the sphere (e.g. Earth) into two equal parts
 - All meridians and the equator are considered to be Great Circles as they all divide the Earth into two equal halves
- An **arc segment of a Great Circle** is the shortest path between two points on a spherical surface
 - To determine the shortest distance between two points on Earth, a Great Circle is constructed passing through the two points, then calculate the distance along that segment $s = R_{\oplus} \Delta\sigma$ (with $\Delta\sigma$ in radians)
- A great circle may be defined equivalently by:
 - a) Its pole vector
 - b) Two distinct points on the circle
 - c) A plane through the origin



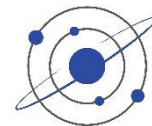


- Azimuth angle

$$az = \text{atan}\left(\frac{\cos(\phi_2) \sin(\lambda_2 - \lambda_1)}{\cos(\phi_1)\sin(\phi_2) - \sin(\phi_1) \cos(\phi_2) \cos(\lambda_2 - \lambda_1)}\right)$$

$$\tan \alpha_0 = \frac{\sin \alpha_1 \cos \phi_1}{\sqrt{\cos^2 \alpha_1 + \sin^2 \alpha_1 \sin^2 \phi_1}}.$$

$$\tan \lambda_{01} = \frac{\sin \alpha_0 \sin \sigma_{01}}{\cos \sigma_{01}},$$



Use of GreatCircle2() and GreatCircle()

% 1) using GreatCircle2 based on two coordinates on the globe $(\lambda_1, \phi_1) \rightarrow (\lambda_2, \phi_2)$

```
GC = GreatCircle2(lat1, lon1, lat2, lon2); % between points; computes  $\lambda_0$  and  $\alpha_0$  (+ other angles)
```

```
GC.Pos = GC.xyz * Earth.r1_km; % scaling to sphere radius
```

```
Plot3(GC.Pos, 'b-', 'Linewidth', 2);
```

```
% Computes ground distance along the path
```

```
deg2m(GC.arclen) / 1000 % [km]  $\approx 7050$  km
```

% Completes Great Circle around the Earth

```
f = linspace(1, 360 / GC.arclen, 101); % spans the whole great circle
```

```
A = sind((1 - f) * GC.arclen) / sind(GC.arclen);
```

```
B = sind(f * GC.arclen) / sind(GC.arclen);
```

```
xyz = Earth.r1_km * [(A * cosd(lat1) * cosd(lon1) + B * cosd(lat2) * cosd(lon2));  
                    (A * cosd(lat1) * sind(lon1) + B * cosd(lat2) * sind(lon2));  
                    (A * sind(lat1) + B * sind(lat2))];
```

```
Plot3(xyz, 'b--', 'Linewidth', 1.5);
```

% 2) Using GreatCircle(az, el) with main angles

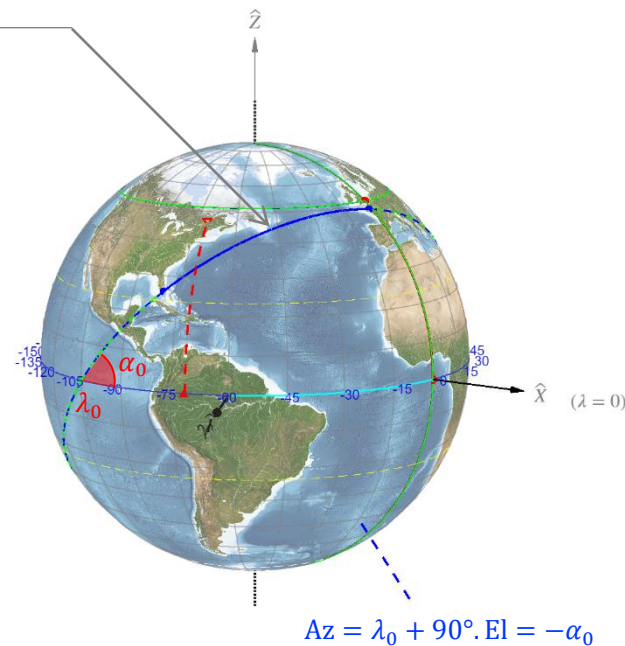
```
% el = declination/latitude/elevation of circle axis intercepting the sphere
```

```
% az = RA/longitude/azimuth of circle axis intercepting the sphere
```

```
[~, ~, xyz] = GreatCircle(GC.lon0 + 90, -GC.alpha0);
```

```
xyz = xyz * Earth.r1_km;
```

```
Plot3(xyz, 'c:', 'Linewidth', 2);
```



$$Az = \lambda_0 + 90^\circ. El = -\alpha_0$$



Great Circle

```
function [lat, lon, xyz] = GreatCircle(az, el, t)
% Generates a 3D Great Circle coordinates on a sphere
% az = RA/longitude/azimuth of circle axis intercepting the sphere
% el = declination/latitude/elevation of circle axis intercepting the sphere
% t = angles along the circle (e.g. 0..360°)

if 1
    % Great Circle through simplified Small Circle equation, using  $\alpha = 90^\circ$  ( $\sin(\alpha) \equiv 1$  &  $\cos(\alpha) \equiv 0$ )
    xyz = [-sind(el) * cosd(az) * cosd(t) - sind(az) * sind(t);
           -sind(el) * sind(az) * cosd(t) + cosd(az) * sind(t);
           cosd(el) * cosd(t)];
    [lon, lat] = Cart2Sph(xyz(1, :), xyz(2, :), xyz(3, :));
else
    lon = (az + 90) - atan2d(-cosd(t) .* sind(el), sind(t));
    %lon = mod(lon - 180, 360) - 180; % for 2D displays
    lat = asind(cosd(t) .* cosd(el));
    xyz = Sph2Cart(lon', lat', 1);
end

end % function
```



Great Circle

```
function GC = GreatCircle2(lat1, lon1, lat2, lon2, t)
% Generates a 3D Great Circle coordinates on a sphere passing through 2 points
% t = angles along the circle (0..360°)
```

```
% angular separation  $\Delta\sigma$ 
GC.arclen = acosd(sind(lat1) * sind(lat2) + cosd(lat1) * cosd(lat2) * cosd(lon1 - lon2));
% Azimuth angle of the GC
GC.az = atan2d(cosd(lat2) .* sind(lon2-lon1), cosd(lat1) .* sind(lat2) - sind(lat1) .* cosd(lat2) .* cosd(lon2 - lon1));
```

```
if nargin < 5
    t = linspace(0, GC.arclen);
end
```

```
[x1, y1, z1] = Sph2Cart(lon1, lat1); % (unitary sphere)
[x2, y2, z2] = Sph2Cart(lon2, lat2);
```

```
% parametric formulas for intermediate points along a great circle:
```

```
c1 = sind(GC.arclen - t) / sind(GC.arclen);
c2 = sind(t) / sind(GC.arclen);
```

```
GC.xyz = [c1 * x1 + c2 * x2;
          c1 * y1 + c2 * y2;
          c1 * z1 + c2 * z2];
[GC.lon, GC.lat] = Cart2Sph(GC.xyz);
```

```
end % function
```

```
% Great circle navigation: https://en.wikipedia.org/wiki/Great-circle\_navigation
lon12 = lon2 - lon1;
GC.alpha_1 = atan2d(cosd(lat2) * sind(lon12), cosd(lat1) * sind(lat2) - sind(lat1) * cosd(lat2) * cosd(lon12));
GC.alpha_2 = atan2d(cosd(lat1) * sind(lon12), -cosd(lat2) * sind(lat1) + sind(lat2) * cosd(lat1) * cosd(lon12));

% Longitude at the node ( $\alpha_0$ )
GC.alpha_0 = atan2d(sind(GC.alpha_1) * cosd(lat1), sqrt(cosd(GC.alpha_1)^2 + (sind(GC.alpha_1) * sind(lat1))^2));
lat01 = atan2d(tand(lat1), cosd(GC.alpha_1));
lon01 = atan2d(sind(GC.alpha_0) * sind(lat01), cosd(lat01));
GC.lon0 = lon1 - lon01;
GC.lat0 = 0; % (by definition)
```



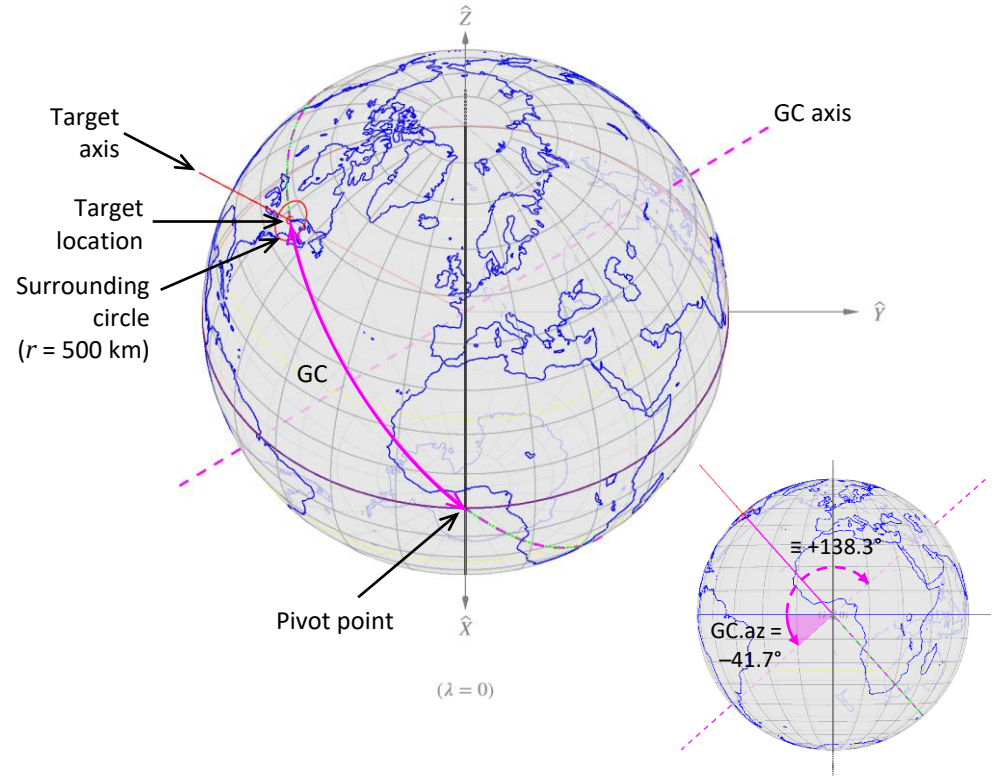
Exercise: Draw Great Circle passing through Quebec

%--- Draws a Great Circle arc relating ($\lambda=0$, $\phi=0$) to Quebec's coordinates

```
XYZ_observer = Sph2Cart(Lon, Lat, Earth.r1_km);
Plot3(XYZ_observer, 'ro');
sf = 1.5; % radius scaling factor
Plot3([[0;0;0], sf * XYZ_observer], 'r-')
```

```
% Draws a 500 km-radius circle centered on location
r = 500; % [km]
alpha = DEG(r / Earth.r1_km); % angle from arc length
[lat, lon, XYZ_SC] = SmallCircle(Lat, Lon, alpha);
Plot3(XYZ_SC * Earth.r1_km, 'r-');
```

```
% a) Using GreatCircle2()
GC = GreatCircle2(0, 0, Lat, Lon);
XYZ_GC2 = Sph2Cart(GC.lon, GC.lat, Earth.r1_km);
Plot3(XYZ_GC2, 'b--', 'Linewidth', 2);
% also computed
XYZ_GC2 = GC.xyz * Earth.r1_km;
Plot3(XYZ_GC2, 'r-', 'Linewidth', 2);
```





Exercise (continued)

```
% b) Using a GreatCircle()
% az & el correspond to the GC axis intersection on the sphere, hence the 90° correction relative to the pivot axis
az = 0 - 90; % along x,  $\lambda = 0^\circ$  on Prime Meridian
if 1
    el = GC.az;
    theta = linspace(0, GC.arclen); % right-hand-rule from az direction
else % equivalent to:
    el = GC.az + 180; % ← axis stays the same when rotated + 180°, simply flipped in direction
    theta = linspace(0, -GC.arclen);
end
% GC axis (plotting in two halves)
XYZ_GC_axis = Sph2Cart(az, el, Earth.r1_km);
Plot3([[0;0;0], +sf * XYZ_GC_axis], 'm--', 'Linewidth',2);
Plot3([[0;0;0], -sf * XYZ_GC_axis], 'm--', 'Linewidth',2);
% GC arc
[lat, lon, XYZ_GC1] = GreatCircle(az, el, theta);
Plot3(Earth.r1_km * XYZ_GC1, 'm-', 'Linewidth',2);
% whole GC
[lat, lon, XYZ_GC1] = GreatCircle(az, el);
Plot3(Earth.r1_km * XYZ_GC1, 'm--', 'Linewidth',2);
```



Exercise (continued)

% c) Draws a rotated Circle on Earth's Equator

% c.1) Initial circle on the Equator

```
theta = linspace(0, 360);  
XYZ_GC = [Earth.r1_km * cosd(theta);  
          Earth.r1_km * sind(theta);  
          zeros(size(theta))]; % (vector of same size as x;  $\equiv 0 * x$ ;  
Plot3(XYZ_GC, 'r-', 'Linewidth',1.5);
```

% c.2) Rotates to intercept Quebec

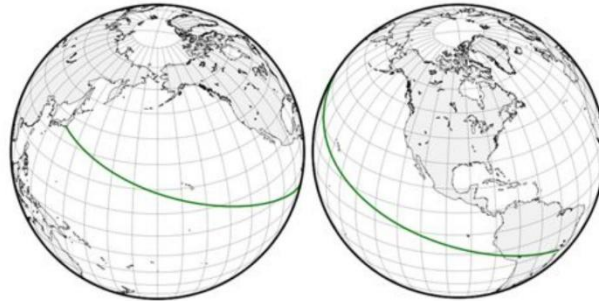
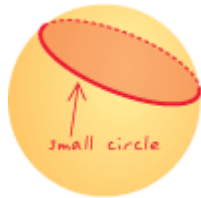
% True GC azimuth angle

```
beta = 90 + GC.az; % complementary angle  
alpha = -beta; % rotation in opposite direction  
XYZ_GC_rot = RotX(alpha) * XYZ_GC; % rotation on axis @  $\lambda = 0^\circ$   
Plot3(XYZ_GC_rot, 'g:', 'Linewidth',1.5);
```

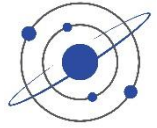

Small Circle



- A *Small Circle* is a circle the results when a plane intersects the Earth but does not go through the center effectively dividing the Earth into two equal parts
 - All parallels, except the equator, and other lines that do not pass through the center of the Earth are small circles
- A small circle is *not the shortest distance between two points* on the spherical surface even though it may appear to be
- The circle made when a flat plane intersects a sphere anywhere but through its center
 - Parallels of latitude other than the equator are small circles



Small Circle – Equation of a circle on a sphere



- Parametric solution in function of t or θ , scanning the circle

t = angle along the circle

ϕ = latitude

λ = longitude

α = central angle, determining the arc length ($x = \alpha R_{\oplus}$)

$$x = -[\sin(\alpha) \sin(\phi) \cos(\lambda)] \cos(t) - [\sin(\alpha) \sin(\lambda)] \sin(t) + \cos(\alpha) \cos(\phi) \cos(\lambda)$$

$$y = -[\sin(\alpha) \sin(\phi) \sin(\lambda)] \cos(t) + [\sin(\alpha) \cos(\lambda)] \sin(t) + \cos(\alpha) \cos(\phi) \sin(\lambda)$$

$$z = +[\sin(\alpha) \cos(\phi)] \cos(t) + \cos(\alpha) \sin(\phi)$$

This reverts to:

for $\varphi = 90^\circ, \lambda = 0^\circ$ (North Pole)

$$x = -\sin(\alpha) \cos(t)$$

$$y = +\sin(\alpha) \sin(t)$$

$$z = \cos(\alpha) \leftarrow \text{spherical cap depth}$$

for $\varphi = 0^\circ, \lambda = 0^\circ$ (on Equator)

$$x = \cos(\alpha) \leftarrow \text{spherical cap depth}$$

$$y = \sin(\alpha) \sin(t)$$

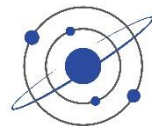
$$z = \sin(\alpha) \cos(t)$$

for $\varphi = 0^\circ, \lambda = 90^\circ$ (on Equator)

$$x = -\sin(\alpha) \sin(t)$$

$$y = \cos(\alpha) \leftarrow \text{spherical cap depth}$$

$$z = \sin(\alpha) \cos(t)$$



Small Circle

```
function [lat, lon, xyz] = SmallCircle(el, az, alpha, t)
% Generates a 3D small circle
% el = declination/latitude/elevation of circle center
% az = RA/longitude/azimuth of circle center
%  $\alpha$  = central angle, radius of the small circle (wrt to center of the sphere)
% t = angles along the circle (0..360°)

if nargin < 4
    t = linspace(0, 360); % → vector can be an arc
end

if 0 % Rotated disk method

    xyz = Sph2Cart(90 - t, 90 - alpha + 0 * t); % generates circle at north pole (unitary sphere)
    xyz = RotZ(az + 90) * RotX(90 - el) * xyz; % rotation into position
    [lon, lat] = RotX_sph(t, 90 - alpha + 0 * t, 90 - el);
    [lon, lat] = RotZ_sph(lon, lat, az + 90);

else % Equivalent expanded algebraic formula

    xyz = [-sind(alpha) * sind(el) * cosd(az) * sind(t) - (sind(alpha) * sind(az)) * cosd(t) + ...
           cosd(alpha) * cosd(el) * cosd(az);
           -sind(alpha) * sind(el) * sind(az) * sind(t) + (sind(alpha) * cosd(az)) * cosd(t) + ...
           cosd(alpha) * cosd(el) * sind(az);
           sind(alpha) * cosd(el) * sind(t) + cosd(alpha) * sind(el)];

    [lon, lat] = Cart2Sph(xyz);

end

end % function
```



Exercise: Great & Small Circles in practice

■ Death Star characteristics

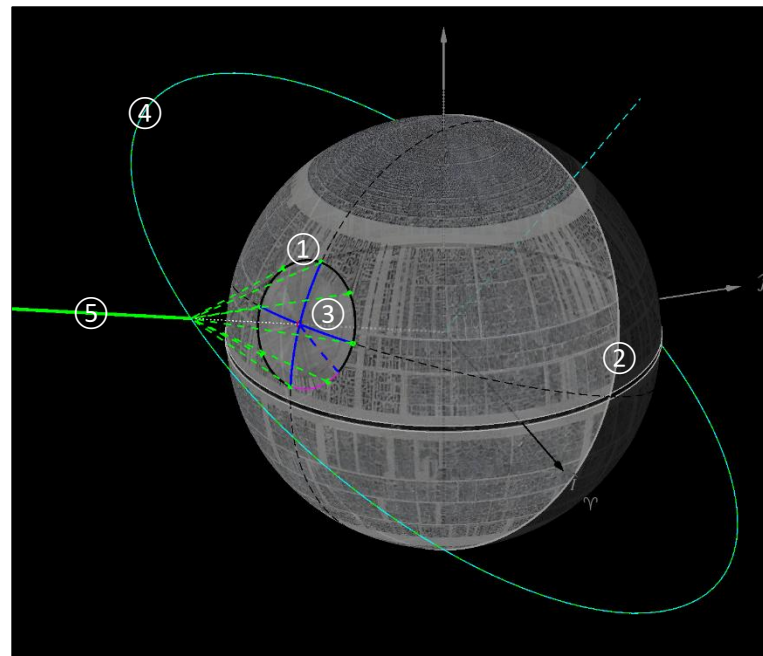
- $\alpha = 16.5$; % [°] Laser Dish angular radius
- $Az = -63$; % [°] Dish center longitude (match bitmap)
- $El = 20$; % [°] Dish center latitude
- $\beta = -30$; % [°] Satellite orbit orientation
% relative to dish
- $H = R * 0.7$; % [km] Focusing beam altitude

■ Draw:

- 1) Draw the “Super-laser” disk (concave dish)
- 2) Great Circle behind, as day-night terminator
- 3) Draws a central cross inside the Small Circle
- 4) Simulates an orbiting satellite passing through beam
- 5) Draws the combined laser beam

■ Compute the satellite FOV

■ Compute orbit inclination



Rhumb Lines & Mercator projection

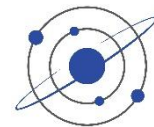


- It was the rhumb line that led to the development of one of the most famous of all **map projections** by Flemish geographer, Gerhardus Mercator in 1568
 - The **Mercator Projection** is frequently thought of as being simply latitude vs. longitude, *but it is not*
 - The horizontal scale is the longitude, but the vertical scale is the *log tangent of the latitude*
 - This tends to substantially **exaggerate the size of features near the poles** (called the *Greenland effect* in maps)
 - However, it has *one very important characteristic* – **a straight line drawn between any two points on a Mercator projection is a rhumb line on the surface of the Earth**
- Rhumb lines are also used from time to time for attitude maneuvers on spacecraft where, for example, we might fire our thrusters at a constant angle relative to the Sun to change the orientation of a spinning spacecraft

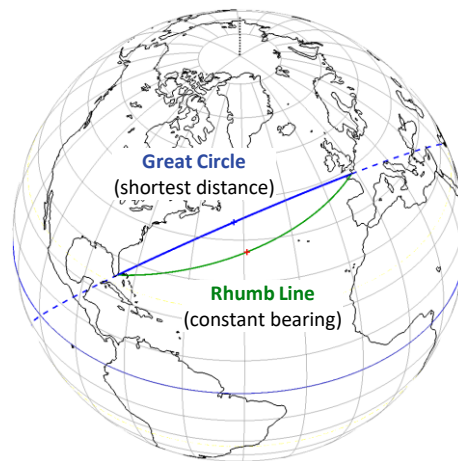


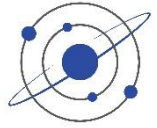
Map projections

Rhumb Line (Loxodrome)



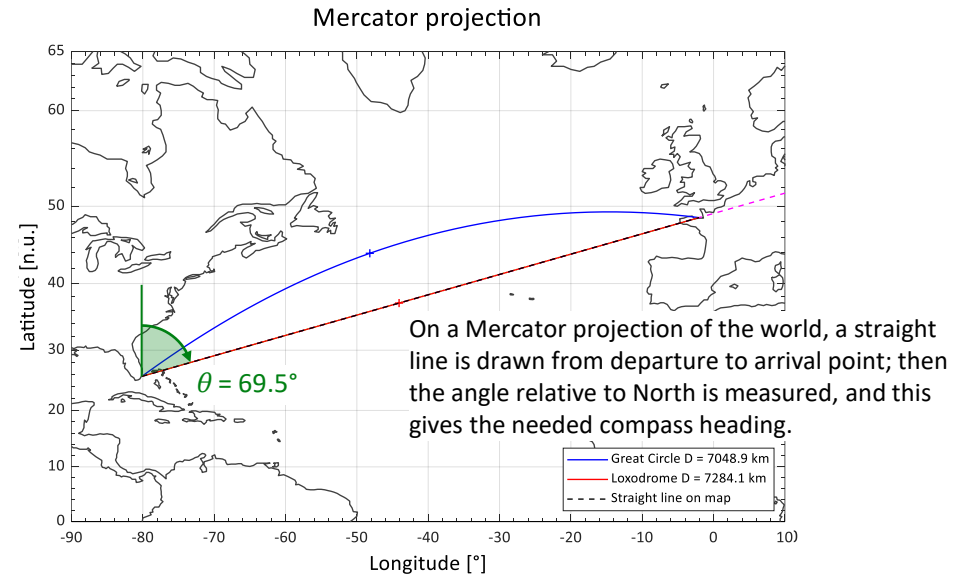
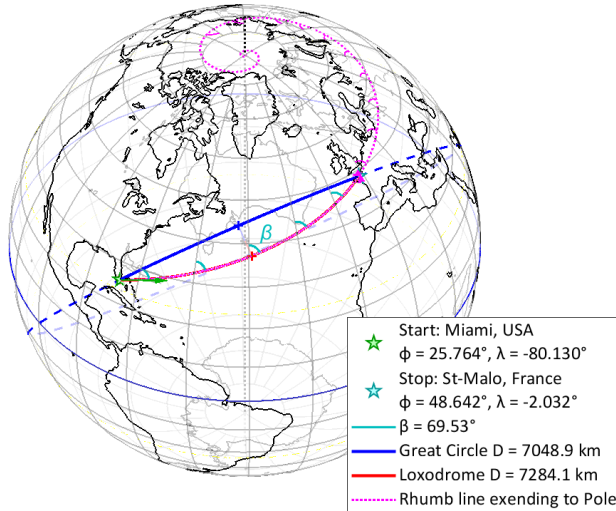
- A *Rhumb Line* is a curve that **crosses each Meridian at the same angle**. This is also known as a *loxodrome*.
 - Rhumb lines are quite useful because they **make navigation easy**; all is required is to maintain constant bearing or azimuth and eliminate the requirement to know one's position along the way
 - However, it does require that you **follow a longer distance** compared to the shortest possible distance which would be a Great Circle
 - The rhumb line doesn't deviate too much from a Great Circle arc in low and moderate latitudes, where it is most important for practical purposes.
 - On the whole, relatively few people are interested in taking ships to the North or South poles, which tends to be inconvenient for other reasons
- Obviously, this was an extremely important characteristic for navigation, prior to the introduction of inertial navigation systems and GPS





Loxodrome path

- Rhumb lines, or *Loxodromes*, are a remarkably convenient way to navigate
 - For example, if we set out from Miami at a compass heading angle of 69.5° , and maintain that azimuth long enough, then even though we don't know where we along the route, we will eventually end up in St-Malo, France



Geocentric vs Geodetic coordinates

- Geodetic coordinate system
- Latitude types
- Sub-satellite point
- Converting between latitude values
 - Iterative vs closed-form solutions

→ Continuation of/based on
Section 2.1.1 [Ellipsoid geometric characteristics](#)

- ▶ WHY THIS?
For computing φ that will be used for ground tracks
- ▶ References
 - YouTube: [Geodesy Digital Academy](#)
 - Celestrack: [Orbital Coordinate Systems, Part III](#)
 - Curtis 2020: 5.5 Topocentric Coordinate System
- ▶ Examples / Exercises / Quiz / Homework
- ▶ Summary / Recap / Takeaway
→ at the end of the section

Need for geodetic system



- The Earth may be approximated as a perfect sphere if the impact of its [equatorial flattening](#) is not significant for a given application – such as interplanetary trajectory analysis
 - Under this assumption, the geocentric and geodetic latitudes become identical, and the position-dependent geocentric radius reduces to a single mean Earth radius
- However, for most practical applications in geodesy and satellite astrodynamics, Earth's non-spherical shape must be accounted for using an oblate spheroid (ellipsoidal) model. This introduces the [distinction between geocentric and geodetic coordinates](#) and requires corresponding ellipsoidal formulations
 - Although the numerical differences are relatively small for Earth, accurate computation of positions, distances, and viewing geometry – particularly in geodesy and Earth observation – requires the use of the geodetic coordinate system
- Although the [oblate Earth model](#) introduces no unique problems with [longitude](#), it requires special handling with [latitude](#), as will be described in this section

Note: When discussing geocentric and geodetic coordinates, be aware of the sometimes confusing notations between different authors who identify angles differently: $\phi/\phi_{gc} \leftrightarrow \varphi/\phi_{gd}/\beta \leftrightarrow \phi' \dots$

→ There is no standard: in the present notes, ϕ is used for the geocentric latitude, and φ is used for the geodetic latitude



Oblate spheroid Earth model

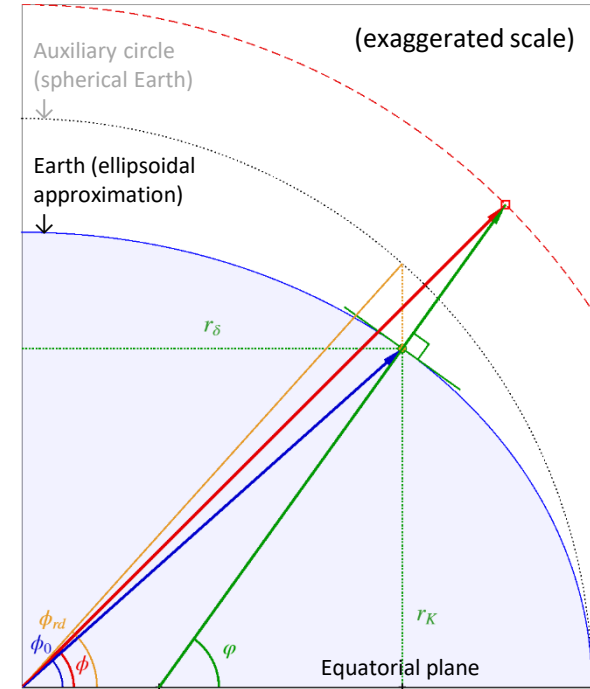
Latitude Geometry



- Consider the Earth's cross-section in the following figure showing the principal types of latitude commonly used in astrodynamics
 - There is also the *astronomic latitude* defined as the angle between the equatorial plane and a plumb-line perpendicular to the geoid, accounting for all local irregularities (not shown here)
- The geocentric latitude ϕ is referenced to the center of the Earth, whereas the geodetic latitude φ makes an angle perpendicular to the surface and the equatorial plane
- The ellipsoidal surface is a surface of revolution obtained by rotating an ellipse around the minor axis (oblate spheroid)
 - Thus, the *geodetic longitude* is the same as the *geographic longitude*, and *only a meridional section can be considered*, because we're only interested in the relationship between latitudes, so we can consider the position as a 2D problem independent of λ

- | | | |
|-------------|---|-----------------------------|
| ϕ | — | geocentric latitude |
| ϕ_0 | — | central geocentric latitude |
| φ | — | geodetic latitude |
| ϕ_{rd} | — | reduced latitude |

Latitudes used in astrodynamics

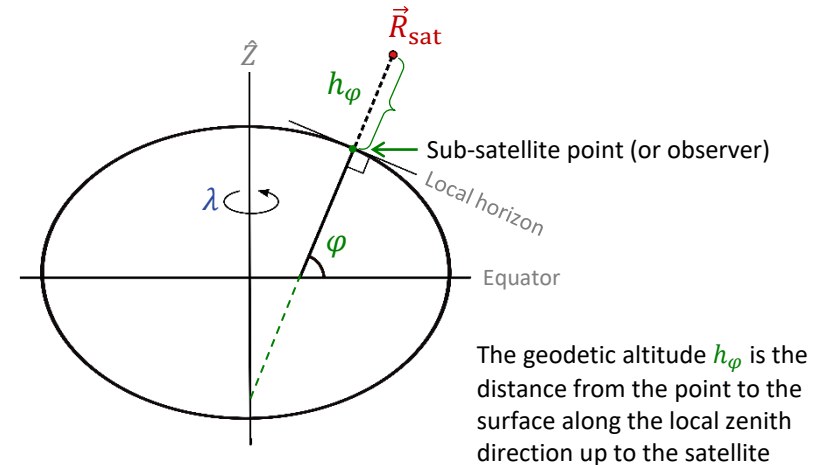


Geocentric vs geodetic for a satellite

Geodetic coordinate system



- **Geodetic coordinates** are a type of curvilinear orthogonal coordinate system used in geodesy *based on a reference ellipsoid*
 - An **Earth ellipsoid** (or oblate spheroid) is a mathematical figure approximating the Earth's shape and size, used as part of a reference frame for coordinates and computations in the geosciences, geodesy, astronomy, and astrodynamics
- The **geodetic coordinate system** defines a position in terms of north/south latitude φ , east/west longitude λ and altitude or ellipsoidal height h_φ above the ellipsoidal surface of the Earth
 - The triad is also known as *Earth ellipsoidal coordinates*
- The **local horizon** is defined as the plane that is tangent to the Earth's surface at a given position
- The surface considered is the **reference ellipsoid** (chosen as a close approximation to the oblate Earth)
- The **local zenith** is the direction away from the point on the Earth's surface perpendicular to the local horizon
 - On a sphere, this direction is always directly away from the Earth's center, but on an ellipsoid, this *is not the case* (except on the equator and at the poles)

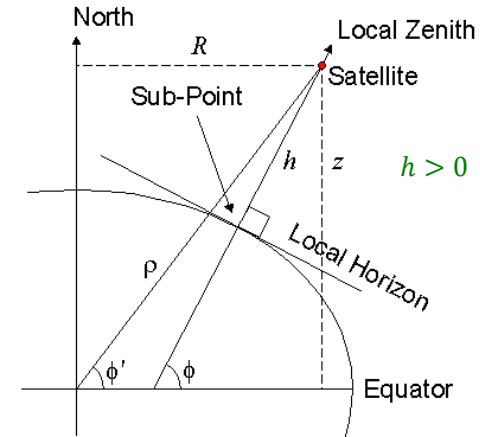


The geodetic altitude h_φ is the distance from the point to the surface along the local zenith direction up to the satellite

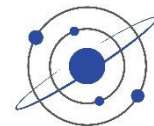
Sub-satellite point (SSP)



- The **sub-satellite point (SSP)** is defined as the intersection of the local vertical (or “nadir” direction) from the satellite with the Earth’s surface. Alternatively, the geodetic subsatellite point, is the point on the reference ellipsoid where the satellite is directly at the local zenith (“straight up”)
- With respect to the ellipsoid, or the geoid specifically, the **sub-satellite point** is related through these principles:
 - The geodetic subsatellite point is defined by the surface normal – a line perpendicular to the ellipsoid and roughly perpendicular to the geoid – intersecting the surface
 - While ground tracks are often calculated on a smooth **reference ellipsoid** (such as WGS84), the **geoid** is an irregular surface representing the gravitational equipotential. In high-precision satellite altimetry, the SSP is defined based on the intersection with the geoid-referenced mean sea level
 - Because the geoid undulates around the reference ellipsoid, the exact SSP definition may vary by a few meters (geoid undulation) depending on whether the height is measured relative to the **ellipsoid** or the **geoid** (orthometric height)



Calculating Sub-satellite point – Oblate Earth



1) Longitude $\lambda_\phi \equiv \lambda_\varphi \rightarrow \lambda$

- **Longitude λ** corresponds to the rotational angle in the equatorial plane from the x-axis (zero meridian) to the meridian plane of the measured point (→ see Section [Terrestrial coordinates, Parallels / Latitude](#))
- In the ECEF reference frame (where most of the computations are performed), it is computed from the cartesian coordinates of the measured point:

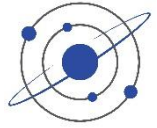
$$\lambda = \text{atan}\left(\frac{y_{\text{ECEF}}}{x_{\text{ECEF}}}\right)$$

- In the ECI reference frame, the offset due to the Earth's rotation at the epoch of measurement must be considered

$$\lambda = \text{atan}\left(\frac{y_{\text{ECI}}}{x_{\text{ECI}}}\right) - \theta_{\text{GMST}} \quad \rightarrow \text{see SOL Part II – Sidereal Time \& GMST}$$

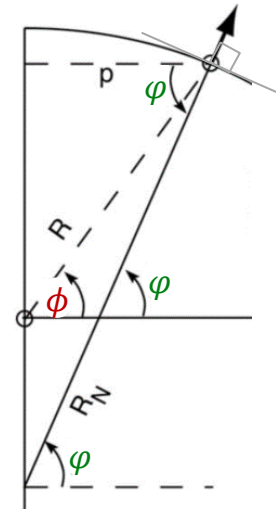
- This is the **same as the spherical longitude** $\lambda_\phi \equiv \lambda_\varphi$ (due to the rotational symmetry), the Earth ellipsoid flattening being along the vertical axis only (North to South)

2a) Geocentric latitude ϕ



- Geocentric latitude ϕ is the angle measured at the Earth's center from the equatorial plane to the point of interest, i.e. the observer or the sub-satellite point on Earth surface, or the satellite position in orbit
- Geocentric coordinates are Earth centered **spherical coordinates used for most space applications**; also, geocentric values ϕ are required for calculations involving gravitational potential operations
- The geocentric latitude of a point ϕ is the angle between the Earth's equatorial plane and the vector from the center of the Earth to the point of interest, expressed in cartesian coordinates

$$\phi = \text{atan} \left(\frac{z}{\sqrt{x^2 + y^2}} \right)$$

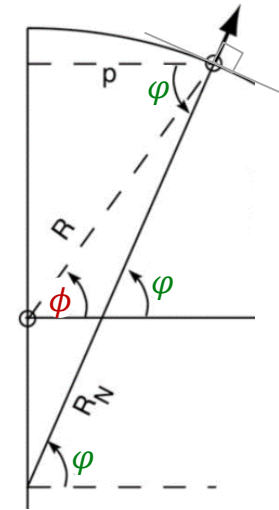


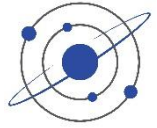
(Converting latitude values $\phi \rightarrow \varphi$)

2b) Geodetic latitude φ



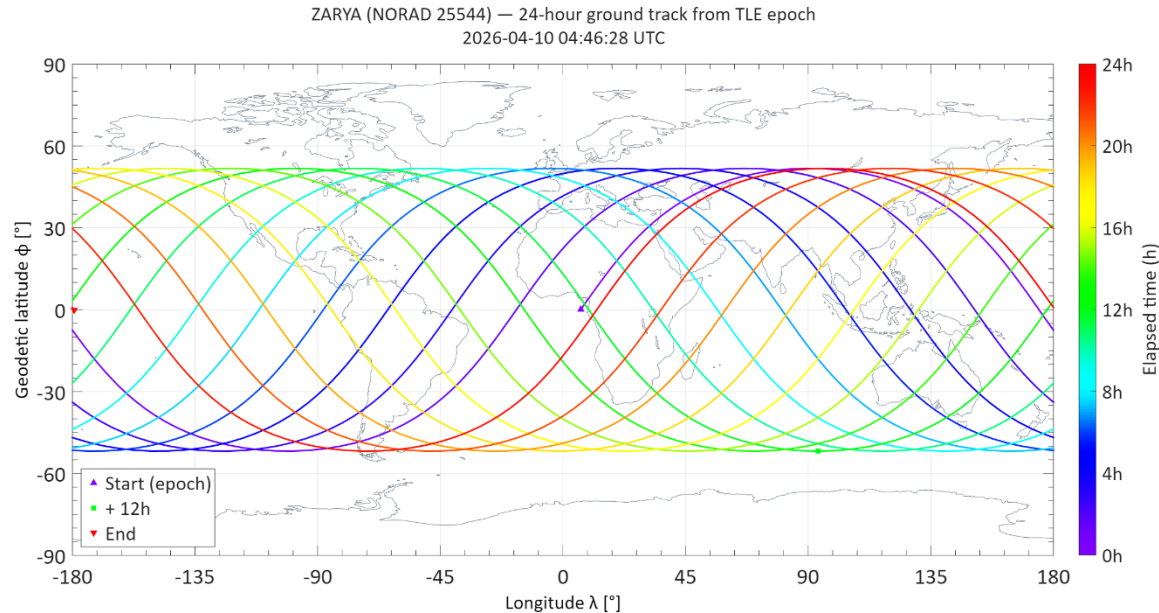
- The **geodetic** or geographic latitude φ , is the angle between the equatorial plane and the normal to the surface of the reference ellipsoid; The most common latitude used in geodesy (hence the name)
- φ measures how close to the poles or equator a point is along a meridian. It is the angle between the equatorial plane and a line that is normal to the reference ellipsoid at the measured point
 - It is represented as an angle from -90° to $+90^\circ$, where 0° is the equator
 - is the angle between the local zenith and the equatorial plane
 - Except at the poles and the equator, φ differs slightly from the geocentric latitude ϕ , depending on the flattening
- Unless otherwise specified, the term “latitude” generally refers to the geodetic latitude φ
 - The latitude used in geographic coordinate systems, maps, and standard station location data is typically the geodetic latitude
 - Consequently, most mapping and geodetic applications use geodetic coordinates.





Ground Tracks are traced in geodetic latitude

- Ground track is the **continuous path** formed by the sequence of **sub-satellite geodetic points** as the satellite moves

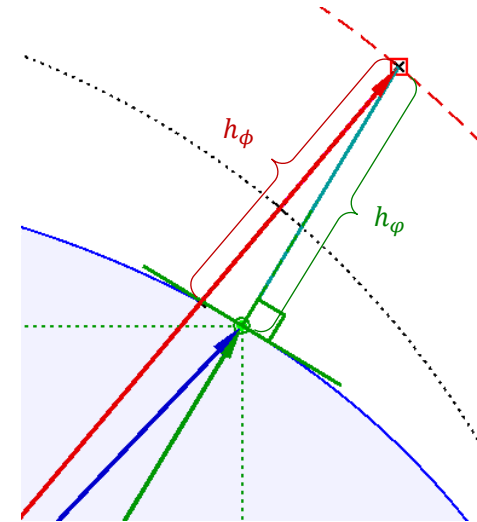


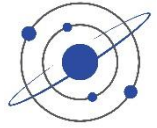
Ground Tracks

3) Altitude – height



- The **geocentric altitude** h_ϕ is the distance to the reference ellipsoid along a radial line to the geocenter
 - This corresponds to the *Orbital Altitude*: $h_\phi \equiv \rho - R_\phi$ where R_ϕ is the ellipsoid radius at latitude ϕ , given by `GeocentricRadius(phi_geoc, a,b)`
- The **geodetic altitude** h_ϕ (also known as *ellipsoidal height*) is the height above the ellipsoid surface, normal to the ellipsoid
 - When used without qualification, as in aviation, the term altitude refers to geodetic altitude
 - It is defined as a signed distance such that points inside the ellipsoid have negative height



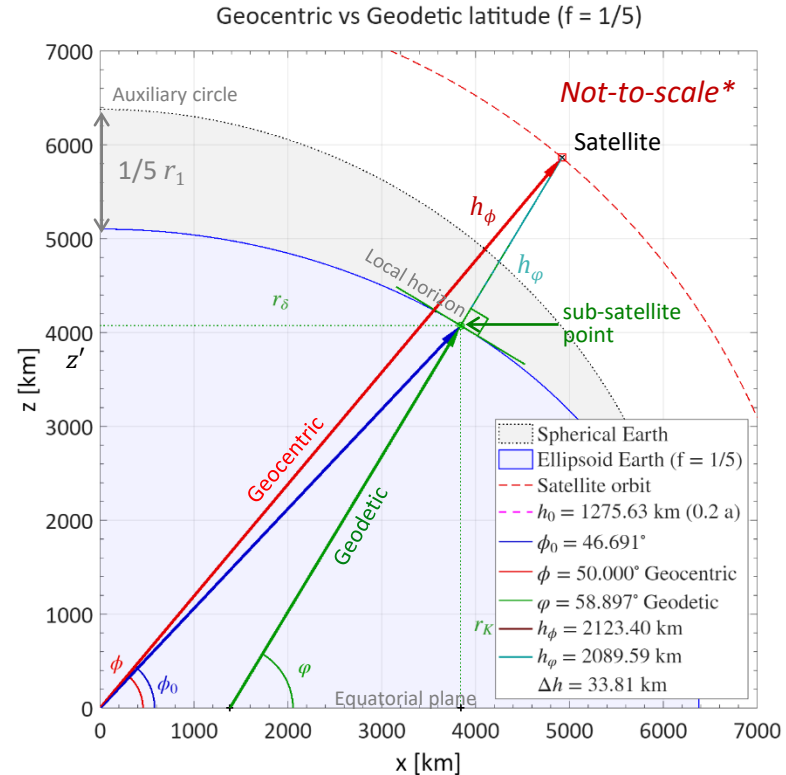


Geocentric vs geodetic for a satellite

- **Central geocentric angle ϕ_0**
 $\angle$ between the Earth's center and a point on the ellipsoid surface (e.g. sub-satellite point)
 – Used for calculations involving R_e and arc-lengths
- **Geocentric latitude ϕ :**
 $\angle$ between the equatorial plane and a radial line connecting the center of the ellipsoid to the satellite
- **Geodetic latitude φ**
 $\angle$ between the equatorial plane and the surface normal at a point exactly below the satellite (at zenith)

$$\phi_0 < \phi < \varphi$$

Note: Many diagrams are drawn with greatly exaggerated scales for clarity (here $b/a = 0.8 \rightarrow e = 0.6, f = 1/5$)



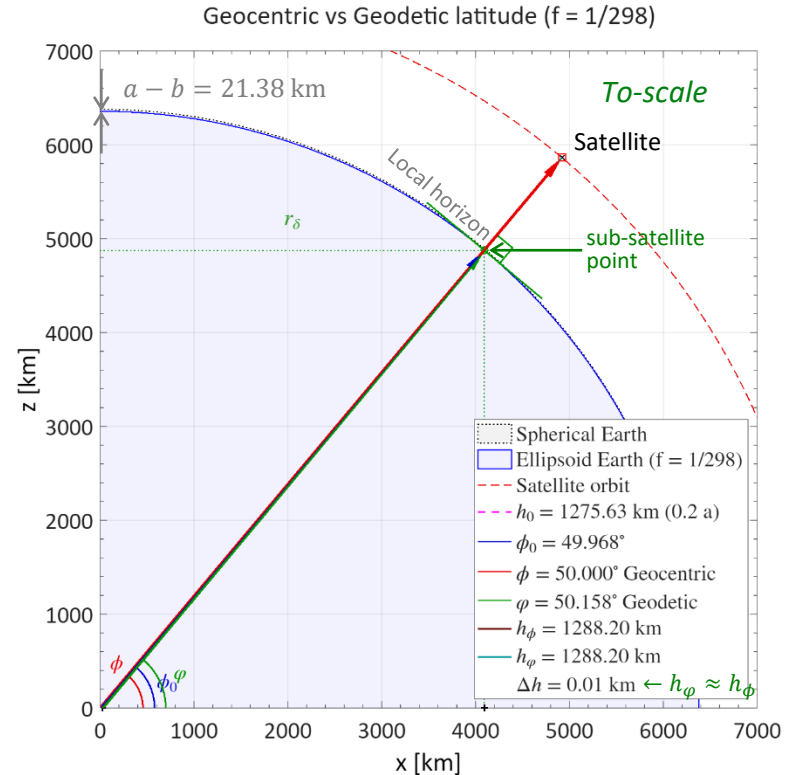
Latitude geometry

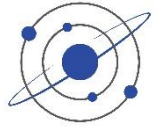


Geocentric vs geodetic ($e_{\oplus} = 0.0818, f_{\oplus} = 1/298$)

- Ellipsoidal representation of the Earth
 - Earth's shape can be approximated with a flattening of $f = 1 - b/a \approx 1/298.257$ to account for the 21.38 km difference in equatorial and polar radii
- At this scale, this flattening is almost indistinguishable from the spherical Earth reference circle

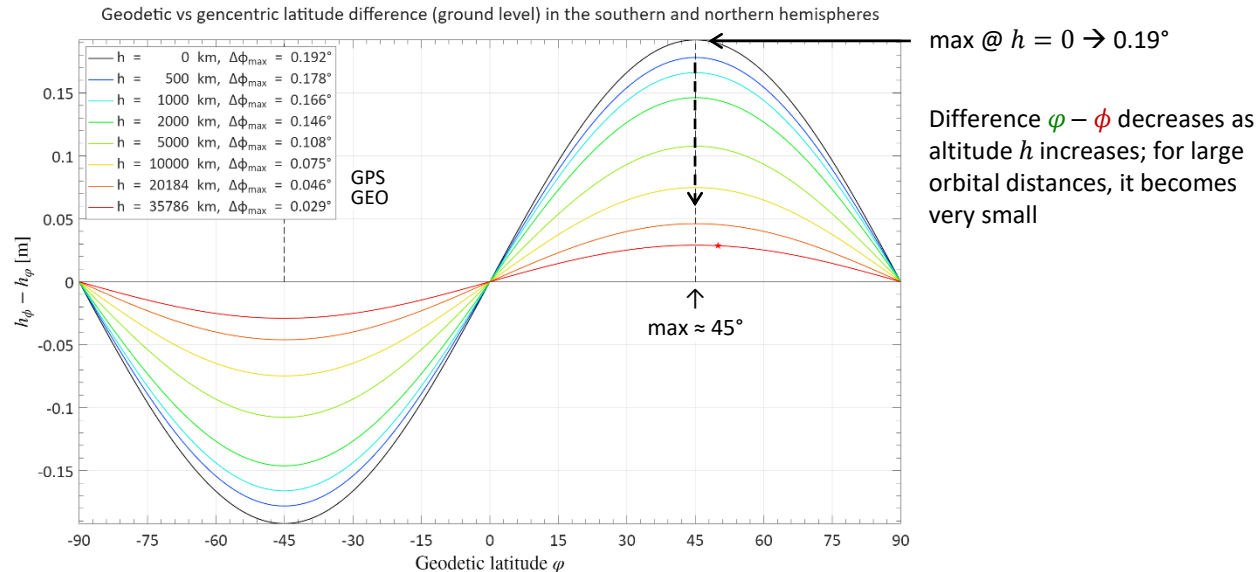
* Keep in mind the true scale of the ellipsoidal Earth model





Geodetic vs geocentric latitude difference

- The difference between geocentric and geodetic latitude is very small: the maximum $\varphi - \phi$ is $< 0.2^\circ$;
→ depending on tracking requirements, some applications must account for this variation
- Difference from using an oblate spheroid Earth vs a purely spherical model is illustrated in the following figure



Exercise



- What is the Δ for a GPS satellite at 50° ?

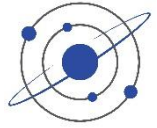
```
phi_geod = 50;  
phi_geoc = geod_to_geoc(phi_geod, a,b, Earth.GEO_H_km);  
phi_geod - phi_geoc % == 0.029°
```

- What is the maximum amount by which a line normal to the reference ellipsoid misses the center of the Earth at a latitude of $\varphi = 45^\circ$

```
[~, ~, ~, ~, ~, ~, z0] = geod_to_pos(45,0, 0, a,b)  
[~, ~, ~, ~, ~, ~, z0] = geod_to_pos(45,0, 1000, a,b) % ← same: independent of h_φ as co-linear  
% Answer: z0 = 30.24 m
```

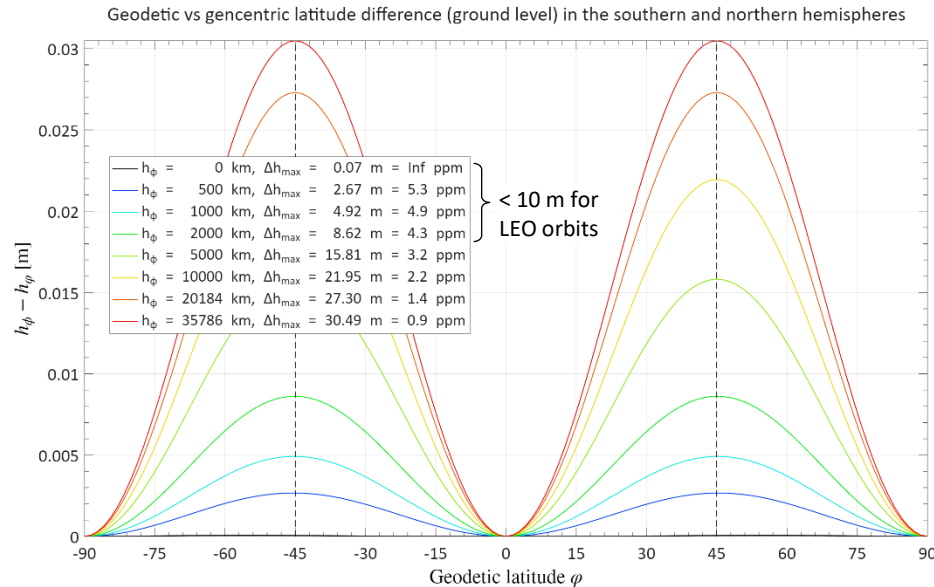
- Now at a latitude of $\varphi = 90^\circ$

```
[~, ~, ~, R_N, ~, ~, z0] = geod_to_pos(90,0, 0, a,b) % Answer: z0 = (-)42.84 m  
[ R_N == a^2/b ] % @ φ = 90°  
[ b - R_N == z0 ]  
% z0 ≡ b - R_N == b - a^2/b == -(a^2/b - b) == -(a^2 - b^2) / b == -(a^2 * e2)/b
```



Geodetic vs geocentric altitude difference

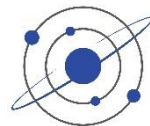
- The difference in altitudes is below 10 m (< 5 ppm) for LEO orbits



Geocentric to position conversion

Conversions $\varphi \leftrightarrow \phi$





Converting latitude values $\phi \rightarrow \varphi$

- Modern astro-geodetic applications require a mutually consistent set of transformations between geocentric and geodetic coordinates, with high numerical accuracy maintained across all latitudes and altitudes
- The basic problem involves transforming *from* cartesian/geocentric $\vec{R}_{sat}$, ϕ *to* geodetic φ latitudes
 - One must be able to represent position vectors in terms of the different latitudes, as well as to transform between each type and computing the corresponding altitudes

Cartesian $\rightarrow$ Geocentric

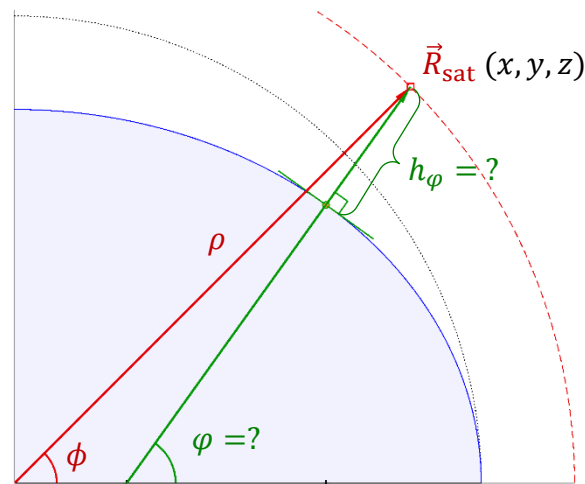
$\vec{R}_{sat} = (x, y, z)$
known from orbital propagation

$$\rho = |\vec{R}|$$

$$\lambda = \text{atan}\left(\frac{y}{x}\right)$$

$$\phi = \text{atan}\left(\frac{z}{\sqrt{x^2 + y^2}}\right) = \text{asin}\left(\frac{z}{\rho}\right)$$

Note: In general the atan is preferred to the asin function to avoid quadrant ambiguity, but in this case this is not applicable as $-90^\circ < \phi < +90^\circ$



Note: Works for any circular or elliptical orbit, as we start from state vector position $\vec{R}_{sat}$



(2a) Geocentric latitude ϕ



Geocentric to position calculation $\phi \rightarrow R$

- Geocentric coordinates are easily transformed into rectangular Cartesian coordinates
- The expression for the cartesian position in terms of the geocentric latitude ϕ is:

$$R_\phi = \frac{a}{\sqrt{1 + \left(\left(\frac{a}{b}\right)^2 - 1\right) \sin(\phi)^2}}$$

$$r_\delta = (R_\phi + h_\phi) \cos(\phi)$$

$$r_K = (R_\phi + h_\phi) \sin(\phi)$$

→ Result is approximate (but still very good ≈ 5 m) when h_ϕ (not measured directly along the geocentric radius) is provided instead of h_ϕ (see figure [Geodetic vs geocentric altitude difference](#))



Geodetic vs geocentric altitude difference



Geocentric to position

```
function [R, r_geoc] = geoc_to_pos(phi_geoc, lambda, h_geoc, a,b)
% Converts geocentric latitude  $\phi$  & longitude  $\lambda$  [°] to position vector [x;y;z],
% optionally at a given altitude h above ellipsoid in the geocentric direction ( $h_\phi$ )
% Although not used as often as geod_to_pos, it has certain advantages in some situations.
% → Result is approximate when h_geod (not measured directly along the geocentric radius) is provided instead of h_geoc
%   (but still very good as  $h_\phi \approx h_\varphi$ )

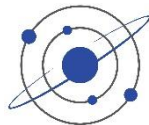
% in function of  $\phi$  ; Vallado 2012, Eq.(3-10)
r_geoc = a ./ sqrt(1 + ((a/b)^2 - 1) * sind(phi_geoc).^2); % ≠ R_N ← ≡ GeocentricRadius(phi_geoc, a,b)
% = a ./ sqrt(1 + e_2 * sind(phi_geoc).^2); % e_2 = (a/b)^2 - 1; % using second eccentricity squared e'^2
% = a * sqrt((1 - e2) ./ (1 - e2 * cosd(phi_geoc).^2)); % ← Vallado (3-10)

% Radius of ellipsoid R( $\phi$ ) from ellipsoid center to surface (h = 0)
% r_geoc = GeocentricRadius(phi_geoc, a,b);
% = a ./ sqrt(1 + ((a/b)^2 - 1) .* sind(phi_geoc).^2);
% = a ./ sqrt(1 + e2 / (1 - e2) .* sind(phi_geoc).^2);
% = a ./ sqrt(1 + e_2 .* sind(phi_geoc).^2);

x = (r_geoc + h_geoc) .* cosd(phi_geoc) * cosd(lambda);
y = (r_geoc + h_geoc) .* cosd(phi_geoc) * sind(lambda);
z = (r_geoc + h_geoc) .* sind(phi_geoc);

R = [x; y; z];

end % function
```



Some useful geometry

- We use r_δ as the horizontal component (this is the *equatorial component* in Fig. 3-1) along the semimajor axis, and r_K as the vertical component parallel to the semiminor axis
→ all at $h = 0$
- These formulas are exact for an ellipsoidal surface, but surface features may cause the site to vary significantly from the reference ellipsoid
 - For example, Death Valley (36.34° N, 117.0° W) is 86 m below sea level; yet, just over 100 km away, Mt. Whitney (36.34° N, 118.18° W) is 4417 m above sea level. These variations don't cause too much difficulty because the altitude can be added into the component equations (→ geod_to_pos: $x = (R_N + h_{\text{geod}}) * \cos(\phi_{\text{geod}}) \dots$)

$$r_\delta = \sqrt{r_I^2 + r_J^2} = r_{\text{sub}} \cos(\phi_0)$$

$$r_K = r_{\text{sub}} \sin(\phi_0)$$

$$\begin{aligned} r_\delta &= R_N \cos(\varphi) \\ r_K &= R_N (1 - e^2) \sin(\varphi) \\ R_e &= R_N \sqrt{1 - e^2 \sin(\varphi)^2} \end{aligned}$$

$$\begin{aligned} r_\delta &= R_\oplus \cos(\phi_{\text{rd}}) \\ r_K &= R_\oplus (1 - e^2) \sin(\phi_{\text{rd}}) \\ R_e &= \sqrt{r_\delta^2 + r_K^2} = R_\oplus \sqrt{1 - e^2 \sin(\phi_{\text{rd}})^2} \end{aligned}$$



Geodetic to cartesian $\varphi, h_\varphi \rightarrow (x, y, z)$ & ϕ, ρ

- The conversion from geodetic coordinates to cartesian coordinates is given by:

$$\begin{aligned} x &= (R_N + h_\varphi) \cos(\varphi) \cos(\lambda) \\ y &= (R_N + h_\varphi) \cos(\varphi) \sin(\lambda) \\ z &= ((1 - e^2)R_N + h_\varphi) \sin(\varphi) = \left((b/a)^2 R_N + h_\varphi \right) \sin(\varphi) \end{aligned}$$

$$r_\delta = \sqrt{x^2 + y^2} = (R_N + h_\varphi) \cos(\varphi)$$

$$r_K = z$$

with h_φ = Altitude (above ellipsoid surface along the geodetic ray vector)

φ = geodetic latitude

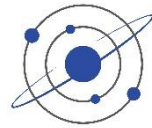
λ = longitude

R_N = [Prime vertical radius of curvature](#)

$\vec{R} = (x, y, z)$ ECEF position of the satellite derived from geodetic coordinates



Inverse transformation: $\phi \rightarrow \varphi$



General latitude relationship

- The relationship between geocentric latitude ϕ and geodetic latitude φ at a height h_φ above the reference ellipsoid (semi-major axis a , semi-minor axis b , eccentricity e) can be expressed using the tangent formula:

$$\phi = \text{atan} \left(\tan(\varphi) \frac{R_N(b/a)^2 + h_\varphi}{R_N + h_\varphi} \right)$$

where $R_N(\varphi)$ is the [radius of curvature in the prime vertical](#)

- When $h_\varphi = 0$, i.e. for points *on the ellipsoid surface*, the relation simplifies to:

$$\phi_0 = \text{atan} \left(\tan(\varphi) \left(\frac{b}{a} \right)^2 \right)$$

$$\varphi = \text{atan} \left(\tan(\phi_0) \left(\frac{a}{b} \right)^2 \right) = \text{atan} \left(\frac{\tan(\phi_0)}{1 - e^2} \right) = \text{atan} \left(\frac{\tan(\phi_0)}{(1 - f)^2} \right)$$

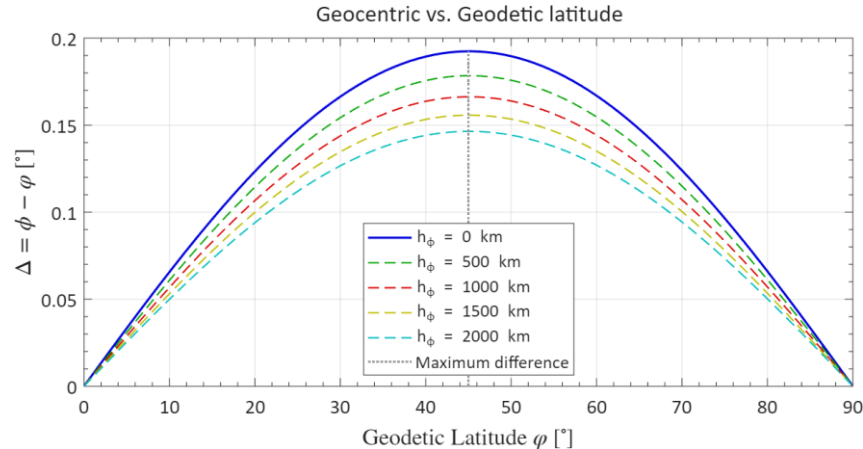


R_N Radius of curvature in the prime vertical



Geocentric vs. Geodetic difference

- Knowing the geodetic latitude φ at a height $h_\varphi \geq 0$, and the ellipsoid parameters (e.g. a & b , or a & e , or a & f), the geocentric latitude ϕ can be determined
- The following figure shows how much of a difference results from using an oblate spheroid, plotting the difference between geodetic and geocentric latitude as a function of geodetic latitude
 - The error decreases as altitude increases, as becomes negligible at large distance from the Earth
 - The largest errors are for observers at $\varphi = 45^\circ$





Geodetic to position (continued)

```
% R_e(φ) = distance from the ellipsoid center to its surface measured along the geocentric radial direction
R_e = a * sqrt(((1 - e2)^2 * sind(phi_geod).^2 + cosd(phi_geod).^2) ./ (1 - e2 * sind(phi_geod).^2));
% == a * sqrt((1 - (2 * e2 - e2^2) * sind(phi_geod)^2) / (1 - e2 * sind(phi_geod)^2))
% == R_N * sqrt(cosd(phi_geod)^2 + (b/a)^4 * sind(phi_geod).^2)
% == sqrt((a^4 * cosd(phi_geod)^2 + b^4 * sind(phi_geod)^2) / (a^2 * cosd(phi_geod)^2 + b^2 * sind(phi_geod)^2))
% == norm(R_sub)

% R_e(φ)
% R_e = a ./ sqrt(1 + ((a/b)^2 - 1) .* sind(phi_geoc0).^2)
% == (a * sqrt(1 - e2)) / sqrt(1 - e2 * cosd(phi_geoc0)^2)
% phi_geoc0 = atand(tand(phi_geod) * (b/a)^2);
% R_e = (a * b) / sqrt(a^2 * sind(phi_geoc0)^2 + b^2 * cosd(phi_geoc0)^2);

%--- Radius of curvature in Meridian (Meridian Radius of Curvature)
R_M = a * (1 - e2) ./ (1 - e2 * sind(phi_geod).^2).^(3/2); % R_M(φ)
% == R_N * (1 - e2) ./ (1 - e2 * sind(phi_geod).^2)
```

end % function

Inverse transformation: cartesian to geodetic $\phi \rightarrow \varphi$



- If geocentric coordinates are known to be directly calculable from geodetic coordinates ($\varphi \rightarrow \phi$) through the use of [simple formulae](#), it is not so with the reverse transformation ($\phi \rightarrow \varphi$) as φ and h_φ are mutually involved through R_N
- For most standard geographic work, latitudes are already geodetic. This conversion is only needed when working with raw Earth-Centered, Earth-Fixed (ECEF) data or specific geophysical models
- There initially prevailed a widespread believe that *the reverse transformation has not any elementary solution*, thus [requiring iterations](#) for extracting φ and h_φ from rectangular coordinates (x, y, z)
 - Thus a rich variety of different algorithms has been devised to find *approximations* to the geodetic latitude φ and altitude h_φ above a reference ellipsoid given the geocentric coordinates ρ and ϕ
- But Paul (1973) seems to have been the first to demonstrate the existence of *exact analytical solutions* to the geodetic coordinates. His algorithm was later considerably improved by [Heikkinen \(1982\)](#) while Hedgley (1976) presented a solution based on an entirely different approach
 - The common feature of these algorithms is [solving of a full quartic \(biquadratic\)](#) equation in some variable



Forward transformation: $\varphi \rightarrow \phi$



Conversion from $\phi \rightarrow \varphi$

Traditional fixed-point **iterative solution**:

- Methods like Bowring's formula typically converge in 3–5 iterations for most Earth-based applications

```

r = hypot(R_sat(1), R_sat(2)); % r_δ equatorial component along the semimajor axis (distance from Z-axis)
z = R_sat(3); % r_K: vertical component parallel to the semiminor axis (distance from equatorial plane)

phi_geod = phi_geoc; % φ initial approximation
phi_geod_ = Inf; % (entry loop value)
while abs(phi_geod - phi_geod_) > 1e-9 % targeted tolerance
    phi_geod_ = phi_geod;
    R_N = a / sqrt(1 - e2 * sind(phi_geod_)^2); % Radius of curvature in prime vertical
    phi_geod = atand((z + R_N * e2 * sind(phi_geod_)) / r);
end

h_geod = r / cosd(phi_geod) - R_N; % h_φ altitude of the satellite above the sub-point
  
```

Closed-form solutions:

- e.g. Heikkinen's Algorithm 1982, Borkowski 1987, Zhu 1993
- Closed-form methods vectorize extremely well in SIMD (Single Instruction, Multiple Data) computer programs

→ See code for different algorithms in `geoc_to_geod.m`



Geocentric to geodetic $x, y, z \rightarrow \varphi, h_\varphi + \lambda, \phi, \rho$

```
function [phi_geod, h_geod, Lon, Lat, Rho] = geoc_to_geod(R_ECEF, a, b, method, tol)
% (ECEF) Geodetic coordinate conversion: (x,y,z) → (φ) φ, h_φ
% h_geod = deviation wrt to perfect sphere (0 km @ equator, 21.38 km at poles)

e2 = 1 - (b/a)^2; % First eccentricity squared
e_2 = (a/b)^2 - 1; % Second eccentricity squared == e2 / (1 - e2)

%--- 0) Computes Geocentric longitude λ & latitude φ and radius length ρ
[Lon, Lat, Rho] = Cart2Sph(R_ECEF); % Lat = phi_geoc

%--- 1) Computes Geodetic latitude φ and altitude
switch method

case 1 % Long 1974, NASA TN D-7522 → FASTEST Max err = 1.2×10-6 °
case 2 % Heikkinen's Algorithm (1982) Δ = 2.8e-14 *** BEST ***
case 3 % Zhu 1993: Direct exact transformation from ECEF coordinates to geodetic coordinates in closed form
case 4 % Borkowski 1987, Transformation of geocentric to geodetic coordinates without approximations

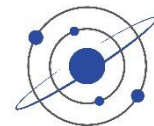
case 10 % Bowring Iterative (convergent) solution
case 11 % Fixed-point iteration of Bowring's formula to calculate the Geodetic Latitude
case 12 % Clynch 2006, Geodetic Coordinate Conversions
case 13 % Fukushima-style implementation

end

R_N = a ./ sqrt(1 - e2 * sind(phi_geod).^2); % R_N(φ)
h_geod = hypot(R_ECEF(1, :), R_ECEF(2, :)) ./ cosd(phi_geod) - R_N;

end % function
```

Heikkinen algorithm $(x, y, z) \rightarrow \varphi, \lambda, h$



- The 1982 Heikkinen algorithm is a classic high-accuracy closed-form (non-iterative) method for transforming Earth-Centered, Earth-Fixed (ECEF) cartesian coordinates (x, y, z) into geodetic coordinates (latitude φ , longitude λ , and height h) on an ellipsoid, i.e. the World Geodetic System ellipsoid
 - Avoids the transcendental iterations found in methods like Bowring's by using a series of algebraic substitutions to solve the quartic equation involved in the transformation
 - Performance: No while loops or convergence checks make it ideal for high-throughput GPU or FPGA implementations
 - Precision: It provides sub-mm accuracy across the entire range of Earth's surface and near-Earth space
 - Stability: Unlike some iterative methods, it does not diverge near the Earth's poles or center
- It is particularly known for:
 - avoiding slow iterative convergence
 - strong numerical stability
 - elegant algebraic structure
- It is somewhat similar in spirit to Borkowski, but formulated differently through auxiliary variables



Heikkinen code

(inside geoc_to_geod)

% r_δ: equatorial component along the semimajor axis (distance from Z-axis)

```
r = hypot(R_ECEF(1, :), R_ECEF(2, :));
```

% r_K: vertical component parallel to the semiminor axis (distance from equatorial plane)

```
z = R_ECEF(3,:);
```

% 1. Initial calculations

```
F = 54 * b^2 * z.^2;
```

```
G = r.^2 + (1 - e2) * z.^2 - e2 * (a^2 - b^2);
```

```
c = (e2^2 * F .* r.^2) ./ (G.^3);
```

% 2. Intermediate variables

```
s = (1 + c + sqrt(c.^2 + 2*c)).^(1/3);
```

```
k = s + 1 + 1./s;
```

```
P = F ./ (3 * k.^2 .* G.^2);
```

```
Q = sqrt(1 + 2 * e2^2 * P);
```

% 3. Calculate Surface Reference r₀

```
term1 = -(P .* e2 .* r) ./ (1 + Q);
```

```
term2 = 0.5 * a^2 * (1 + 1./Q);
```

```
term3 = (P .* (1 - e2) .* z.^2) ./ (Q .* (1 + Q));
```

```
term4 = 0.5 * P .* r.^2;
```

% max(0, ...) to prevent imaginary results from floating point noise

```
r0 = term1 + sqrt(max(0, term2 - term3 - term4));
```

% 4. Final Geodetic Coordinates

```
U = sqrt((r - e2 * r0).^2 + z.^2);
```

```
V = sqrt((r - e2 * r0).^2 + (1 - e2) * z.^2);
```

```
z0 = (b^2 * z) ./ (a * V);
```

```
h_geod = U .* (1 - b^2 ./ (a * V)); % h
```

```
phi_geod = atan2d(z + e_2 * z0, r);
```

```
% lambda = atan2d(y, x);
```



Geodetic to geocentric $\varphi, h_0 \rightarrow \phi, h_\varphi$

```
function [phi_geoc, h_geod] = geod_to_geoc(phi_geod, a,b, h0)
% Retrieves geocentric latitude  $\phi$  and geodetic altitude  $h_\varphi$  (above a) for the geodetic latitude  $\varphi$ 
```

```
rho = a + h0;
```

```
% a) Computes sub-satellite point on the ellipsoid surface
```

```
[R_sub, ~, ~, R_N] = geod_to_pos(phi_geod,0, 0, a,b); % (h = 0)
```

```
x_sub = R_sub(1, :);
```

```
z_sub = R_sub(3, :);
```

```
% b) Computes normal line perpendicular to ellipsoid at sub-satellite point
```

```
m = cotd(90 - phi_geod);
```

```
b_ = z_sub - m .* x_sub; % ordinate at origin  $\neq$  ellipsoid parameter b
```

```
% c) Intersecting normal vector at R_sub with spherical Earth @  $\rho$ 
```

```
% Quadratic equation solution
```

```
x_sat = (-b_ .* m + sqrt(rho.^2 * (m.^2 + 1) - b_.^2)) ./ (m.^2 + 1);
```

```
z_sat = m .* x_sat + b_;
```

```
h_geod = x_sat ./ cosd(phi_geod) - R_N; % Vermeille's algorithm
```

```
phi_geoc = atan2d(z_sat, x_sat); %  $\phi$ 
```

```
% Handles NaNs
```

```
k = ismember(phi_geod, [-90, 0, 90]);
```

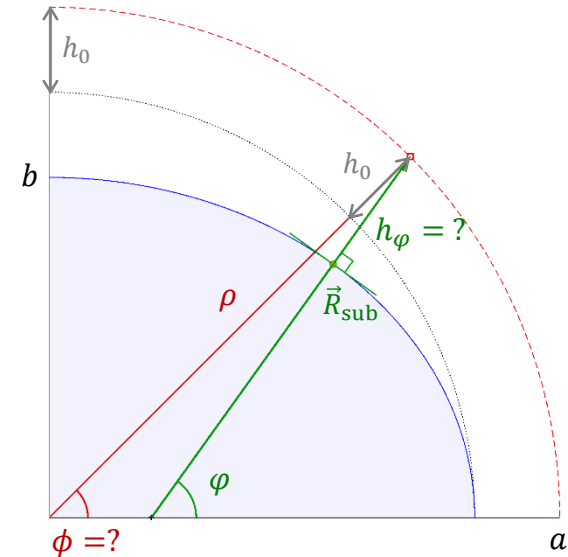
```
phi_geoc(k) = phi_geod(k);
```

```
h_geod(phi_geod == 0) = h0;
```

```
h_geod(abs(phi_geod) == 90) = a + h0 - b;
```

```
end % function
```

Special case of retrieving ϕ, h_φ from φ, h_0





Exercise

- Find site coordinates for the observatory near the top of the Mt. Fujii in Japan
 → Mt. Fujii in Japan, is located at $\varphi = 5.3606^\circ$ N, $\lambda = +138.7274^\circ$ E, $H_{MSL} = 3776$ m

```
phi_geod = 5.3606;    % [°]
lambda = +138.7274;  % [°]
H_MSL = 3776;    % [m]
```

```
R_top = geod_to_pos(phi_geod, lambda, H_MSL/1000); % m → km
% Answer:
% x: -4775.6777 km
% y: 4191.4906 km
% z: 592.2501 km
```

```
r = hypot(R_top(1), R_top(2)) %  $r_\delta \approx 6354.187$  km
z = R_top(3) %  $r_K \approx 592.25$  km
```

Earth Rotation & Revolution

- Earth rotational velocity variation
- Foucault's Pendulum





Earth's Rotation & Revolution

Rotation refers to an object's spinning motion about its own axis → *counter clockwise rotation* (right-hand rule)

- **Earth rotation** describes the spin of our planet, which defines the its orientation in space and Universal Time ([UT1](#))
- **Mean Solar Time:** The Earth rotates on its axis relative to the Sun every 24 hours
 - Representing an *average* of the variations caused by Earth's non-circular orbit
- **Sidereal time:** Earth rotation relative to “fixed” stars
 - is ≈ 4 minutes shorter than the mean solar day, the equivalent of one solar day per year

Revolution refers the object's orbital motion around another object.

- Earth revolves about the Sun, producing the ≈ 365 -day year
 - The [Sidereal Period](#) is 365.2564 days with reference to the stars, at a speed ≈ 30 km/s

$$\text{Earth_speed} = (2 * \pi * \text{Earth.a}) / (\text{Earth.Sidereal_year} * 24 * 60^2); \% \approx 29.784 \text{ km/s}$$
 - The [Tropical Year](#) is 365.2422 days, practically identical to the [Calendar Year](#) 365.2425 days including **leap years**
- Earth's orbit is elliptical and reaches its closest approach to the Sun, a perihelion of 147 M km, on about January fourth of each year. Aphelion comes six months later at 152 M km

→ more on this later in SOL Part I – [Time](#)

www.britannica.com/video/151528/Earth-rotation-axis-revolution-Sun

Earth rotational velocity variation



- Earth rotational velocity $\omega_{\oplus}$ **is not constant** and varies over timescales ranging from hours to millions of years due to several physical processes
- Any motion of mass in or on Earth causes a slowdown or speedup of the rotation speed (fluctuations of the order of milliseconds), or a change of rotation axis
 - Some motions involving movements of very large masses can produce discernable changes in Earth's rotation, and can affect very precise astronomical observations
 - Through interactions of individual components, angular momentum can be transferred between the atmosphere, ocean, and solid Earth, or between the Earth's core and mantle due to the coupling mechanisms.

1. Long-term slowing (*secular* trend)

Earth's rotation gradually slows because of **tidal friction caused by the Moon** resulting in an increase of day length by ≈ 2.4 ms/century ($\rightarrow$ see [Tidal dissipation](#))

2. Short-term variations (days to years)

- **Geophysical processes** like **atmospheric winds**, tides and ocean currents (hydrosphere redistribution)
- Interactions between mantle and core



UT1

Earth rotational velocity variation (continued)



3. Polar motion and wobble

The rotation axis itself moves relative to Earth's crust, changing the orientation and affecting rotational dynamics

- Chandler wobble (≈ 433 days); Annual wobble ($\rightarrow$ see Section Polar Motion)

4. Sudden events

Large earthquakes can slightly alter Earth's moment of inertia

- Example: 2004 Sumatra earthquake shortened the day by microseconds

$\rightarrow$ All of these effects produce changes in Length of Day (LOD), and variations are tracked through UT1

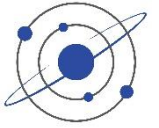
- These variations are quantified through [Earth Orientation Parameters](#) (EOP), which link the celestial and terrestrial reference frames, and include [Polar motion](#), [UT1–UTC](#) (LOD changes), and [Precession-Nutation](#)
 - Global simulations of atmosphere, ocean, and land dynamics are used to create effective angular momentum (EAM) functions that can be used to predict changes in EOP

Winds can change the length of a day



- The movement of Earth's atmosphere can make days very slightly longer or shorter. Although this may seem surprising, the way air circulates around the planet can subtly alter the speed at which Earth rotates (affect UT1)
- Earth's rotation produces the cycle of day and night, with one rotation lasting about 24 hours. However, the exact length of a day is not perfectly constant. Small variations – typically only a few milliseconds – occur as a result of tides, changing winds, and other natural processes
- When strong winds circulate around the globe, particularly in the jet stream and the stratosphere, they carry angular momentum, which reflects the overall rotation of the atmosphere. Because the atmosphere and the solid Earth are dynamically coupled, any increase in atmospheric angular momentum must be balanced by a slight decrease in Earth's rotational speed. For example, when westerly winds intensify, Earth's rotation slows imperceptibly; when those winds weaken, Earth's rotation speeds up slightly
- This exchange of angular momentum occurs through surface friction and pressure forces exerted against major mountain ranges such as the Rockies and the Himalayas
- Even though the effect is very small – *fractions of a millisecond* – it can be measured precisely using satellites and atomic clocks

Winds can change the length of a day (continued)



- Large-scale climate phenomena, including El Niño, can modify global wind patterns over periods of months or even years. [These atmospheric shifts produce measurable variations in the length of a day](#)
- Studying this interaction between atmospheric dynamics and Earth's rotation helps scientists improve climate modeling and refine satellite-based measurements, where even millisecond-level changes are significant

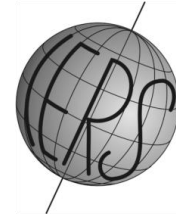


The pressure of strong winds against major mountain chains like the Rockies can subtly influence Earth's rotation. *Credit: Kyle Brittain*



EOP – Earth Orientation Parameters

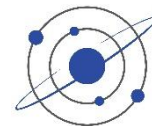
- In geodesy and astrometry, **Earth Orientation Parameters (EOP)** describe **irregularities in the rotation of the Earth**
 - EOP describe the **change of the orientation of the Earth's surface**, on which the observatories are located, **with respect to a space fixed reference frame**
 - EOP provide the rotational transform from ECI $\leftrightarrow$ ECEF as a function of time
- Published by the **International Earth Rotation and Reference Systems Service (IERS)**
 - EOPs are continuously monitored using precise geodetic techniques, combining data from four global space geodetic networks: **VLBI** (**Very Long Baseline Interferometry**), **SLR** (Satellite Laser Ranging), **GNSS** (Global Navigation Satellite Systems), and **DORIS** (Doppler Orbitography and Radiopositioning Integrated by Satellite)
- Relevance of the knowledge of accurate EOP:
 - Realization of **reference frames** on Earth and in space
 - **Precise positioning** and **satellite navigation**
 - **Precise orbit** determination (POD)
 - Realization of **time systems** (e.g. monitoring of the duration of days)
 - **Earth system studies** (e.g. climate change)



IERS – Int. Earth Rotation and Reference Systems Service
<https://geodesy.science/item/iers>

→ Latest EOP data table can be downloaded from [Celestrak.org](https://celestrak.org/SpaceData/eop19620101.txt)
<https://celestrak.org/SpaceData/eop19620101.txt>; [EOP Data Documentation](#)
 (contains data for dates = 1962-01-01 to current date + $\approx$ 6 months)

Earth Orientation Parameter (EOP) Data	
CSV starting 1962 Jan 01 (2,349,673 bytes)	CSV starting 2020 Jan 01 (236,207 bytes)
Legacy starting 1962 Jan 01 (2,447,532 bytes)	Legacy starting 2020 Jan 01 (244,395 bytes)



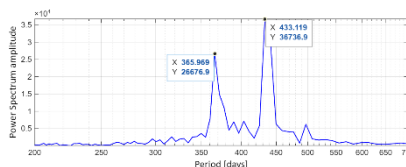
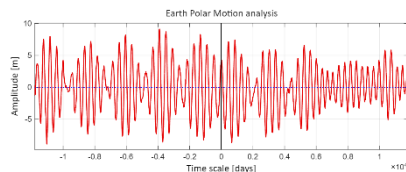
EOP data study

- The following figures present the data contained in the EOP data file

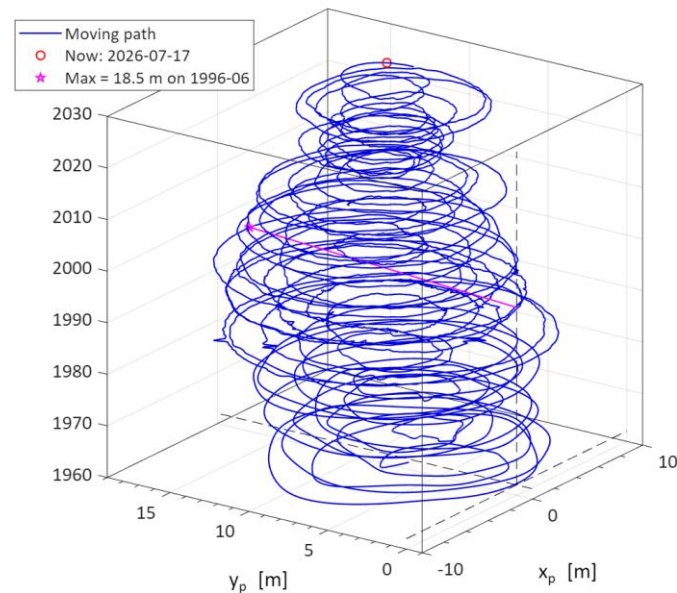
- <https://celestrak.org/SpaceData/EOP-format.php>

- Polar Motion x, y:

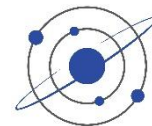
- The coordinates of the Earth's instantaneous rotational axis with respect to the solid Earth's crust (the Conventional International Origin)
- Tracks the shift of the Earth's rotational axis across the physical crust, indicating where the North Pole drifts across the globe
- Pole position is not steady due to the Earth mass inhomogeneity
366 – 433 days period with an amplitude of < 10 m



Polar motion, 1962-01 to 2026-08, from IERS EOP series

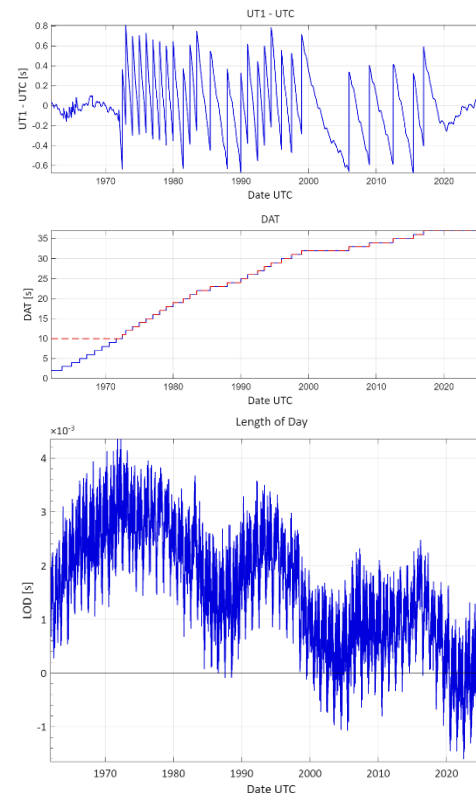


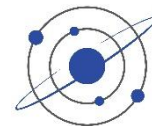
Polar wandering



EOP data study

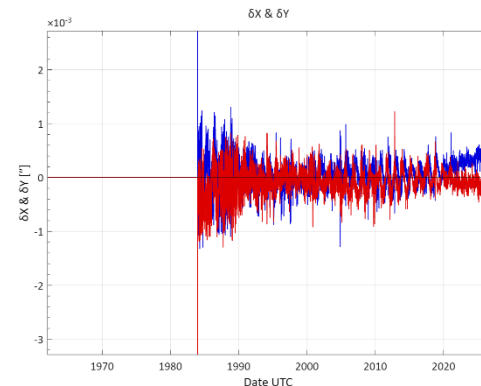
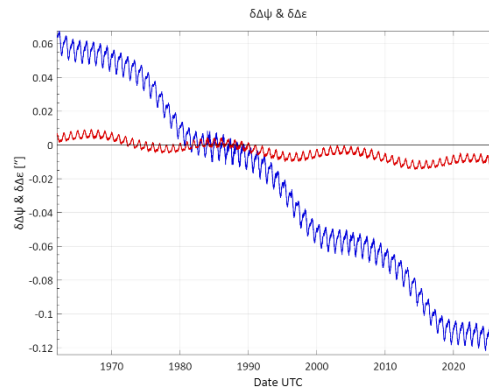
- **UT1-UTC (Earth Rotation Angle)**
 - The difference between Universal Time (based on the Earth's actual rotation) and Coordinated Universal Time (based on atomic clocks)
 - Measures the discrepancy between uniform atomic time and the Earth's actual, irregular rotation angle
 - It tells the exact angular offset of the Earth due to its slight speed variations
- **LOD (Length of Day)**
 - represents the tiny, fluctuating difference between the actual observed duration of Earth's rotation and the standard nominal day of 86,400 atomic seconds (exactly 24 hours)
- **DAT (Delta Atomic Time)**
 - The difference between International Atomic Time (TAI) and UTC (also known as the number of leap seconds)





EOP data study

- $\delta\Delta\psi, \delta\Delta\varepsilon$ (Nutation Corrections):
 - Found in older datasets (using IAU-76/FK5 theory), these are the corrections to the celestial pole's longitude (ψ) and obliquity (ε), used to account for the gravitational “wobble” of the Earth's axis caused by the Moon and Sun
 - $< 0.1''$ are on top of Precession & Nutation $< 10''$
- $\delta X, \delta Y$ (Celestial Pole Offsets):
 - Monitors the long-term corrections to precession and nutation models in space
 - Used in modern conventions (IAU-2000/2010), these replace $\delta\Delta\psi$ and $\delta\Delta\varepsilon$. They represent the small, observed adjustments to the location of the Celestial Intermediate Pole (CIP) in the celestial reference frame



Nutation in obliquity vs longitude

Foucault's Pendulum



- The apparent rotation of the celestial sphere can be explained either by a daily revolution of the heavens around a stationary Earth or by the rotation of Earth itself. Since the 17th century, the latter interpretation has been widely accepted; however, it was not until the nineteenth century that the French physicist Jean Foucault provided a direct and unambiguous demonstration of Earth's rotation
- In 1851, Foucault suspended a pendulum approximately 60 m long and weighing about 25 kg from the dome of the Panthéon in Paris and set it into uniform oscillation
- If Earth were motionless, the pendulum's plane of oscillation would have remained fixed, tracing the same path indefinitely. Yet, after only a few minutes, Foucault observed that the plane of motion slowly rotated
- He explained that the pendulum itself did not change direction; rather, the Earth was turning beneath it. As Earth rotates, the plane of oscillation of the Foucault pendulum appears to shift progressively, so that over roughly twelve hours, the pendulum successively knocks down all the targets arranged in a circle around its platform



Foucault's Pendulum



- The angular velocity ω_p of the precession (apparent rotation) of the pendulum's plane is given by:

$$\omega_p = \omega_E \sin(\phi)$$

where: $\omega_E = 2\pi/T_E$ is Earth's angular velocity,
 $T_E = 23$ h 56 min is the sidereal day, and
 ϕ is the latitude

- The period of rotation of the pendulum's plane is thus:

$$T_p = \frac{T_E}{\sin(\phi)}$$

- At:
 - the North Pole ($\phi = 90^\circ$): $T_p \approx 24$ h
 - the Equator ($\phi = 0^\circ$): $T_p = \infty$ (no precession),
 - mid-latitudes (e.g., $\phi = 46.8^\circ$): $T_p \approx 33$ h

```
>> Earth.Sidereal_day / sind(46.774)
32.847 % [hr] @ Québec
```

Ecliptic plane & Obliquity

- Celestial sphere
- Ecliptic plane
- Axial tilt & seasons
- First Point of Aries / Vernal Equinox
- Milankovitch cycles

Further reading:

Rice University 2022 – **Astronomy 2e**
2.1 The Sky Above (pp. 34–41)

NASA Basics of Spaceflight:
[Chapter 2: Reference Systems](#)

Some definitions



■ North

- For planets and natural satellites, the spin-pole direction extending to the positive side of the plane defined by the solar systems' positive angular momentum vector. For asteroids and comets, north is along the body's spin-axis positive angular momentum direction

■ East

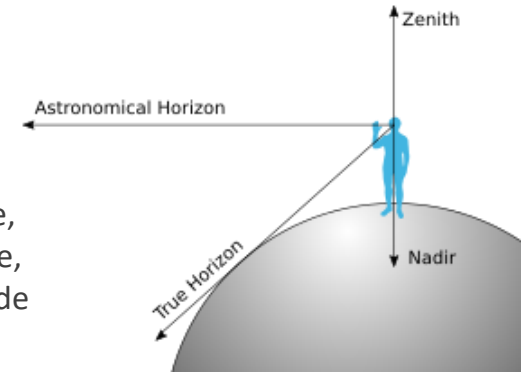
- The **counter-clockwise** (RHR) direction around the north-pole of rotation

■ Zenith

- The point in the sky **directly above an observer's location**, at 90° elevation from the horizon. It is the highest point in the celestial sphere relative to the observer

■ Nadir <https://en.wikipedia.org/wiki/Nadir>

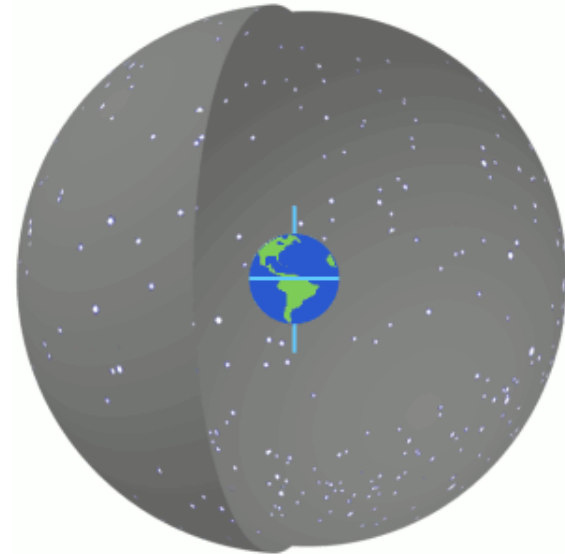
- The point directly opposite zenith, **located beneath the observer**, pointing straight down toward the center of the Earth
- A **satellite nadir view** is a top-down, straight-down perspective of the Earth's surface, where the sensor points directly beneath the satellite (the nadir point) at a 90° angle, providing accurate geometric imagery for mapping and land use but often lacking side details. It contrasts with **oblique views (off-nadir)**, which capture slanted, angled perspectives that reveal features like building sides but introduce geometric distortions



Celestial Sphere



- An **imaginary sphere that surrounds Earth**, representing the appearance of the sky from our planet
 - Like a globe with stars, planets, and other celestial objects drawn on it
- It is an useful tool for describing the sky and for establishing coordinate systems for stars and planets, and to model the positions of celestial objects
- SEE:
 - Edward Murphy: Lecture 05 – The Celestial Sphere
<https://youtu.be/-tsxLKGFWiY?si=gN9E6eItfsGSLa8A> [1:07:10]
 - Edward Murphy: Lecture 06 – As the World Turns
<https://youtu.be/JQaeJrSbWS8?si=mpt5hGqPF7U9IUSh>

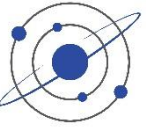


<https://cosmosatyourdoorstep.com/2014/09/07/the-celestial-sphere/>

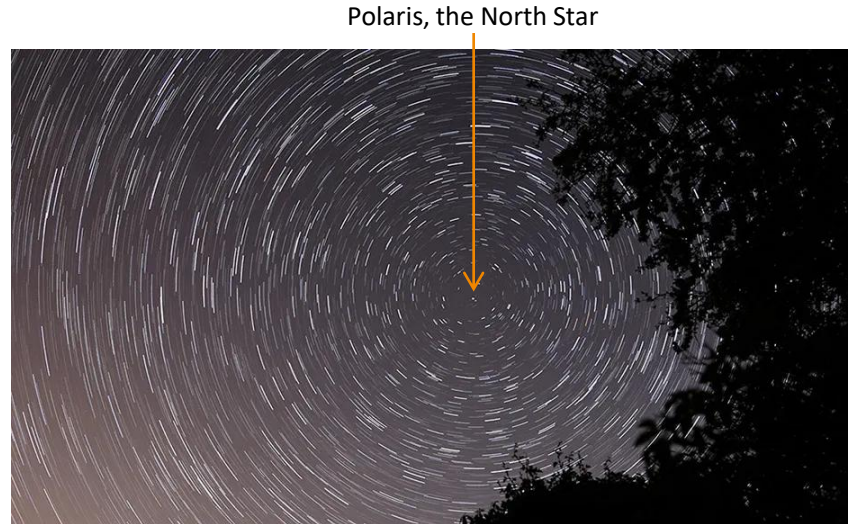
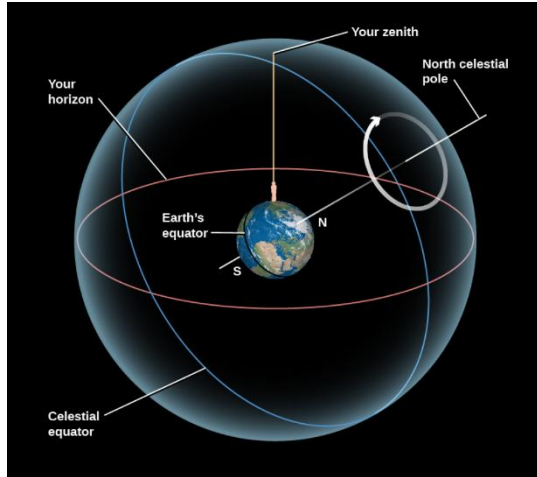


Celestial Sphere (continued)

Celestial Sphere



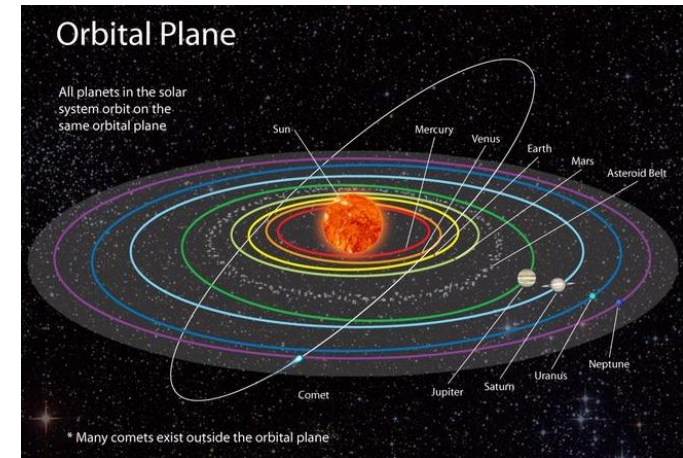
- This long-exposure photo shows trails left by stars as a result of the apparent rotation of the celestial sphere around the north celestial pole. → *In reality, it is Earth that rotates*
 - https://en.wikipedia.org/wiki/Star_trail



Planet formation and orbits stabilization

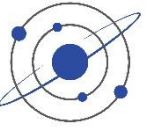


- The planets of the Solar System originated from the **solar accretion disc**, a flattened, rotating structure of gas and dust surrounding the young Sun. Within this disc, local concentrations of material underwent progressive accretion through collisions and gravitational coalescence, forming planetesimals and, ultimately, the planets
 - In the early Solar System, dynamical interactions among growing bodies produced a **highly chaotic environment**
 - Frequent collisions, near misses, and **gravitational perturbations modified orbital eccentricities and inclinations**, occasionally ejecting material or altering trajectories
 - Over time, mutual interactions, dissipative processes, and long-term gravitational resonances led to a gradual damping of orbital instabilities
 - The resulting system evolved toward dynamical equilibrium, yielding the **relatively stable** and **approximately coplanar planetary orbits that characterize the Solar System today**
- All major planets in the Solar System orbit in approximately the same plane; their orbits are not perfectly aligned, with Mercury (7°) and Venus (3.4°) deviating the most. → The dwarf planet Pluto is inclined by $\approx 17^\circ$



Planet's orbit inclinations

Saturn's rings



- An analogous process can be observed in [Saturn's rings](#), where countless icy and rocky particles [orbit within an almost perfectly coplanar configuration](#), illustrating the natural tendency of rotating systems to flatten under angular momentum conservation

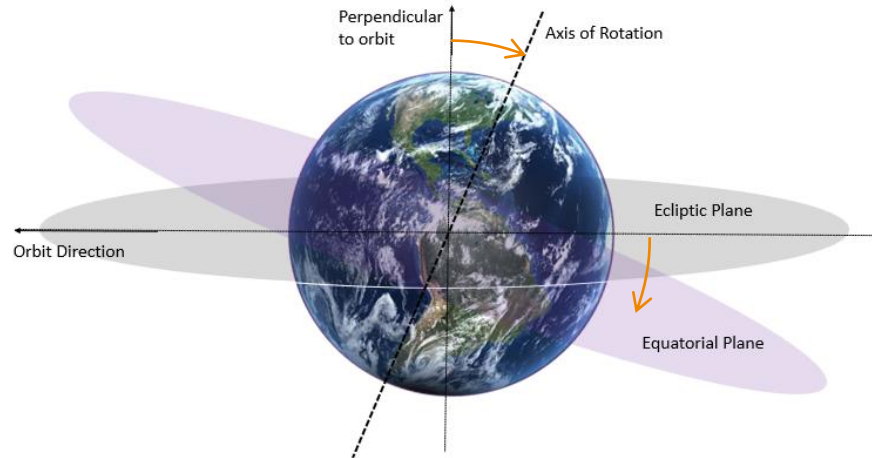


www.sciencefriday.com/segments/saturn-rings-moon-origin (Credit: [Shutterstock](#))

Ecliptic plane



- The **Ecliptic Plane** is the orbital plane in which the Earth orbits around the Sun
 - It serves as a fundamental reference plane in celestial mechanics and astronomy
 - Most planets in the Solar System have orbits lying close to it; the Earth orbits the Sun at 0° (by definition)
 - The **Ecliptic** is the **Great Circle on the celestial sphere** that represents the apparent annual path of the Sun as observed from Earth
- It is where the Sun and Moon converge during their eclipses (*hence the name*)

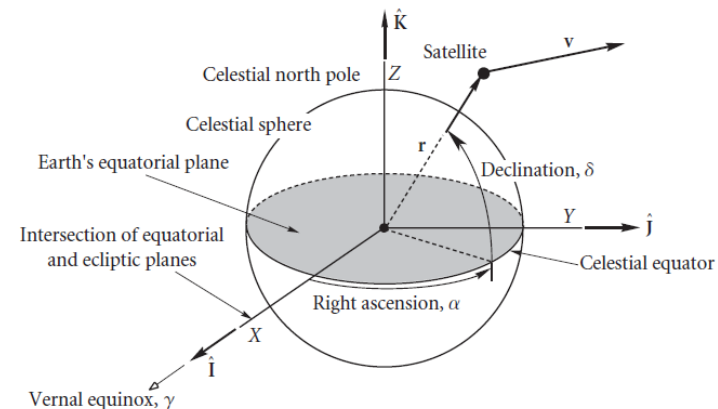


Ecliptic reference



- The **Heliocentric Ecliptic (ECL)** reference frame is an inertial system in astrodynamics centered on the Sun, using the ecliptic plane (Earth's orbit plane) as the fundamental reference plane.
 - The x - y plane is based on the orbit of the Earth around the Sun, with the x -axis origin toward the **First Point of Aries**, the z -axis normal to the ecliptic, and the y -axis to complete a right-handed system.
- Since the Earth's orbit plane fluctuates slightly due to gravitational perturbations, the ecliptic chosen can be either:
 - At a **fixed standard time** (such as the **J2000.0 epoch**) providing an *inertial frame*, or instead
 - The **time of interest** ("**of-date**"), providing a *dynamic frame*, depending on the purpose.
- Both amount to adopting a standard due to definitional ambiguities caused by the Earth and Moon's mutual motion affecting the Earth's orbital plane

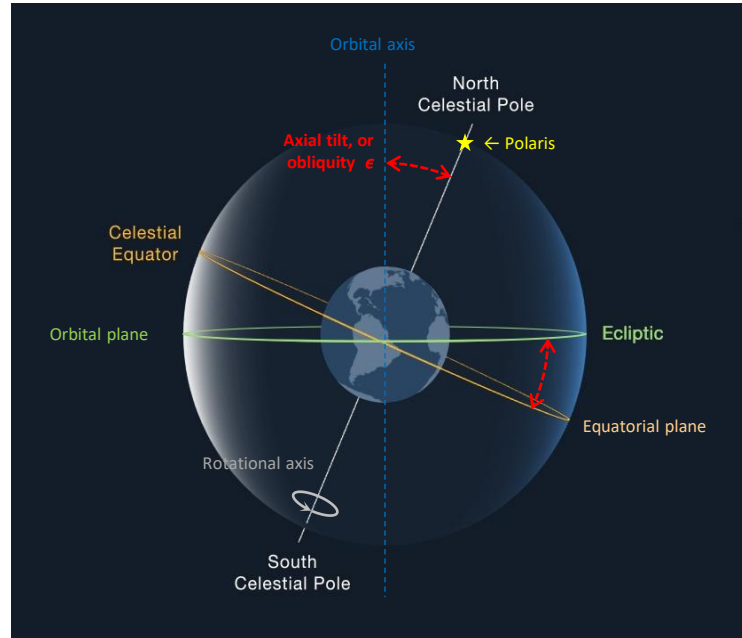
<https://ssd.jpl.nasa.gov/horizons/manual.html>



Axial tilt the Earth



- **Axial tilt** is the angle between an object's rotational axis and its orbital axis, which is the line perpendicular to its orbital plane; equivalently, it is the angle between its equatorial plane and orbital plane (ecliptic)



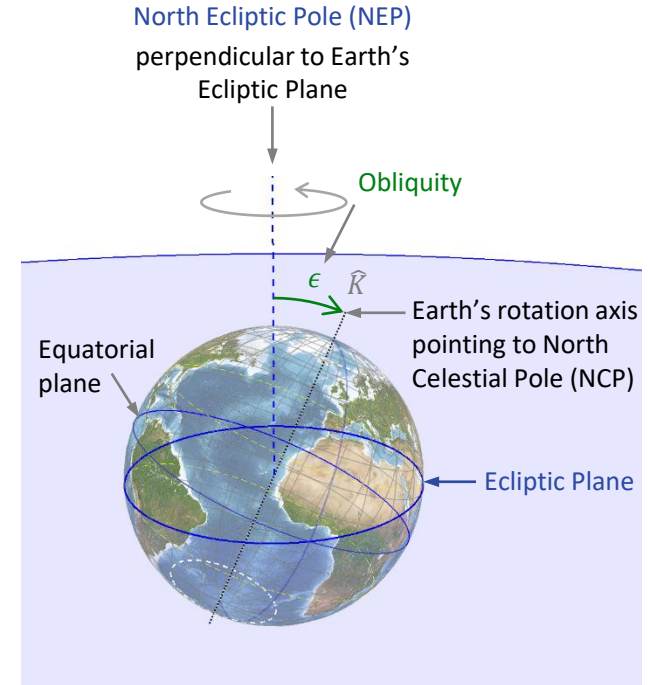
Obliquity of the ecliptic ϵ



- The **Earth ecliptic** is inclined by $\epsilon \approx 23.44^\circ$ with respect to Earth's equatorial plane, an inclination known as the **obliquity of the ecliptic**
→ *Obliquity is why Earth has seasons*
- This value remains about the same relative to a stationary orbital plane throughout the cycles of **axial precession**, the gradual shift in the orientation of Earth's axis of rotation in a cycle of $\approx 26,000$ years
- ϵ changes only by $\pm 0.00013^\circ$ per year (see [Milankovitch Cycles](#))

```
EarthObliquity() % = 23.4359°  
EarthObliquity(now)
```

```
EarthObliquity(datetime(2000,01,01)) % = 23.4394°  
diff(EarthObliquity(datetime(2025:2026,01,01))) % = -0.00013
```



Milankovitch Cycles

Cause of Earth obliquity



The Earth's present obliquity angle is the result of its early history and subsequent gravitational dynamics

1. Giant Impact Hypothesis

- ≈ 4.5 billion years ago, Earth is thought to have [collided with a Mars-sized protoplanet \(often called *Theia*\)](#).
- This impact likely tilted Earth's rotation axis away from being perpendicular to its orbital plane, and ejected material that [later formed the Moon](#). / Also alternative theory where the Moon was separated from Earth due to gravitational stresses and cosmic forces, occurring when Earth was still hot and unstable
- Numerical models show that such a collision could easily impart a $20\text{--}30^\circ$ tilt

2. [Stabilizing Role](#) of the Moon

- The Moon's strong gravitational influence stabilizes Earth's obliquity
- Without the Moon, Earth's tilt would [vary chaotically](#) between 0° and 85° on million-year timescales, due to gravitational torques from Jupiter and other planets [Laskar *et al.* 1980s–2000s]
- With the Moon, Earth's tilt oscillates in a narrow range (roughly $22\text{--}24.5^\circ$) with a 41,000-year cycle (one of the [Milankovitch cycles](#))

3. Planetary Perturbations

- Gravitational interactions with the Sun, Jupiter, and other planets [cause long-term oscillations of the tilt](#).
- These oscillations, [combined with precession of the equinoxes](#), drive cyclic climate changes such as ice ages

How scientists reconstruct Earth's obliquity history



1. Mathematical / Orbital Mechanics Models

- Earth's obliquity evolution is governed by:
 - **Precession**: the slow conical wobble of Earth's axis ($\approx 26,000$ years).
 - **Planetary perturbations**: mainly from Jupiter, Saturn, and Venus, which shift Earth's orbital plane.
 - **Lunar torque**: the Moon pulls on Earth's equatorial bulge, damping chaotic variations
- Using modern celestial mechanics, scientists integrate the **N-body** equations of motion backwards and forwards in time for tens to hundreds of millions of years

2. Paleoclimate Evidence

- Earth's axial tilt affects how solar radiation is distributed across latitudes (seasonality).
This leaves fingerprints in climate records:
 - **Ice cores** (Antarctica, Greenland): Show glacial–interglacial cycles linked to $\approx 41,000$ -year obliquity oscillations.
 - **Sediment cores** (ocean drilling): Contain fossil shells (foraminifera) and chemical ratios ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$) that track ocean temperature and ice volume, matching obliquity-driven cycles
 - **Loess (windblown dust) and lake records**: Dust and pollen deposition also reflect climate rhythms tied to tilt variations. Together, these allow reconstruction of Earth's obliquity back several million years, and indirectly even further

How scientists reconstruct Earth's obliquity history

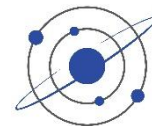


3. Fossil Day-Length Records

- Ancient corals preserve growth bands that record Earth's day length hundreds of millions of years ago
- Since day length is linked to Earth-Moon tidal evolution (which also affects obliquity), these provide indirect constraints on how tilt stability evolved through geologic time

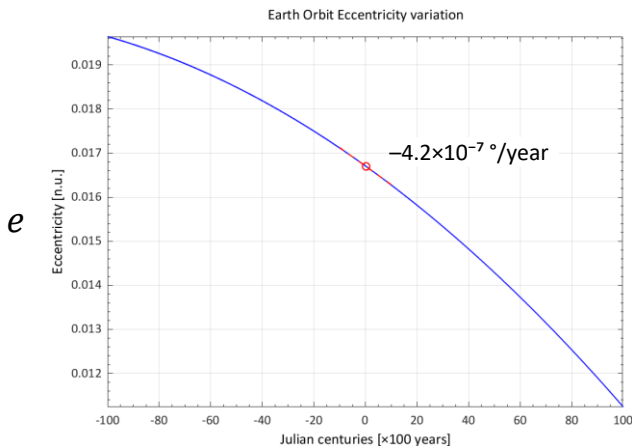
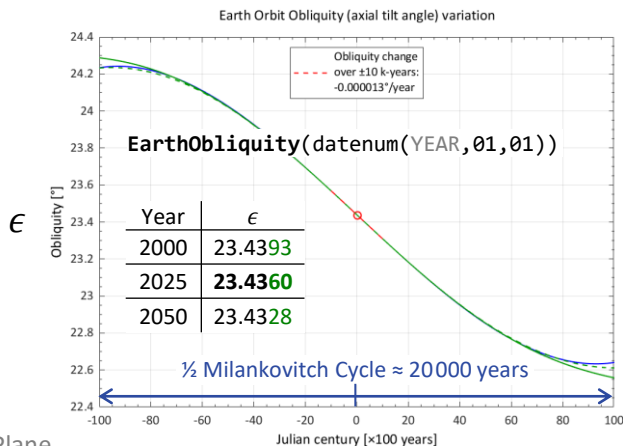
4. Numerical Simulations of Early Earth

- Supercomputer models simulate giant impacts and proto-Earth accretion
- They show that Earth's tilt after the [Theia](#) impact could have been much larger initially (even $> 60^\circ$), then [tidal interactions with the Moon damped it down](#) to today's $\approx 23.44^\circ$ range over billions of years

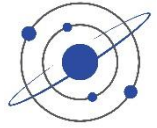


Earth's orbit variation over time

- **Milankovitch Cycles** describe the collective effects of changes in the Earth's movements on its climate over tens of thousands of years
- Earth's orbit **Obliquity angle ϵ** slowly varies **over millennia**
 - Over the last million years, obliquity has varied between 22.1° and 24.5° with respect to Earth's orbital plane.
- Earth's orbit **Eccentricity e** also slowly varies with time, due to gravitational interaction with other celestial bodies
 - The eccentricity value can get as low as 0.0034 and up to 0.058

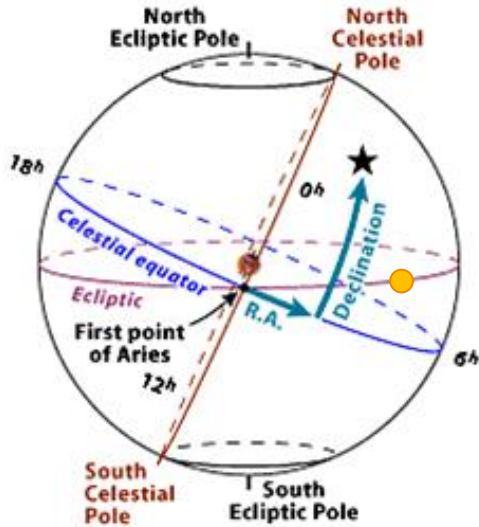


Earth in Ecliptic Plane



Celestial sphere and coordinates

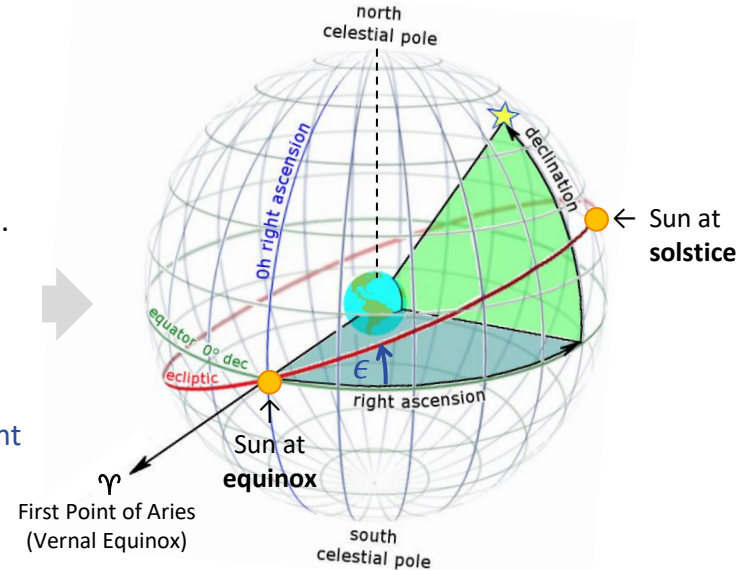
- Equatorial (Celestial/Earth) vs Ecliptic reference planes



An equivalent way to look at our path around the Sun each year is to *imagine that the Sun moves around Earth on the ecliptic circle.*

Because Earth's axis is tilted, the ecliptic is tilted by $\epsilon \approx 23.44^\circ$ relative to the celestial equator aligned with Earth's equator.

As a result, *we see the Sun's height in the sky change during the year.*

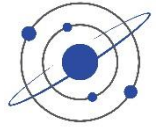


IMP: Right Ascension and Declination are global coordinates, *independent of observer location.*

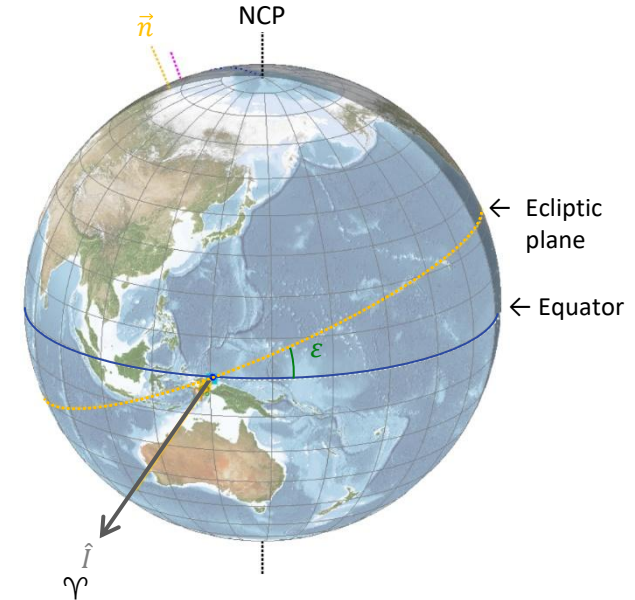


Celestial Sphere

Υ First Point of Aries / Vernal Equinox



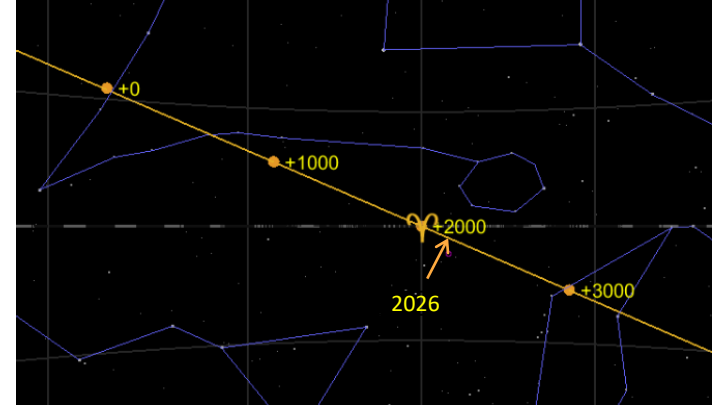
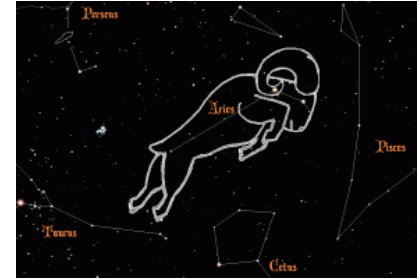
- Just as the Greenwich Prime meridian has been arbitrarily chosen as the zero point for measuring longitude on the surface of the Earth, the **First Point of Aries** (1stA, Υ) has been chosen as the zero point in the celestial sphere calculations of **right ascension**
- Υ is the **intersection of the ecliptic and the celestial equator planes**
 - When the Sun reaches this point going from south to north, as it does once a year, the **Vernal Equinox** occurs, and this is defined as the first day of spring (*vernus/vernalis* in Latin)
- This fundamental reference direction in the sky is chosen because, as opposed to the Sun, which is (only) ≈ 150 M km away from the Earth, it is measured relative to far away constellations, for which none lie less than 100 lightyears away from our solar system
 - As the Earth travels around the Sun throughout the year, the 1stA direction remains essentially fixed in the same direction, allowing astrodynamics to use this as an inertial coordinate frame



“First Point of Pisces” ?

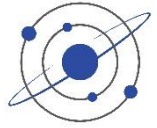


- It is interesting to note that the “First Point of ARIES” is actually a bit misleading today
- It is so named because it was originally in the zodiac constellation, **ARIES**, “The Ram”
 - Consequently the zodiacal symbol for the Ram, ♈ , or sometimes the gamma letter γ (which has a similar appearance), is used for the vernal equinox.
- This is where The Point was located in 150 B.C. when Ptolemy first mapped the constellations.
- However, due to **Precession of the Equinoxes**, it moved from ARIES into the adjacent constellation of PISCES in about 70 B.C.
 - So it should actually be called the “first point of PISCES”!
- The Point’s journey is now about 75% complete through PISCES; in another 600 years, it will pass into AQUARIUS.



Precession of the Equinox

Sun declination δ



- δ is the angle between the equator and a line drawn from the center of the Earth to the center of the Sun, at a given moment of the year
- This angle, which is the **primary cause of Earth's seasons**, varies throughout the year due to the Earth's axial tilt, or obliquity ϵ , as it orbits the Sun ($-\epsilon \leq \delta(t) \leq +\epsilon$)
- The declination is:
 - **Positive** in the Northern Hemisphere's summer,
 - **Negative** in its winter, and
 - **Zero at the equinoxes** (March 22 and September 22),
 - with a maximum of $+23.44^\circ$ at the June solstice and a minimum of -23.44° at the December solstice

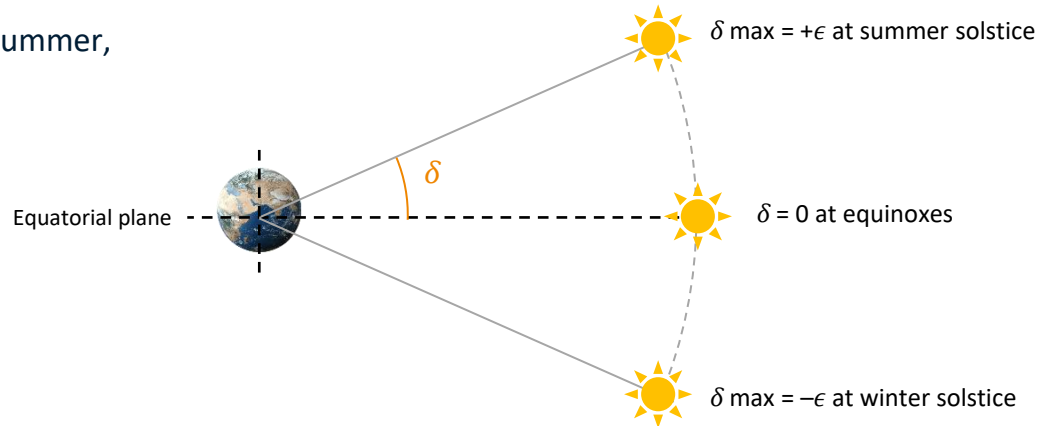
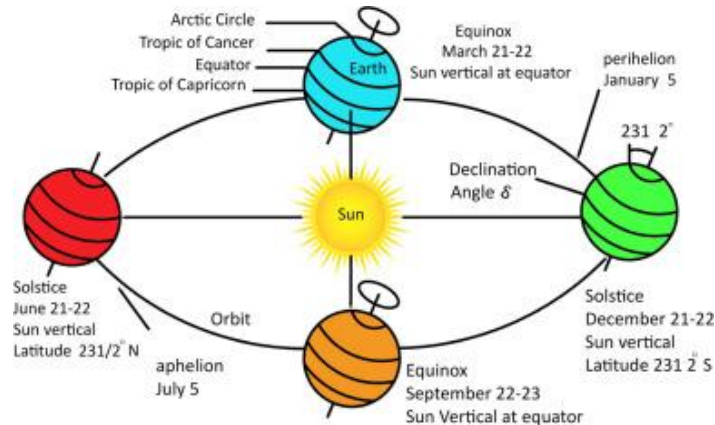


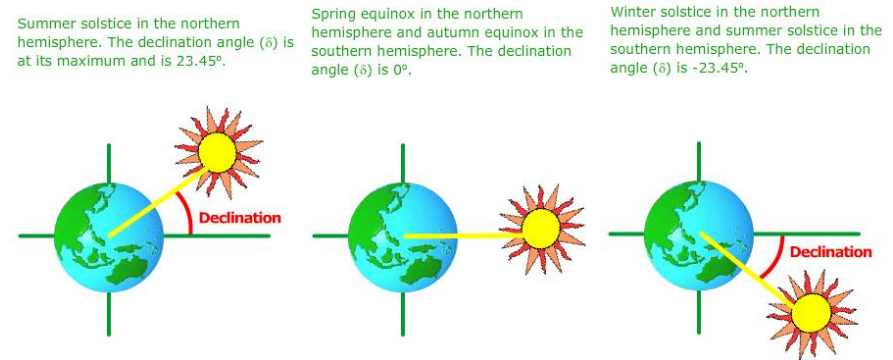
Illustration of Sun's declination



View from a fixed Sun



View from a fixed Earth



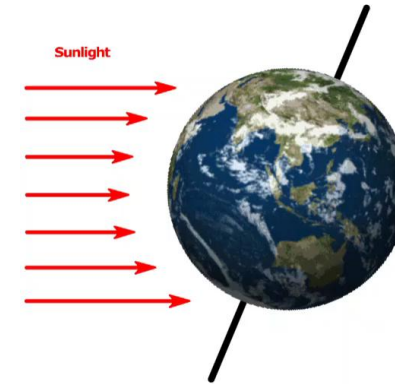
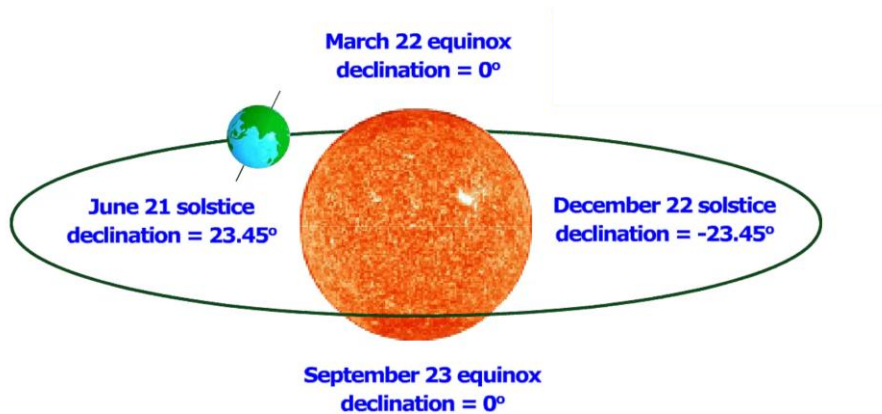
IMP: Despite the fact that the Earth revolves around the sun, it is simpler to think of the Sun revolving around a stationary Earth. This requires a coordinate transformation. Under this alternative coordinate system, the Sun moves around the Earth.

www.pveducation.org/pvcdrom/properties-of-sunlight/declination-angle

Seasonal variation of the declination angle

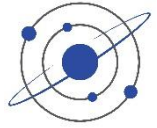


- The following animations show how the tilt angle changes from the summer solstice in the northern hemisphere (or winter in the southern hemisphere) to the northern hemisphere winter solstice (summer in the south)

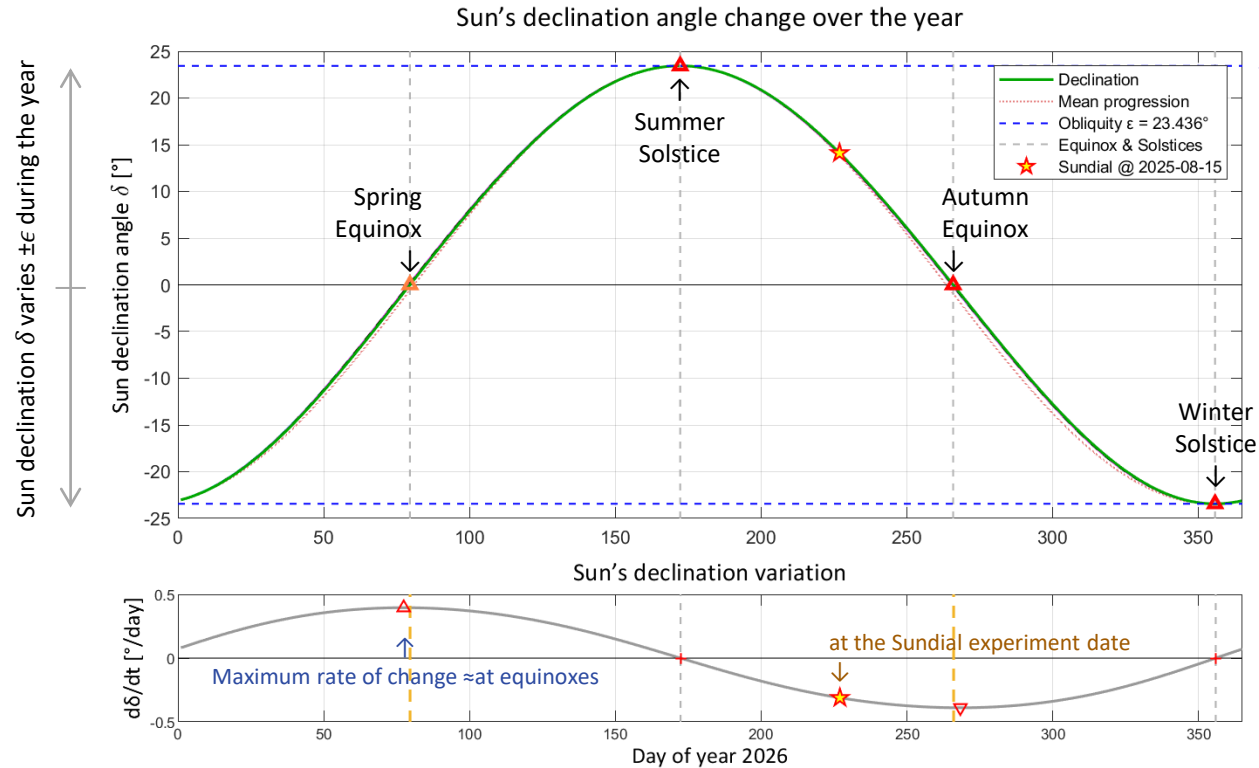


The Earth does 365 revolutions as it goes around the Sun in one year. Many more than are shown here.

www.pveducation.org/pvcdrom/properties-of-sunlight/declination-angle



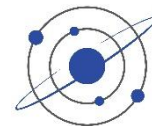
Sun declination yearly variation



← -23.436° is the current value of Earth obliquity ϵ (changes over millennia)

δ is independent of observer's location on Earth

Note: $\frac{d^2\delta}{dt^2} = 0$ does not perfectly coincide with equinox dates



δ calculation

Angle δ can be calculated by different methods (see code SOL_Ephemerides):

- **Cosine model:** mean progression of Earth's orbit → Declination approximation (assuming a circular orbit)

$$\delta \approx +23.436^\circ \times \cos\left(360 \times \frac{d + \Delta d}{365.25}\right)$$

with d is the day of the year ($d = 1$ on Jan 1), and Δd the offset corresponding to the winter solstice occurrence, ≈ 193 days before the start of the year

- **Sine model** (Solar Radiation Atlas, Section 3.4)

$$j = d \times 360^\circ / 360.2425$$

day number expressed as a day angle

$$\delta \approx \text{asin}(0.3978 \sin(j - 80.2 + 1.92 \sin(j - 2.8)))$$

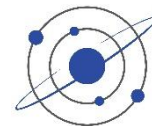
- **Series approximation** (European Solar Radiation Atlas equations)

$$\delta \approx b_1 + b_2 \sin(\omega) + b_3 \sin(2\omega) + b_4 \sin(3\omega) + b_5 \cos(\omega) + b_6 \cos(2\omega) + b_7 \cos(3\omega)$$

- **Ecliptic obliquity method** (Meeus Astronomical Algorithms, ESRL/NOAA)

$$\delta \approx \text{asin}(\sin(\epsilon) \sin(L_\odot))$$

SunEphemeris(Dates)



δ calculation

- **Chebyshev fitting method** (JPL Ephemeris / NASA Horizons)

- JPL's planetary ephemerides (DE series) and the NASA Horizons system both rely on a [highly efficient numerical representation of orbits using Chebyshev polynomial fits](#) to very accurate numerical integrations of the Solar System's equations of motion.
- JPL numerically integrates Solar System motion, then fits each body's trajectory in short time segments with Chebyshev polynomials, which Horizons evaluates on demand.

```
DE_440 = JPL_Ephemerides(Dates);  
[SV, OBS] = QueryHorizons(10, Dates(1), Dates(end), length(Dates));
```

→ See SOL Part II – Ephemerides

Declination table (for navigation)

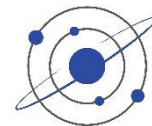


Table of the Sun's Declination
Mean Value for the Four Years of a Leap Year Cycle

Day	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
1	S23° 04'	S17° 20'	S7° 49'	N4° 18'	N14° 54'	N21° 58'	N23° 09'	N18° 10'	N8° 30'	S2° 57'	S14° 14'	S21° 43'
2	S22° 59'	S17° 03'	S7° 26'	N4° 42'	N15° 12'	N22° 06'	N23° 05'	N17° 55'	N8° 09'	S3° 20'	S14° 34'	S21° 52'
3	S22° 54'	S16° 46'	S7° 03'	N5° 05'	N15° 30'	N22° 14'	N23° 01'	N17° 40'	N7° 47'	S3° 44'	S14° 53'	S22° 01'
4	S22° 48'	S16° 28'	S6° 40'	N5° 28'	N15° 47'	N22° 22'	N22° 56'	N17° 24'	N7° 25'	S4° 07'	S15° 11'	S22° 10'
5	S22° 42'	S16° 10'	S6° 17'	N5° 51'	N16° 05'	N22° 29'	N22° 51'	N17° 08'	N7° 03'	S4° 30'	S15° 30'	S22° 18'
6	S22° 36'	S15° 52'	S5° 54'	N6° 13'	N16° 22'	N22° 35'	N22° 45'	N16° 52'	N6° 40'	S4° 53'	S15° 48'	S22° 25'
7	S22° 28'	S15° 34'	S5° 30'	N6° 36'	N16° 39'	N22° 42'	N22° 39'	N16° 36'	N6° 18'	S5° 16'	S16° 06'	S22° 32'
8	S22° 21'	S15° 15'	S5° 07'	N6° 59'	N16° 55'	N22° 47'	N22° 33'	N16° 19'	N5° 56'	S5° 39'	S16° 24'	S22° 39'
9	S22° 13'	S14° 56'	S4° 44'	N7° 21'	N17° 12'	N22° 53'	N22° 26'	N16° 02'	N5° 33'	S6° 02'	S16° 41'	S22° 46'
10	S22° 05'	S14° 37'	S4° 20'	N7° 43'	N17° 27'	N22° 58'	N22° 19'	N15° 45'	N5° 10'	S6° 25'	S16° 58'	S22° 52'
11	S21° 56'	S14° 18'	S3° 57'	N8° 07'	N17° 43'	N23° 02'	N22° 11'	N15° 27'	N4° 48'	S6° 48'	S17° 15'	S22° 57'
12	S21° 47'	S13° 58'	S3° 33'	N8° 28'	N17° 59'	N23° 07'	N22° 04'	N15° 10'	N4° 25'	S7° 10'	S17° 32'	S23° 02'
13	S21° 37'	S13° 38'	S3° 10'	N8° 50'	N18° 14'	N23° 11'	N21° 55'	N14° 52'	N4° 02'	S7° 32'	S17° 48'	S23° 07'
14	S21° 27'	S13° 18'	S2° 46'	N9° 11'	N18° 29'	N23° 14'	N21° 46'	N14° 33'	N3° 39'	S7° 55'	S18° 04'	S23° 11'
15	S21° 16'	S12° 58'	S2° 22'	N9° 33'	N18° 43'	N23° 17'	N21° 37'	N14° 15'	N3° 16'	S8° 18'	S18° 20'	S23° 14'
16	S21° 06'	S12° 37'	S1° 59'	N9° 54'	N18° 58'	N23° 20'	N21° 28'	N13° 56'	N2° 53'	S8° 40'	S18° 35'	S23° 17'
17	S20° 54'	S12° 16'	S1° 35'	N10° 16'	N19° 11'	N23° 22'	N21° 18'	N13° 37'	N2° 30'	S9° 02'	S18° 50'	S23° 20'
18	S20° 42'	S11° 55'	S1° 11'	N10° 37'	N19° 25'	N23° 24'	N21° 08'	N13° 18'	N2° 06'	S9° 24'	S19° 05'	S23° 22'
19	S20° 30'	S11° 34'	S0° 48'	N10° 58'	N19° 38'	N23° 25'	N20° 58'	N12° 59'	N1° 43'	S9° 45'	S19° 19'	S23° 24'
20	S20° 18'	S11° 13'	S0° 24'	N11° 19'	N19° 51'	N23° 26'	N20° 47'	N12° 39'	N1° 20'	S10° 07'	S19° 33'	S23° 25'
21	S20° 05'	S10° 52'	S0° 00'	N11° 39'	N20° 04'	N23° 26'	N20° 36'	N12° 19'	N0° 57'	S10° 29'	S19° 47'	S23° 26'
22	S19° 52'	S10° 30'	N0° 24'	N12° 00'	N20° 16'	N23° 26'	N20° 24'	N11° 59'	N0° 33'	S10° 50'	S20° 00'	S23° 26'
23	S19° 38'	S10° 08'	N0° 47'	N12° 20'	N20° 28'	N23° 26'	N20° 12'	N11° 39'	N0° 10'	S11° 12'	S20° 13'	S23° 26'
24	S19° 24'	S9° 46'	N1° 11'	N12° 40'	N20° 39'	N23° 25'	N20° 00'	N11° 19'	S0° 14'	S11° 33'	S20° 26'	S23° 26'
25	S19° 10'	S9° 24'	N1° 35'	N13° 00'	N20° 50'	N23° 24'	N19° 47'	N10° 58'	S0° 37'	S11° 54'	S20° 38'	S23° 25'
26	S18° 55'	S9° 02'	N1° 58'	N13° 19'	N21° 01'	N23° 23'	N19° 34'	N10° 38'	S1° 00'	S12° 14'	S20° 50'	S23° 23'
27	S18° 40'	S8° 39'	N2° 22'	N13° 38'	N21° 12'	N23° 21'	N19° 21'	N10° 17'	S1° 24'	S12° 35'	S21° 01'	S23° 21'
28	S18° 25'	S8° 17'	N2° 45'	N13° 58'	N21° 22'	N23° 19'	N19° 08'	N9° 56'	S1° 47'	S12° 55'	S21° 12'	S23° 19'
29	S18° 09'	S8° 03'	N3° 09'	N14° 16'	N21° 31'	N23° 16'	N18° 54'	N9° 35'	S2° 10'	S13° 15'	S21° 23'	S23° 16'
30	S17° 53'		N3° 32'	N14° 35'	N21° 41'	N23° 13'	N18° 40'	N9° 13'	S2° 34'	S13° 35'	S21° 33'	S23° 12'
31	S17° 37'		N3° 55'		N21° 50'		N18° 25'	N8° 52'		S13° 55'		S23° 08'

Exemple: 11 août
N15° 27' $\equiv$ +15.45°



Sundial experiment

The seasons

- How the tilt of Earth's axis causes the seasons
- How seasonal differences on Earth vary with latitude
- Solstices and Equinox
- Tropics of Cancer and Capricorn
- Seasonal lag
- Atmospheric refraction and scattering effects



The seasons

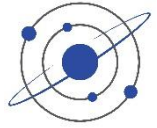


- The amount of solar energy received at Earth's surface varies markedly over the course of a year
- This annual cycle gives rise to *the seasons*, each characterized by distinct durations and intensities of sunlight
- The contrast between seasons becomes increasingly pronounced with latitude – regions farther from the equator experience greater differences in temperature and daylight – while the Northern and Southern Hemispheres experience *opposite seasonal phases*
 - In the Southern hemisphere, the June Sun is low in the sky, meaning fewer daylight hours.
In December, when the Sun is south of the celestial equator, the situation is reversed

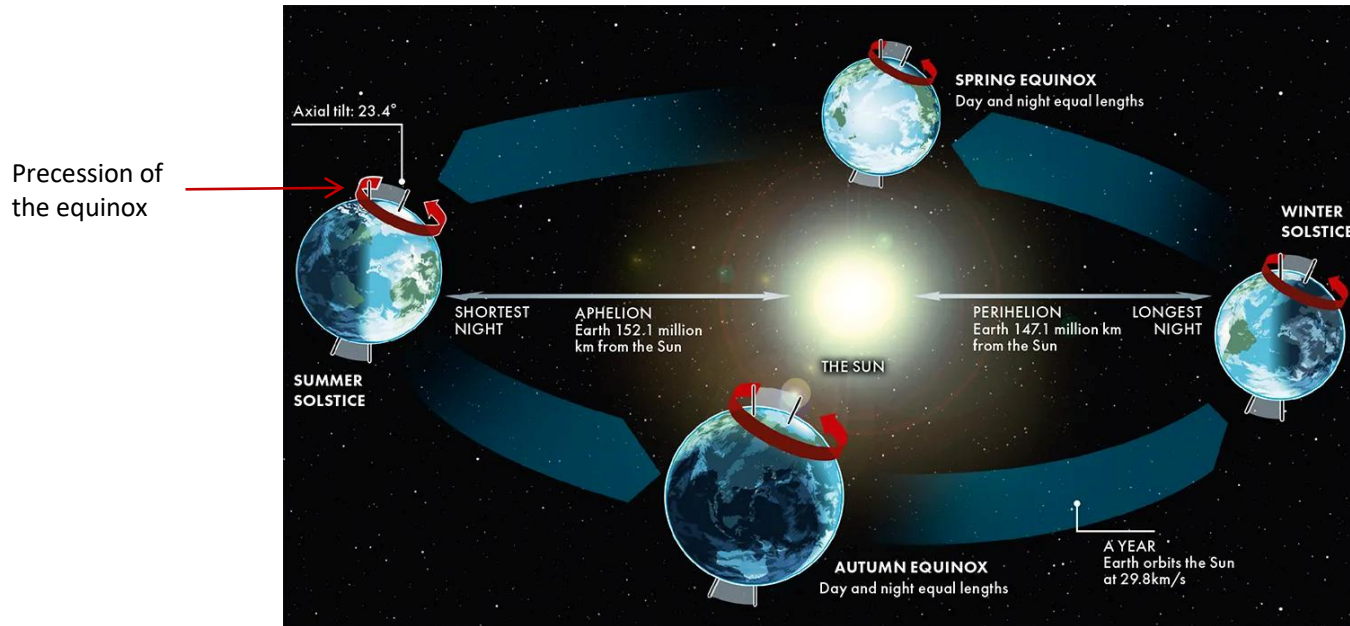
Misconception

- A common misconception *attributes the seasons to changes in Earth's distance from the Sun*
 - Although Earth follows a slightly elliptical orbit, the variation in distance amounts to only about 3%, insufficient to produce significant temperature differences.
 - Moreover, Earth is actually closest to the Sun (at perihelion) in early January, during the Northern Hemisphere's winter
- Clearly, distance cannot account for the alternation of opposite seasons in the two hemispheres
- The true cause lies in the 23.44° tilt of Earth's rotational axis relative to the plane of its orbit, which changes the angle and duration of sunlight received at different latitudes throughout the year

The seasons



- The Earth orbits around the Sun maintaining its spin orientation (obliquity $\epsilon = 23.44^\circ$) relative to the stars

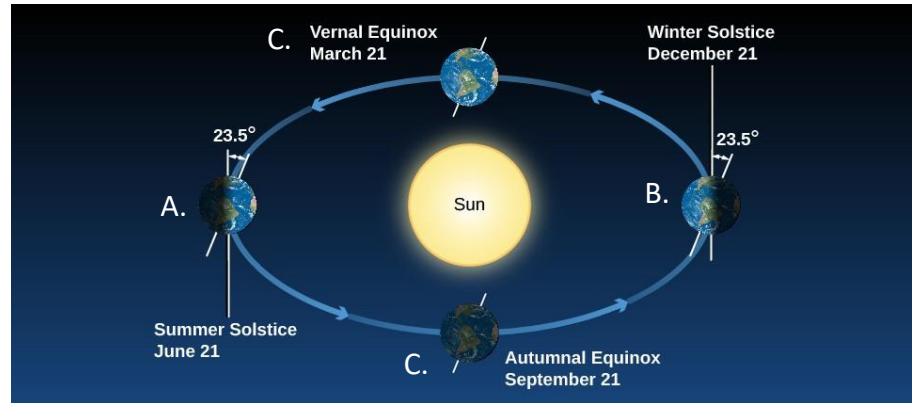


The seasons



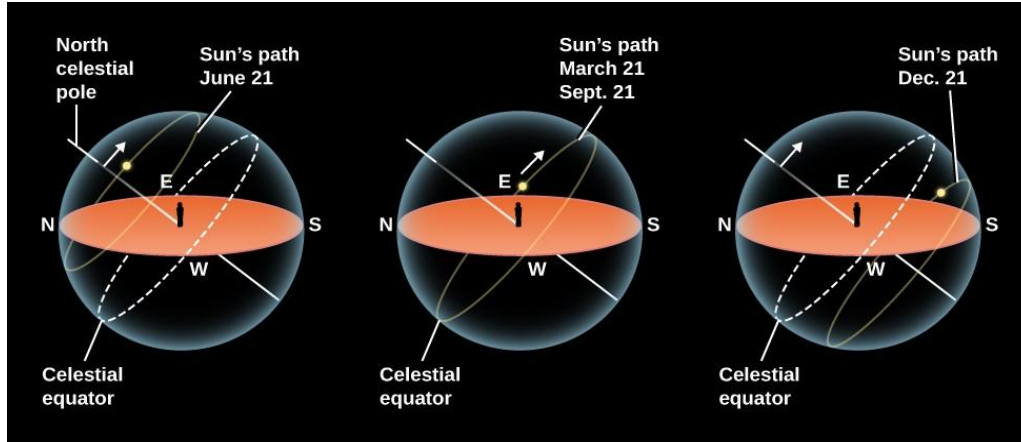
As Earth orbits the Sun, its tilted axis orientation causes variations in solar illumination throughout the year

- A. In the **June Summer solstice**, the Northern Hemisphere is inclined toward the Sun, receiving more direct sunlight and longer days
- B. In **December Winter solstice**, the situation is reversed – the Southern Hemisphere is tilted toward the Sun while the Northern Hemisphere tilts away, leading to shorter days and reduced solar heating in the north
- C. During the **Equinoxes in March and September**, Earth's axis is oriented such that neither hemisphere is tilted toward or away from the Sun, resulting in nearly equal daylight and sunlight distribution across the globe

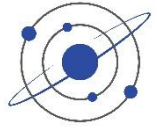


The Earth's axis is fixed ($\epsilon = 23.44^\circ$) and continues to point the same direction throughout the year

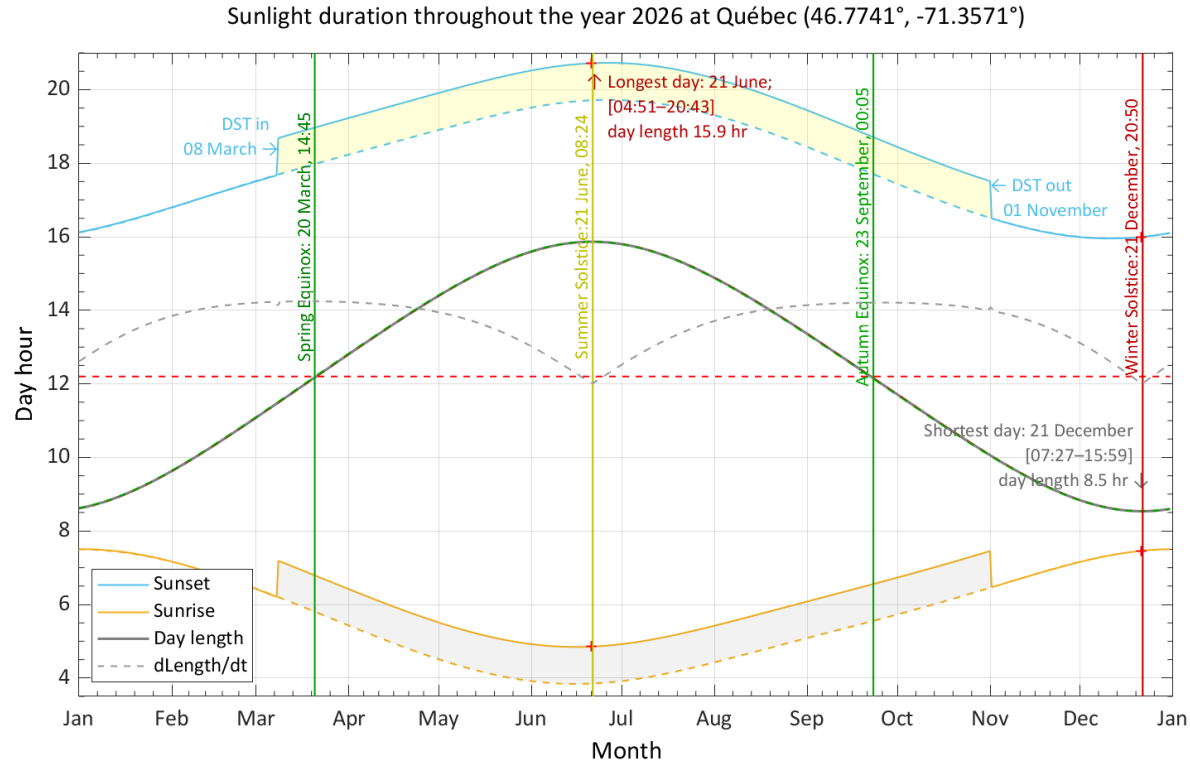
Sun's path in the sky for different seasons



- **Summer Solstice:** The Sun rises north of east and sets north of west. For observers in the Northern Hemisphere of Earth, the Sun spends **15.9 hours above the horizon** in Québec, meaning more hours of daylight
- **Winter Solstice:** The Sun rises south of east and sets south of west. It spends **8.5 hours** above the horizon in Québec
- **Equinoxes:** On March 21 and September 21, the Sun spends equal amounts of time above and below the horizon in both hemispheres



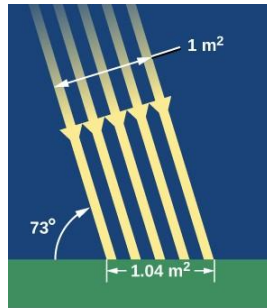
Day length along the year (2026)



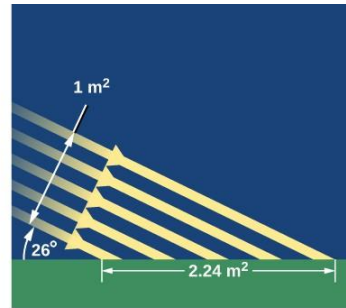
Sun's warming effect



- The Sun's preferential illumination of one hemisphere result in warmer temperatures at Earth's surface. Two main factors are involved:
 1. When a hemisphere tilts toward the Sun, **sunlight strikes the surface at a steeper, more direct angle**, concentrating solar energy over a smaller area and thereby increasing heating efficiency. In June, for example, the Northern Hemisphere receives this more direct and intense sunlight, leading to greater surface warming



(a)



(b)

The Sun's Rays in Summer and Winter.

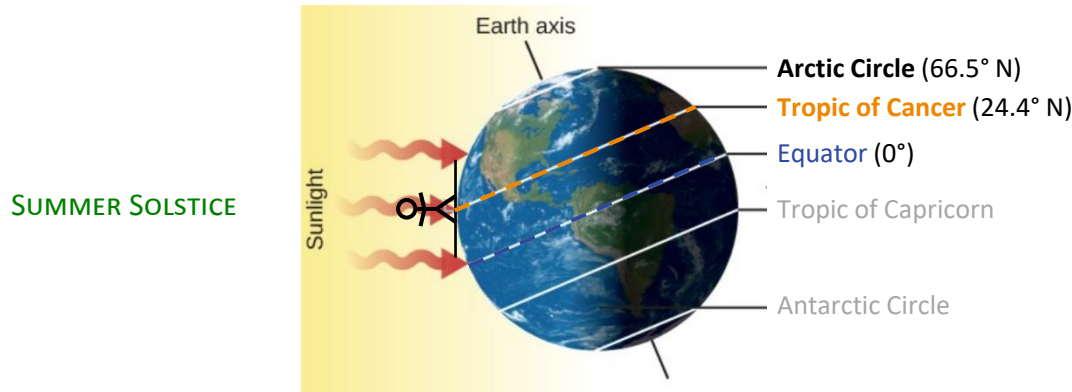
- (a) During summer, the Sun appears higher in the sky, and its rays strike Earth's surface at a steeper angle, concentrating energy over a smaller area
- (b) In winter, the Sun remains lower in the sky, and its rays arrive at a shallower angle, spreading over a larger region and thus providing less effective surface heating

2. When a hemisphere is tilted toward the Sun, the Sun **remains above the horizon for a longer portion of the day**, allowing more time for solar energy to accumulate. Conversely, during winter, the shorter daylight hours reduce the total energy received, resulting in cooler conditions

Summer Solstice



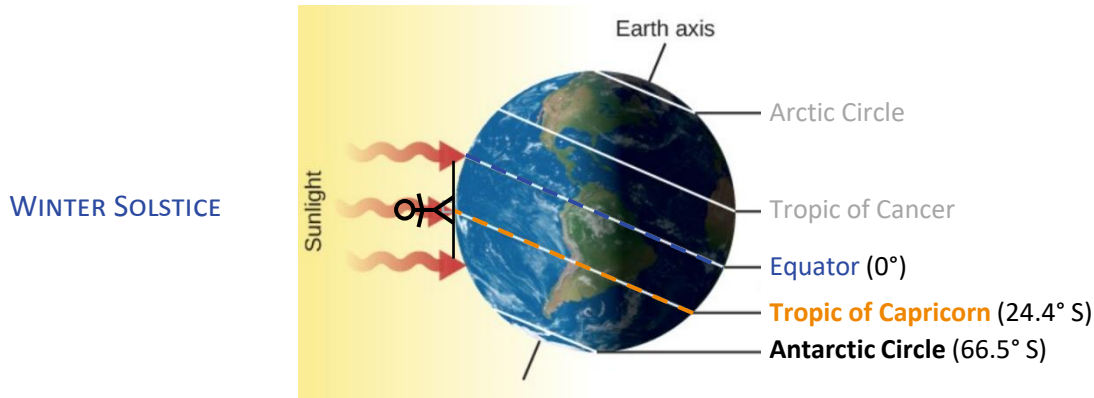
- **June 21** is the date of the summer solstice in the Northern Hemisphere
- The North Pole is in constant sunlight while the South Pole is veiled in 24 hours of darkness
 - The Sun is as far north on this date as it can get; thus, $90^\circ - 23.44^\circ = 66.56^\circ$ N is the northernmost latitude where the Sun can be seen for a full 24-hour period (sometimes called the “land of the midnight Sun”)
 - That circle of latitude is called the *Arctic Circle*
- At latitude $+23.44^\circ$ N, called the Tropic of Cancer, the Sun passes through the zenith at noon



Winter Solstice



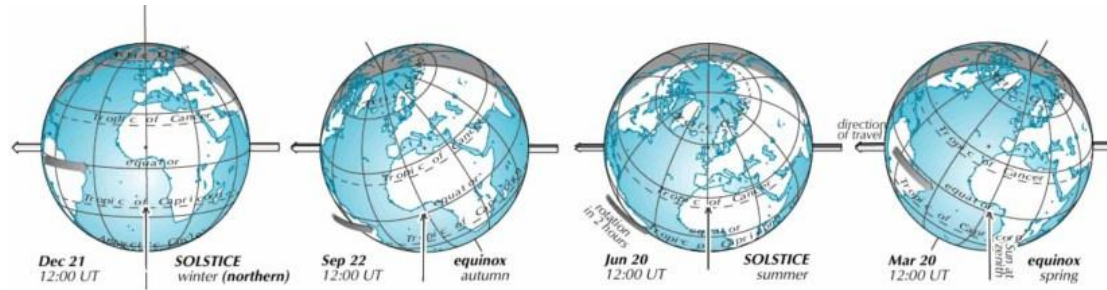
- **December 21** is the date of the winter solstice, or the first day of winter in the Northern Hemisphere
- The North Pole is in darkness for 24 hours and the South Pole is illuminated
- At latitude 23.44°S , called the [Tropic of Capricorn](#), the Sun passes through the zenith at noon
- Days are longer in the Southern Hemisphere and shorter in the north
 - In the United States and Canada, there may be only 9 or 10 hours of sunshine during the day.
 - It is winter in the Northern Hemisphere and summer in the Southern Hemisphere



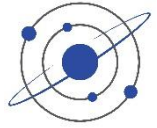
Equinoxes



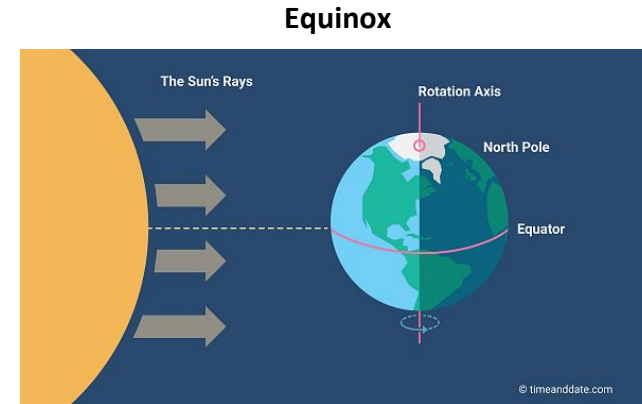
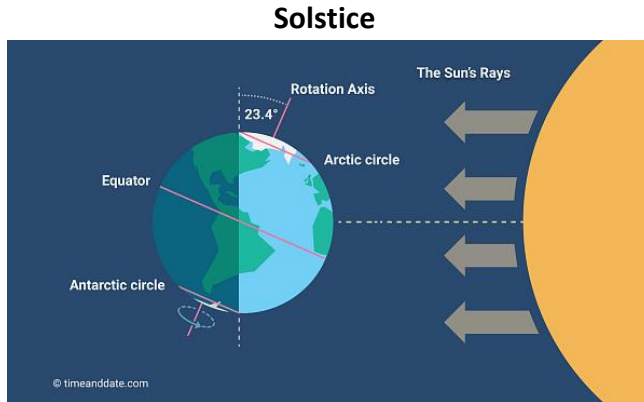
- Halfway between the solstices ($\approx$ March 21 and September 21), the Sun is on the celestial equator
- From Earth, it appears above our planet's equator and favors neither hemisphere
- Every place on Earth then receives roughly 12 hours of sunshine and 12 hours of night
- The points where the Sun crosses the celestial equator are called the *vernal* (spring) and *autumnal* (fall) equinoxes

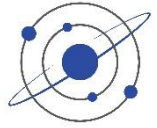


Solstices and Equinox



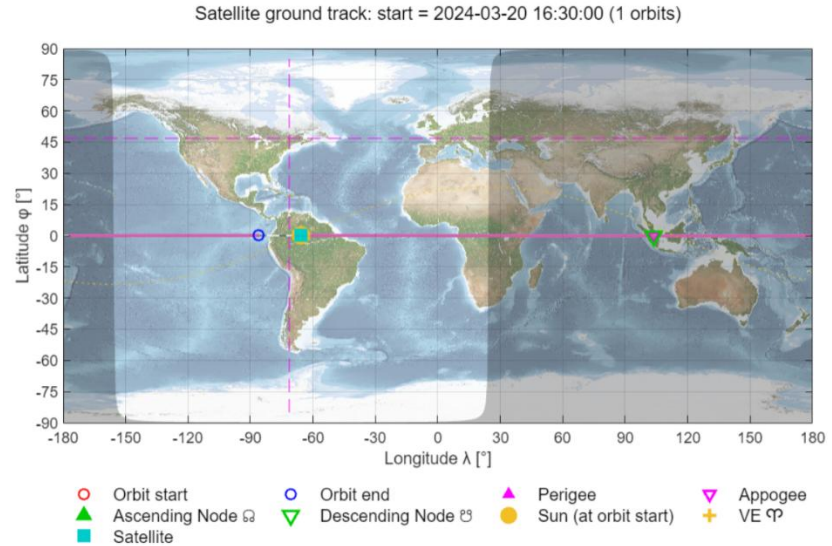
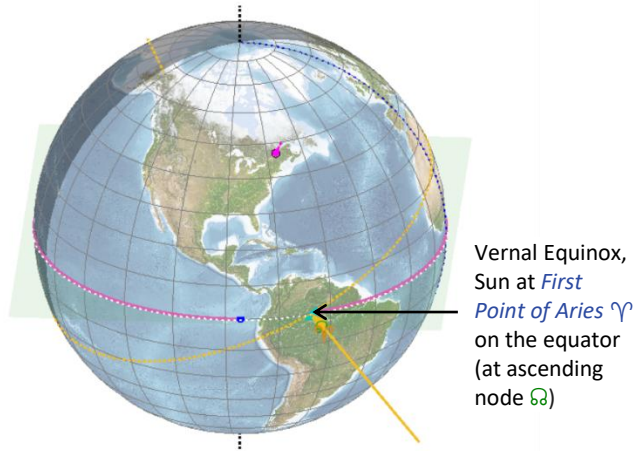
- **Solstice:** A solstice is a moment in time when the Sun's path is farthest from Earth's equator, either north or south. There are two solstices each year, the summer solstice and the winter solstice
 - The axial tilt of the Earth is maximum ($= \pm \epsilon$) wrt to the ecliptic plane
- **Equinox:** The equinox is the time of year when the length of day and night are nearly equal. It marks the beginning of spring and fall in the Northern Hemisphere
 - The axial tilt of the Earth is null with respect to the ecliptic plane

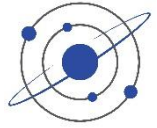




Spring Equinox

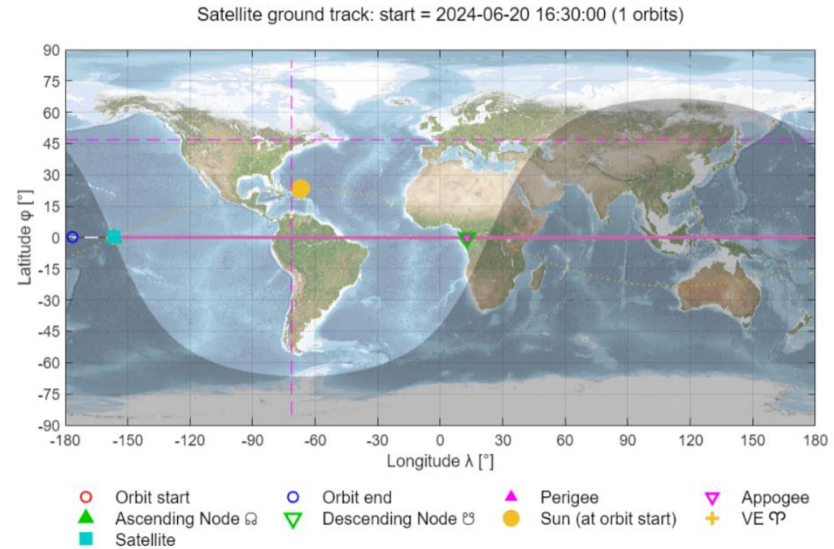
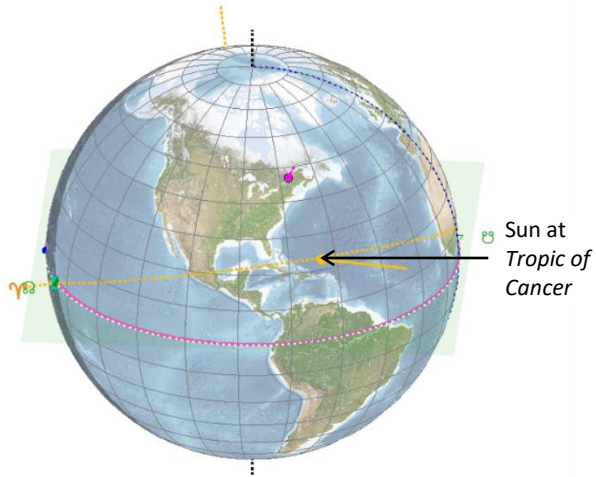
Winter solstice → ❄️ ♒️ ← **Spring equinox**
 Summer solstice → ☀️ ♋️ ← Autumn equinox

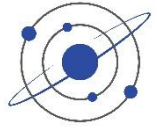




Summer Solstice

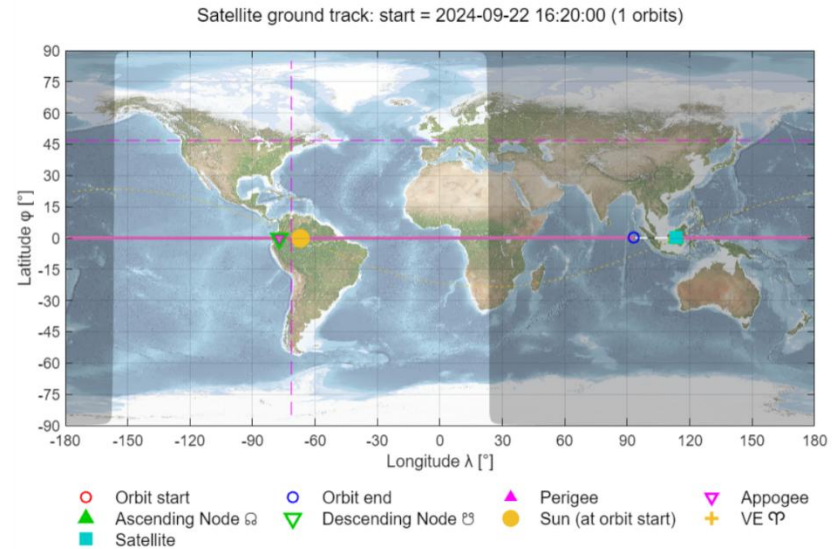
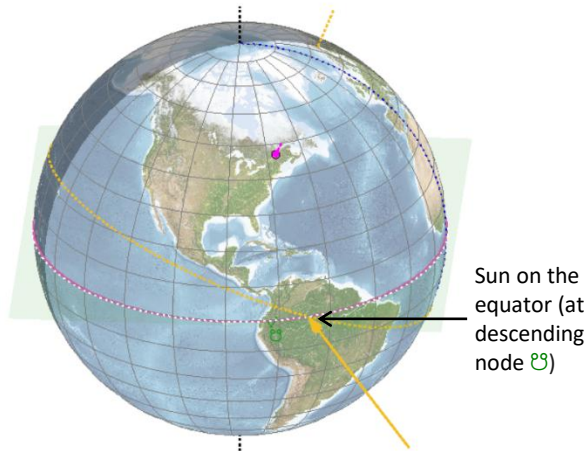
Winter solstice → ❄️ ☾ ← Spring equinox
Summer solstice → ☀️ ♈ ← Autumn equinox

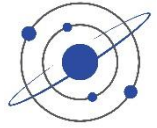




Autumn Equinox

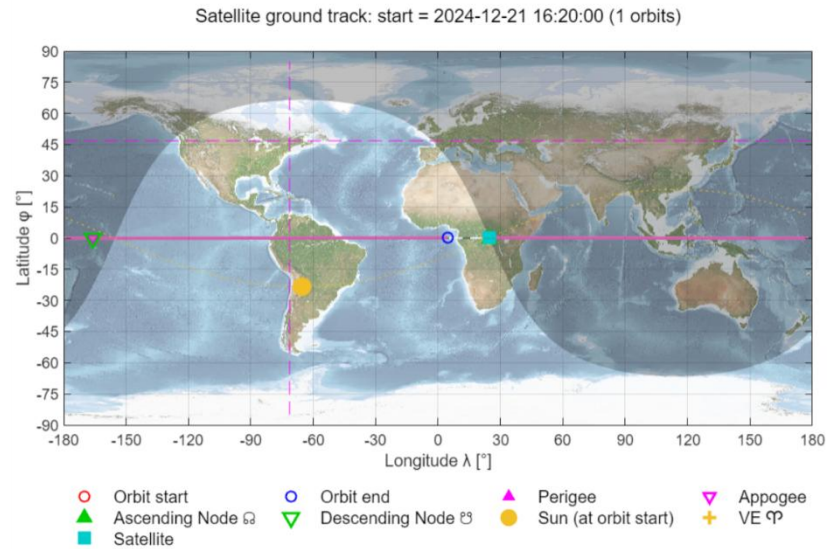
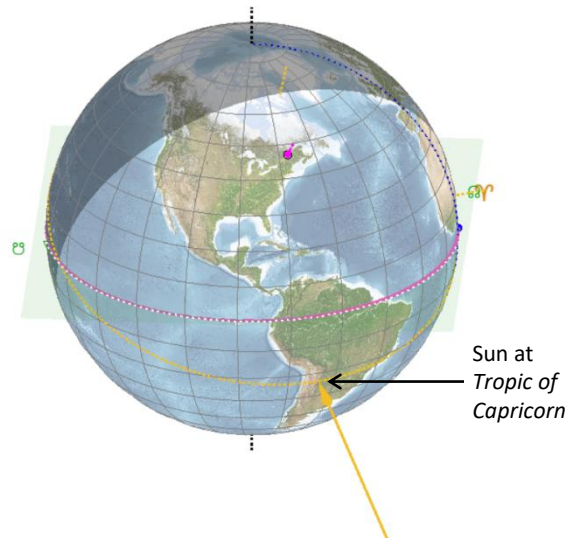
Winter solstice → ❄️ ☾ ← Spring equinox
 Summer solstice → ☀️ ♊ ← **Autumn equinox**





Winter Solstice

Winter solstice → ☸ ☿ ← Spring equinox
 Summer solstice → ☼ ♏ ← Autumn equinox

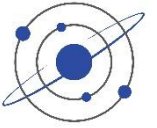


The seasons at different latitudes



- The seasonal effects are different at different latitudes on Earth
- Near the equator, for instance, all seasons are much the same
 - Every day of the year, the Sun is up half the time, so there are ≈ 12 hours of sunshine and 12 hours of night
 - Local residents define the seasons by the amount of rain (wet season and dry season) rather than by the amount of sunlight
- As we travel north or south, the seasons become more pronounced, until we reach extreme cases in the Arctic and Antarctic
 - At the North Pole, all celestial objects that are north of the celestial equator are always above the horizon and, as Earth turns, circle around parallel to it
 - The Sun is north of the celestial equator from about March 21 to September 21, so at the North Pole, the Sun rises when it reaches the vernal equinox and sets when it reaches the autumnal equinox
 - Each year there are 6 months of sunshine at each pole, followed by 6 months of darkness

Seasonal lag



- The coldest and warmest months often don't align with the winter and summer solstices: this phenomenon is called **seasonal lag**
- While sunshine hours influence temperatures, there's a **delay between changes in sunlight and changes in temperature**
- This delay **occurs because water**, which covers 71% of Earth's surface, **heats and cools more slowly than land** due to its higher heat capacity – the amount of energy needed to change its temperature
 - Water absorbs solar radiation, gradually releasing this heat into the air through processes like evaporation. This slow heating and cooling process delays the impact of the solstices, meaning the warmest or coldest temperatures typically occur weeks after the peak or minimum sunlight

Real world effects



■ Atmospheric refraction effects

- Affects apparent position of celestial bodies such as the Sun, Moon, Planets, and stars
- In the preceding analysis, we have described the rising and setting of the Sun as they would appear if Earth possessed no atmosphere. In reality, the terrestrial atmosphere alters the apparent positions of celestial objects by refracting their light rays, effectively **lifting them above the true geometric horizon**
- This effect is a **result of refraction**, the bending of light passing through air (like in water). This continuous variation of the air's refractive index with altitude causes light to follow a curved path as it propagates through the atmosphere.
- Because of this atmospheric refraction (and the fact that the Sun is not a point of light but a disk), the Sun appears to rise earlier and to set later than it would if no atmosphere were present

■ Atmospheric scattering

- Moreover, the atmosphere scatters light and provides some twilight illumination even when the Sun is below the horizon. Astronomers define **morning twilight** as beginning when the Sun is 18° below the horizon, and **evening twilight** extends until the Sun sinks more than 18° below the horizon

■ These atmospheric effects **require small corrections** to many standard descriptions of the seasons.

- At the equinoxes, for example, the Sun appears to be above the horizon **for a few minutes longer** than 12 hours, and below the horizon for fewer than 12 hours. These effects are most dramatic at Earth's poles, where the Sun actually can be seen more than a week before it reaches the celestial equator

Time-lapse seasons demonstrator



- <https://astro.unl.edu/classaction/animations/coordsmotion/transitmovie.html>

